TM Forum Specification

Resource Pool Management User Guide

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Introduction

The following document is the user guide of the REST API for Resource Pool management. It includes the model definition as well as all available operations.

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Sample Use Cases

The following scenario is used in the further chapters to demonstrate the use of the API model structure.

UC: PhoneNumber Pool Management

In this scenario, the CSP obtained a range of Phone Numbers from the regulatory agency.

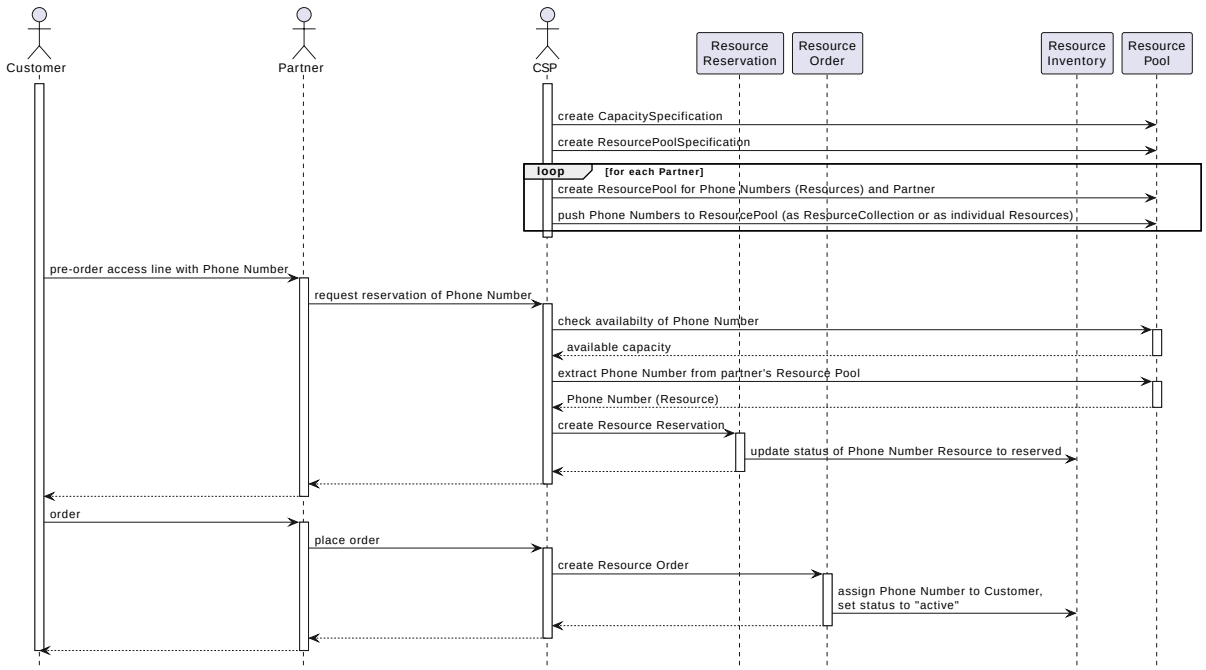
The CSP splits that range into smaller parts, each dedicated to one of CSP's partners. For each partner, CSP creates a Resource Pool for Phone Numbers (attaching Phone Number ResourceSpecification to the Pool), with the corresponding Capacity, and fills the respective Phone Number Resources into the Pool by a Push task.

Before the CSP can create the individual Resource Pools, he sets up a ResourcePoolSpecification; to be able to attach capacities, a CapacitySpecification is needed.

A partner can then (via the Resource Reservation API, as part of the pre-order process) request reservation of individual Phone Numbers.

Precondition:

- There is a ResourceSpecification in the Resource Catalog that defines how Resources representing a Phone Number are structured.
- The partners exist as PartyRole with attached Organization (Party)



Note: The inclusion of Resource Reservation and Order in the diagram is out of scope of the functionality of the Resource Pool, but shown here only for reasons of illustration.

Remark: Although CSP and/or Partner might not be persons, but e.g. automated

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applications on CSP resp. Partenr side, they are shown as persons, to symbolize that from the Enterprise's perspective the occurrence of their requests is out of the Enterprise's control.

A Resource Pool represents a collection of Resources of some type, modelled by a Resource Specification. The Pool can contain specific Capacities of such Resources, without the need to instanciate the Resources. For example, a range of Phone Number can be modeled as Capacity in a Pool with having to create individual Resources for the numbers.

The Pool can be filled by either pushing Resource instances into it, or by updating the Pool with additional Capacities.

The Pool can be queried for a specific amount (capacityDemand) of Resources, and returns possible options to fulfill the demand. The availabilty check only returns available capacities, not any individual resources. This avoids that the Resources are used by some follow-up process; the Pool would then loose control of whether the Resources are still available in the Pool of have been used elsewhere.

To get individual Resources from the Pool, they can be extracted. Resources that are extracted from the Pool may or may not be stored in the Resource inventory. How instantiation of Resources is handled, and in particular if and by which IT component their ID is generated, is out of scope of the management of the Resource Pool.

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Support of polymorphism and extension patterns

Support of polymorphic collections and types and schema based extension is provided by means of a list of generic meta-attributes that we describe below. Polymorphism in collections occurs when entities inherit from base entities, for instance a BillingAccount and SettlementAccount inheriting properties from the abstract Account entity.

Generic support of polymorphism and pattern extensions is described in the TMF API Guidelines, Part 2 (TMF630).

The @type attribute provides a way to represent the actual class type of an entity. For example, within a list of Account instances some may be instances of BillingAccount where other could be instances of SettlementAccount. The @type gives this information. All resources and sub-resources of this API have a @type attributes that can be provided when this is useful.

The @referredType can be used within reference entities (like for instance an AccountRef object) to explicitly denote the actual entity type of the referred class. Notice that in reference entities the @type, when used, denotes the class type of the reference itself, such as BillingAccountRef or SettlementAccountRef, and not the class type of the referred object. However since reference classes are rarely sub-classed, @type is generally not useful in reference objects.

The @schemaLocation property can be used in resources to allow specifying user-defined properties of an Entity or to specify the expected characteristics of an entity.

The @baseType attribute gives a way to provide explicitly the base of class of a given resource that has been extended.

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RESOURCE MODEL

Managed Entity and Task Resource Models

CapacitySpecification resource

Capacity is a specific amount of an entity measured in quantity and units of quantity over an extended period. In essence, a CapacitySpecification defines the common attributes and relationships of a set of related Capacities, while Capacity defines a specific instance that is based on a particular CapacitySpecification.

Resource model

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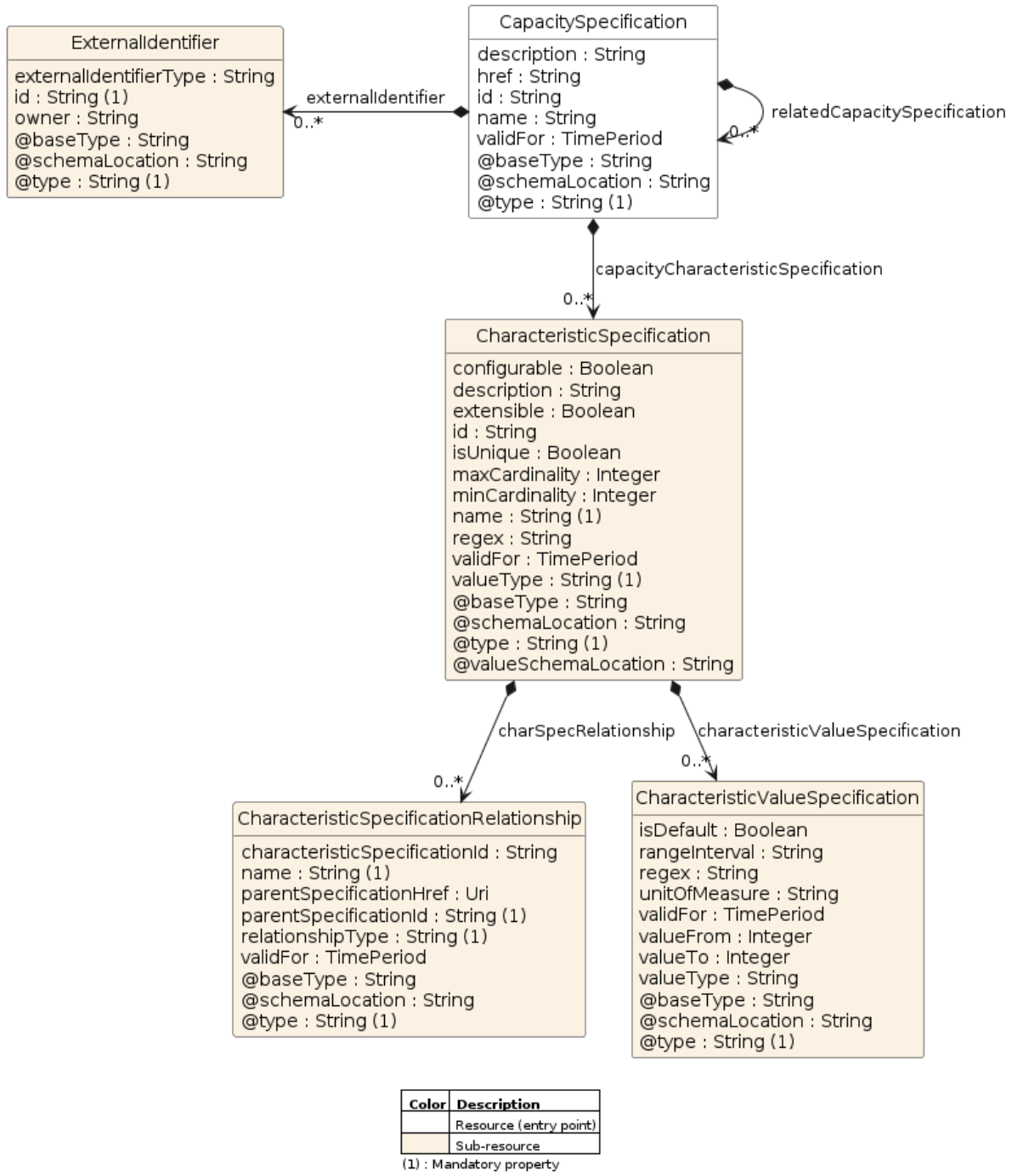


Figure 1 - CapacitySpecification

Field descriptions

CapacitySpecification fields

capacityCharacteristicSpecification	A CharacteristicSpecification. This class defines a characteristic specification.
description	A String. Description of this REST resource.

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externalIdentifier	An ExternalIdentifier. An identification of an entity that is owned by or originates in a software system different from the current system, for example a ProductOrder handed off from a commerce platform into an order handling system. The structure identifies the system itself, the nature of the entity within the system (e.g. class name) and the unique ID of the entity within the system. It is anticipated that multiple external IDs can be held for a single entity, e.g. if the entity passed through multiple systems on the way to the current system. In this case the consumer is expected to sequence the IDs in the array in reverse order of provenance, i.e. most recent system first in the list.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name given to this REST resource.
relatedCapacitySpecification	A CapacitySpecification. Capacity is a specific amount of an entity measured in quantity and units of quantity over an extended period. In essence, a CapacitySpecification defines the common attributes and relationships of a set of related Capacities, while Capacity defines a specific instance that is based on a particular CapacitySpecification.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ExternalIdentifier sub-resource fields

externalIdentifierType	A String. Type of the identification, typically would be the type of the entity within the external system.
id	A String. Identification of the entity within the external system.
owner	A String. Name of the external system that owns the entity.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

CharacteristicSpecification sub-resource fields

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charSpecRelationship	A CharacteristicSpecificationRelationship. An aggregation, migration, substitution, dependency or exclusivity relationship between/among Characteristic specifications. The specification characteristic is embedded within the specification whose ID and href are in this entity, and identified by its ID.
characteristicValueSpecification	A CharacteristicValueSpecification. Specification of a value (number or text or an object) that can be assigned to a Characteristic. This is an abstract base class, the actual value is in one of the strongly-typed subclasses : BooleanArrayCharacteristicValueSpecification, BooleanCharacteristicValueSpecification, FloatArrayCharacteristicValueSpecification, FloatCharacteristicValueSpecification, IntegerArrayCharacteristicValueSpecification, IntegerCharacteristicValueSpecification, MapArrayCharacteristicValueSpecification, MapCharacteristicValueSpecification, NumberArrayCharacteristicValueSpecification, NumberCharacteristicValueSpecification, ObjectArrayCharacteristicValueSpecification, ObjectCharacteristicValueSpecification, StringArrayCharacteristicValueSpecification, StringCharacteristicValueSpecification…
configurable	A Boolean. If true, the Boolean indicates that the target Characteristic is configurable.
description	A String. A narrative that explains the CharacteristicSpecification.
extensible	A Boolean. An indicator that specifies that the values for the characteristic can be extended by adding new values when instantiating a characteristic for a resource.
id	A String. Unique ID for the characteristic.
isUnique	A Boolean. Specifies if the value of this characteristic is unique across all entities instantiated from the specification that uses this characteristic. For example, consider a ProductSpecification for a set-top box, with a CharacteristicSpecification cardID. Each set-top box must have a different value for cardID, so this isUnique attribute would be set to true for the characteristic.
maxCardinality	An Integer. The maximum number of instances a CharacteristicValue can take on. For example, zero to five phone numbers in a group calling plan, where five is the value for the maxCardinality.
minCardinality	An Integer. The minimum number of instances a CharacteristicValue can take on. For example, zero to five phone numbers in a group calling plan, where zero is the value for the minCardinality.
name	A String. A word, term, or phrase by which this characteristic specification is known and distinguished from other characteristic specifications.
regex	A String. A rule or principle represented in regular expression used to derive the value of a characteristic value.

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validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
valueType	A String. A kind of value that the characteristic can take on, such as numeric, text and so forth.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.
@valueSchemaLocation	A String. This (optional) field provides a link to the schema describing the value type.

CharacteristicSpecificationRelationship sub-resource fields

characteristicSpecificationId	A String. Unique identifier of the characteristic within the specification.
name	A String. Name of the target characteristic within the specification.
parentSpecificationHref	A Uri. Hyperlink reference to the parent specification containing the target characteristic.
parentSpecificationId	A String. Unique identifier of the parent specification containing the target characteristic.
relationshipType	A String. Type of relationship such as aggregation, migration, substitution, dependency, exclusivity.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

CharacteristicValueSpecification sub-resource fields

isDefault	A Boolean. If true, the Boolean Indicates if the value is the default value for a characteristic.
rangeInterval	A String. An indicator that specifies the inclusion or exclusion of the valueFrom and valueTo attributes. If applicable, possible values are "open", "closed", "closedBottom" and "closedTop".
regex	A String. A regular expression constraint for given value.
unitOfMeasure	A String. A length, surface, volume, dry measure, liquid measure, money, weight, time, and the like. In general, a determinate quantity or magnitude of the kind designated, taken as a standard of comparison for others of the same kind, in assigning to them numerical values, as 1 foot, 1 yard, 1 mile, 1 square foot.

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validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
valueFrom	An Integer. The low range value that a characteristic can take on.
valueTo	An Integer. The upper range value that a characteristic can take on.
valueType	A String. A kind of value that the characteristic value can take on, such as numeric, text and so forth.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Json representation sample(s)

The following example demonstrates the details of the CapacitySpecification resource.


```
{
    "@type": "CapacitySpecification",
    "id": "3d01344d-4409-4b15-8fcb-b8802314ff9c",
    "href": "https://{server}:{port}/tmf-
api/resourcePoolManagement/v5/capacitySpecification/3d01344d-4409-4b15-8fcb-
b8802314ff9c",
    "name": "Phone Number range",
    "description": "This CapacitySpecification defines the details of capacity for
Phone Number ranges.",
    "capacityCharacteristicSpecification":
    [
        {
            "@type": "CharacteristicSpecification",
            "name": "lowerNumber",
            "description": "the lower boundary of a block of phone numbers; phone
numbers consist of a country code (cc), national destination code (ndc), subscriber
number (sn), and optionally a direct dial in (ddi), all sections separated by '-'",
            "valueType": "string",
            "characteristicValueSpecification":
            [
                {
                    "@type": "CharacteristicValueSpecification",
                    "valueType": "string",
                    "regex": "^\\+(?<cc>[1-9]\\d{0,2})-(?<ndc>[1-9]\\d*)-(?<sn>[1-
9]\\d*)(?:-?(?<ddi>\\d+))?$"
                }
            ]
        },
        {
            "@type": "CharacteristicSpecification",
            "name": "upperNumber",
            "description": "the upper boundary of a block of phone numbers; phone
numbers consist of a country code (cc), national destination code (ndc), subscriber
number (sn), and optionally a direct dial in (ddi), all sections separated by '-'",
            "valueType": "string",
            "characteristicValueSpecification":
            [
                {
```


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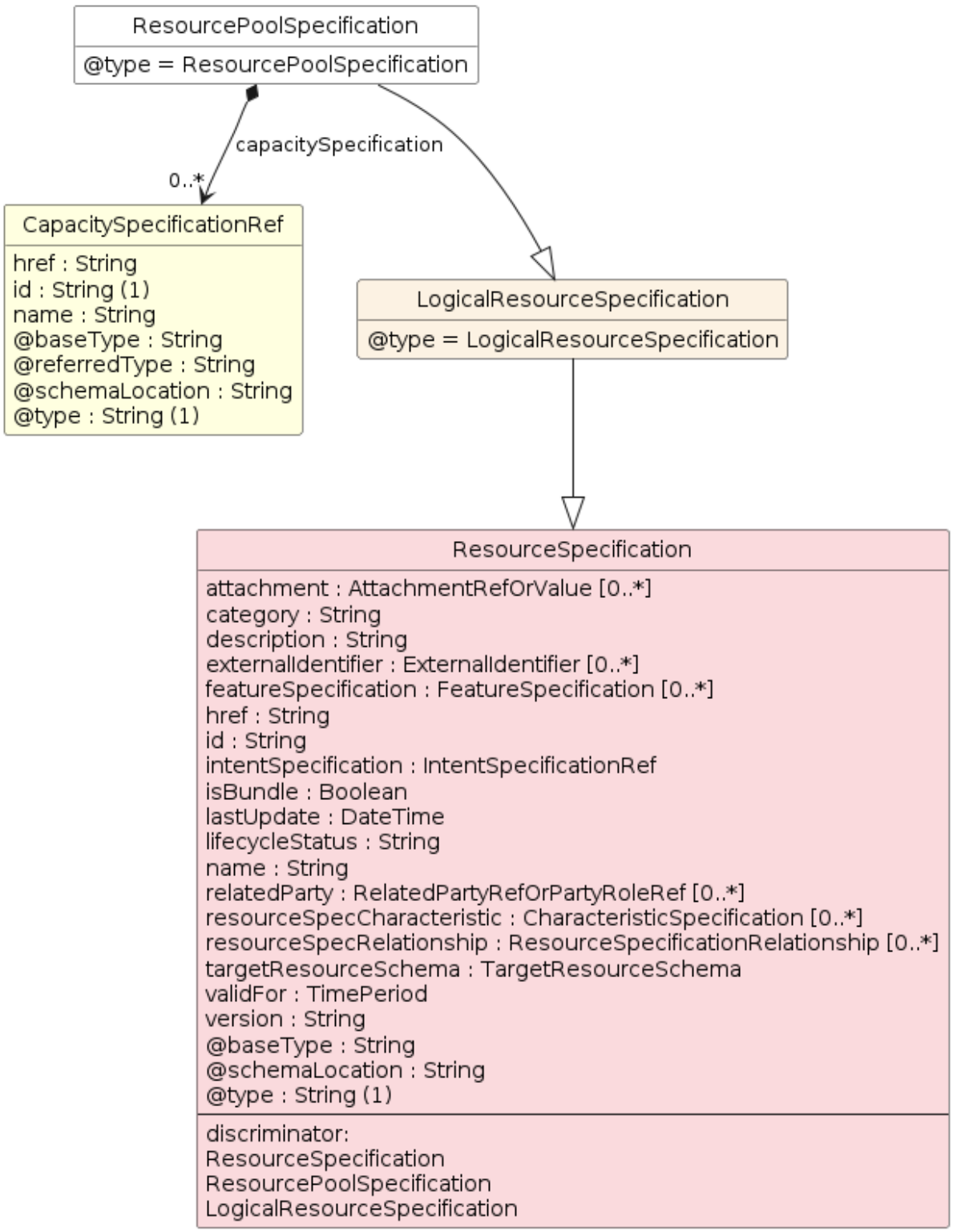
```
                    "@type": "CharacteristicValueSpecification",
                    "valueType": "string",
                    "regex": "^\\+(?<cc>[1-9]\\d{0,2})-(?<ndc>[1-9]\\d*)-(?<sn>[1-
9]\\d*)(?:-?(?<ddi>\\d+))?$"
                }
            ]
        }
    ]
}
```

ResourcePoolSpecification resource

Defines the common characteristics of a specific kind of Resource Pools, including the types of Capacities that can be handled by Pools of this type. Resource Pools are managed as (Logical) Resources, and accordingly Resource Pool Specifications are managed the same way as (Logical) Resource Specification, e.g. put into the Resource Catalog.

Resource model

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Color	Description
	Resource (entry point)
	Sub-resource
	Sub-resource (reference entity)
	Sub-resource with details in separate diagram

(1) : Mandatory property

Figure 2 - ResourcePoolSpecification

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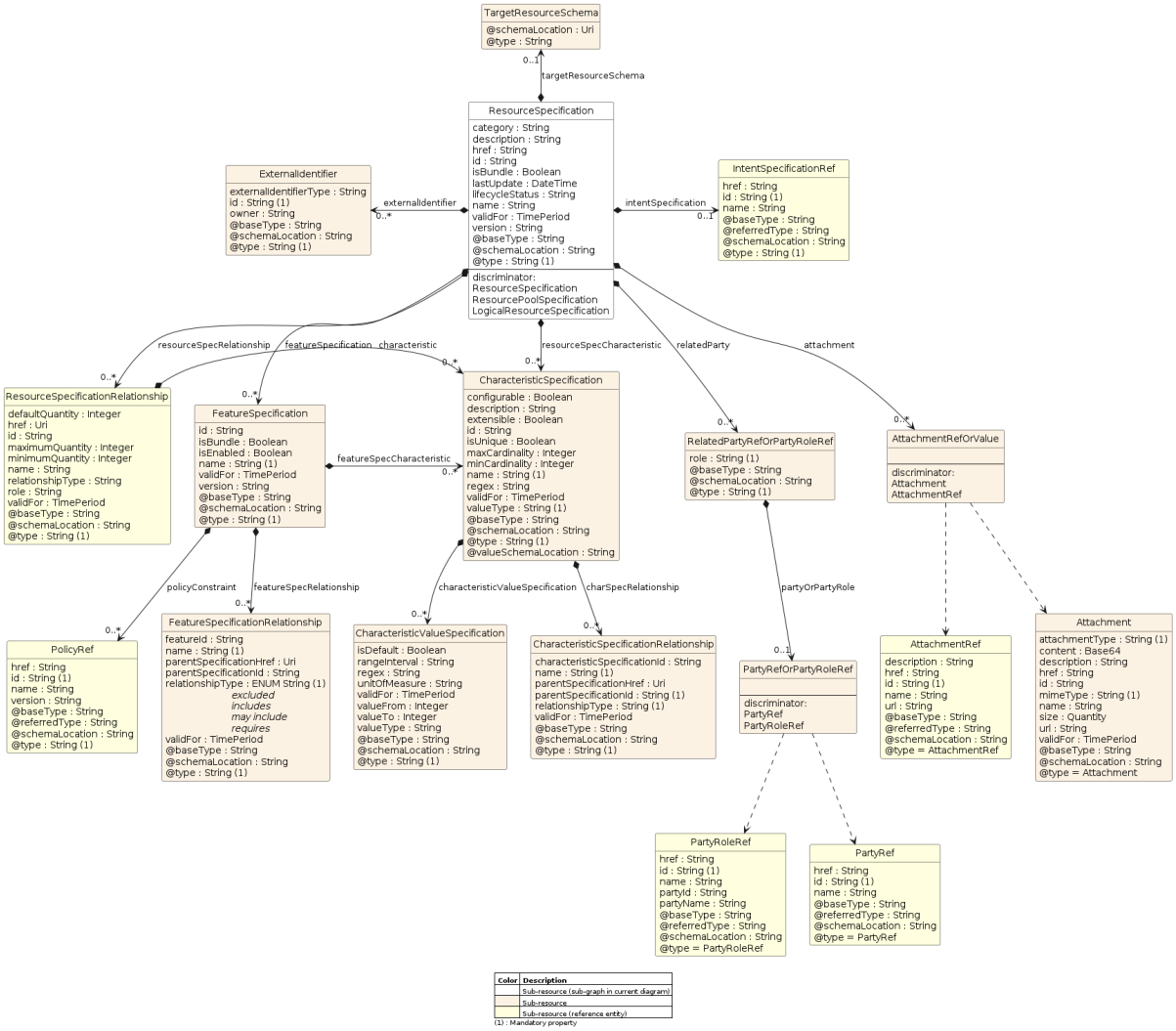


Figure 3 - ResourceSpecification

Field descriptions

ResourcePoolSpecification fields

attachment	An AttachmentRefOrValue. The polymorphic attributes @type, @schemaLocation & @referredType are related to the Attachment entity and not the AttachmentRefOrValue class itself.
capacitySpecification	A CapacitySpecificationRef. CapacitySpecification reference, for when CapacitySpecification is used by other entities.
category	A String. Category of the target resource like NetworkConnectivity, PhysicalLinks, Generic, L2Network and so on.
description	A String. Description of this REST resource.

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externalIdentifier	An ExternalIdentifier. An identification of an entity that is owned by or originates in a software system different from the current system, for example a ProductOrder handed off from a commerce platform into an order handling system. The structure identifies the system itself, the nature of the entity within the system (e.g. class name) and the unique ID of the entity within the system. It is anticipated that multiple external IDs can be held for a single entity, e.g. if the entity passed through multiple systems on the way to the current system. In this case the consumer is expected to sequence the IDs in the array in reverse order of provenance, i.e. most recent system first in the list.
featureSpecification	A FeatureSpecification. Specification for an entity features.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
intentSpecification	An IntentSpecificationRef.
isBundle	A Boolean. A flag indicates that if this resource specification is a bundled specification (true) or single (false).
lastUpdate	A DateTime. Date and time of the last update of this REST resource.
lifecycleStatus	A String. Used to indicate the current lifecycle status of the resource specification.
name	A String. Name given to this REST resource.
relatedParty	A RelatedPartyRefOrPartyRoleRef. RelatedParty reference. A related party defines party or party role or its reference, linked to a specific entity.
resourceSpecCharacteristic	A CharacteristicSpecification. This class defines a characteristic specification.
resourceSpecRelationship	A ResourceSpecificationRelationship. A migration, substitution, dependency or exclusivity relationship between/among resource specifications.
targetResourceSchema	A TargetResourceSchema. The reference object to the schema and type of target resource which is described by resource specification.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
version	A String. Resource Specification version.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

CapacitySpecificationRef sub-resource fields

href	A String. Hyperlink reference.

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id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ResourceSpecification sub-resource fields

attachment	An AttachmentRefOrValue. The polymorphic attributes @type, @schemaLocation & @referredType are related to the Attachment entity and not the AttachmentRefOrValue class itself.
category	A String. Category of the target resource like NetworkConnectivity, PhysicalLinks, Generic, L2Network and so on.
description	A String. Description of this REST resource.
externalIdentifier	An ExternalIdentifier. An identification of an entity that is owned by or originates in a software system different from the current system, for example a ProductOrder handed off from a commerce platform into an order handling system. The structure identifies the system itself, the nature of the entity within the system (e.g. class name) and the unique ID of the entity within the system. It is anticipated that multiple external IDs can be held for a single entity, e.g. if the entity passed through multiple systems on the way to the current system. In this case the consumer is expected to sequence the IDs in the array in reverse order of provenance, i.e. most recent system first in the list.
featureSpecification	A FeatureSpecification. Specification for an entity features.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
intentSpecification	An IntentSpecificationRef.
isBundle	A Boolean. A flag indicates that if this resource specification is a bundled specification (true) or single (false).
lastUpdate	A DateTime. Date and time of the last update of this REST resource.
lifecycleStatus	A String. Used to indicate the current lifecycle status of the resource specification.
name	A String. Name given to this REST resource.
relatedParty	A RelatedPartyRefOrPartyRoleRef. RelatedParty reference. A related party defines party or party role or its reference, linked to a specific entity.

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resourceSpecCharacteristic	A CharacteristicSpecification. This class defines a characteristic specification.
resourceSpecRelationship	A ResourceSpecificationRelationship. A migration, substitution, dependency or exclusivity relationship between/among resource specifications.
targetResourceSchema	A TargetResourceSchema. The reference object to the schema and type of target resource which is described by resource specification.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
version	A String. Resource Specification version.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.
	ResourceSpecification can be instanciated as * LogicalResourceSpecification

LogicalResourceSpecification sub-resource fields

attachment	An AttachmentRefOrValue. The polymorphic attributes @type, @schemaLocation & @referredType are related to the Attachment entity and not the AttachmentRefOrValue class itself.
category	A String. Category of the target resource like NetworkConnectivity, PhysicalLinks, Generic, L2Network and so on.
description	A String. Description of this REST resource.
externalIdentifier	An ExternalIdentifier. An identification of an entity that is owned by or originates in a software system different from the current system, for example a ProductOrder handed off from a commerce platform into an order handling system. The structure identifies the system itself, the nature of the entity within the system (e.g. class name) and the unique ID of the entity within the system. It is anticipated that multiple external IDs can be held for a single entity, e.g. if the entity passed through multiple systems on the way to the current system. In this case the consumer is expected to sequence the IDs in the array in reverse order of provenance, i.e. most recent system first in the list.
featureSpecification	A FeatureSpecification. Specification for an entity features.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
intentSpecification	An IntentSpecificationRef.
isBundle	A Boolean. A flag indicates that if this resource specification is a bundled specification (true) or single (false).

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lastUpdate	A DateTime. Date and time of the last update of this REST resource.
lifecycleStatus	A String. Used to indicate the current lifecycle status of the resource specification.
name	A String. Name given to this REST resource.
relatedParty	A RelatedPartyRefOrPartyRoleRef. RelatedParty reference. A related party defines party or party role or its reference, linked to a specific entity.
resourceSpecCharacteristic	A CharacteristicSpecification. This class defines a characteristic specification.
resourceSpecRelationship	A ResourceSpecificationRelationship. A migration, substitution, dependency or exclusivity relationship between/among resource specifications.
targetResourceSchema	A TargetResourceSchema. The reference object to the schema and type of target resource which is described by resource specification.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
version	A String. Resource Specification version.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.
	LogicalResourceSpecification can be instanciated as * ResourcePoolSpecification
capacitySpecification	This property is present in subclasses

Json representation sample(s)

The following example demonstrates the details of the ResourcePoolSpecification resource.

```
{
    "@type": "ResourcePoolSpecification",
    "id": "b0e0827c-5693-4e41-8445-22438abb33cb",
    "href": "https://{server}:{port}/tmf-
api/resourcePoolManagement/v5/resourcePoolSpecification/b0e0827c-5693-4e41-8445-
22438abb33cb",
    "name": "Phone Number pool",
    "description": "Specifies the details of ResourcePools for Phone Number ranges",
    "capacitySpecification":
    [
        {
            "@type": "CapacitySpecificationRef",
            "@referredType": "CapacitySpecification",
            "id": "3d01344d-4409-4b15-8fcb-b8802314ff9c",
            "href": "https://{server}:{port}/tmf-
api/resourcePoolManagement/v5/capacitySpecification/3d01344d-4409-4b15-8fcb-
```

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```
b8802314ff9c",
            "name": "Phone Number range"
        }
    ],
    "resourceSpecCharacteristic":
    [
        {
            "@type": "CharacteristicSpecification",
            "name": "cc",
            "description": "Phone Number Country Code",
            "valueType": "string",
            "characteristicValueSpecification":
            [
                {
                    "@type": "CharacteristicValueSpecification",
                    "valueType": "string",
                    "regex": "^[1-9]\\d{0,2}$"
                }
            ]
        },
        {
            "@type": "CharacteristicSpecification",
            "name": "ndc",
            "description": "Phone Number National Destination Code",
            "valueType": "string",
            "characteristicValueSpecification":
            [
                {
                    "@type": "CharacteristicValueSpecification",
                    "valueType": "string",
                    "regex": "^[1-9]\\d*?$"
                }
            ]
        }
    ]
}
```

ResourcePool resource

Represents a set of Resource of a given type (represented by a ResourceSpecification), without the need to instantiate all Resource entities, e.g. all phone numbers in a given range. The amount of Resources in th pool is modelled as Capacity. The Pool can be querried for some amount (Capacity) of availabile resources; actual Resource instances can be pushed into the pool, and contained Resources can be extracted. The Pool itself is understood to be a (Logical) Resource in itself, so it can be managed the same way as any other Resource, e.g. put into the Resource inventory, and can be associated with some Party and located at some Place.

Resource model

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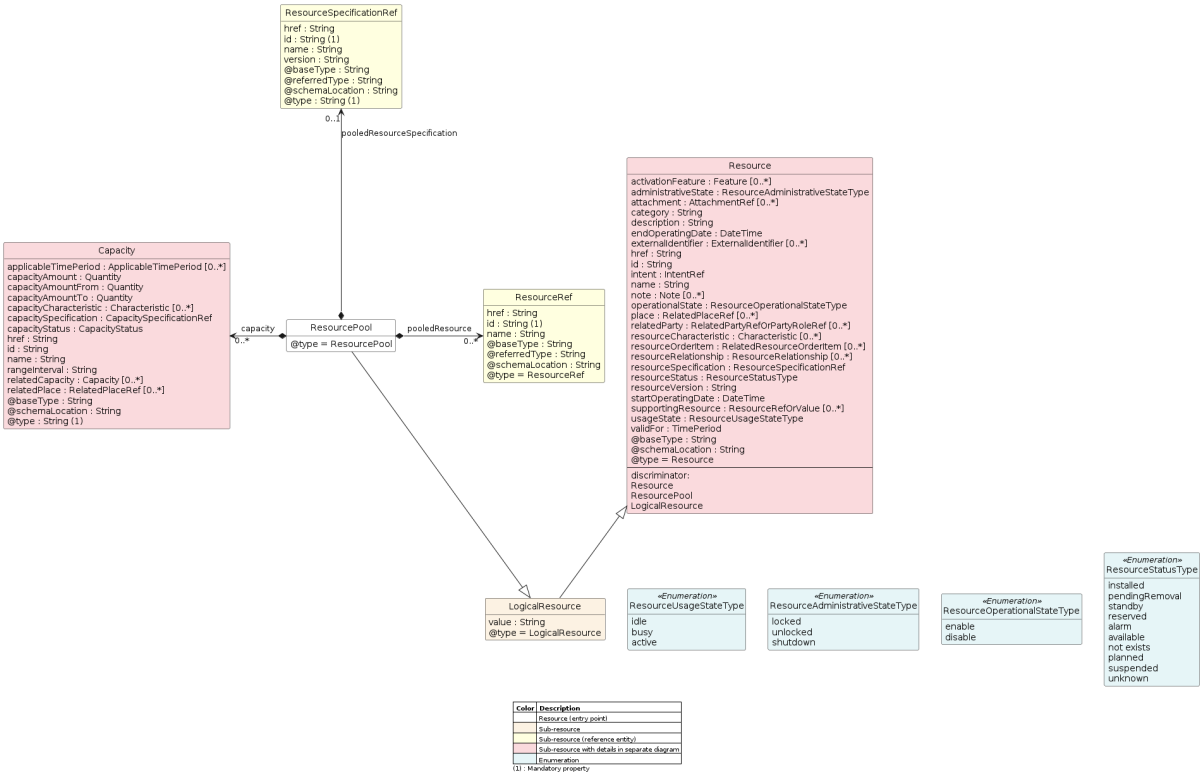


Figure 4 - ResourcePool

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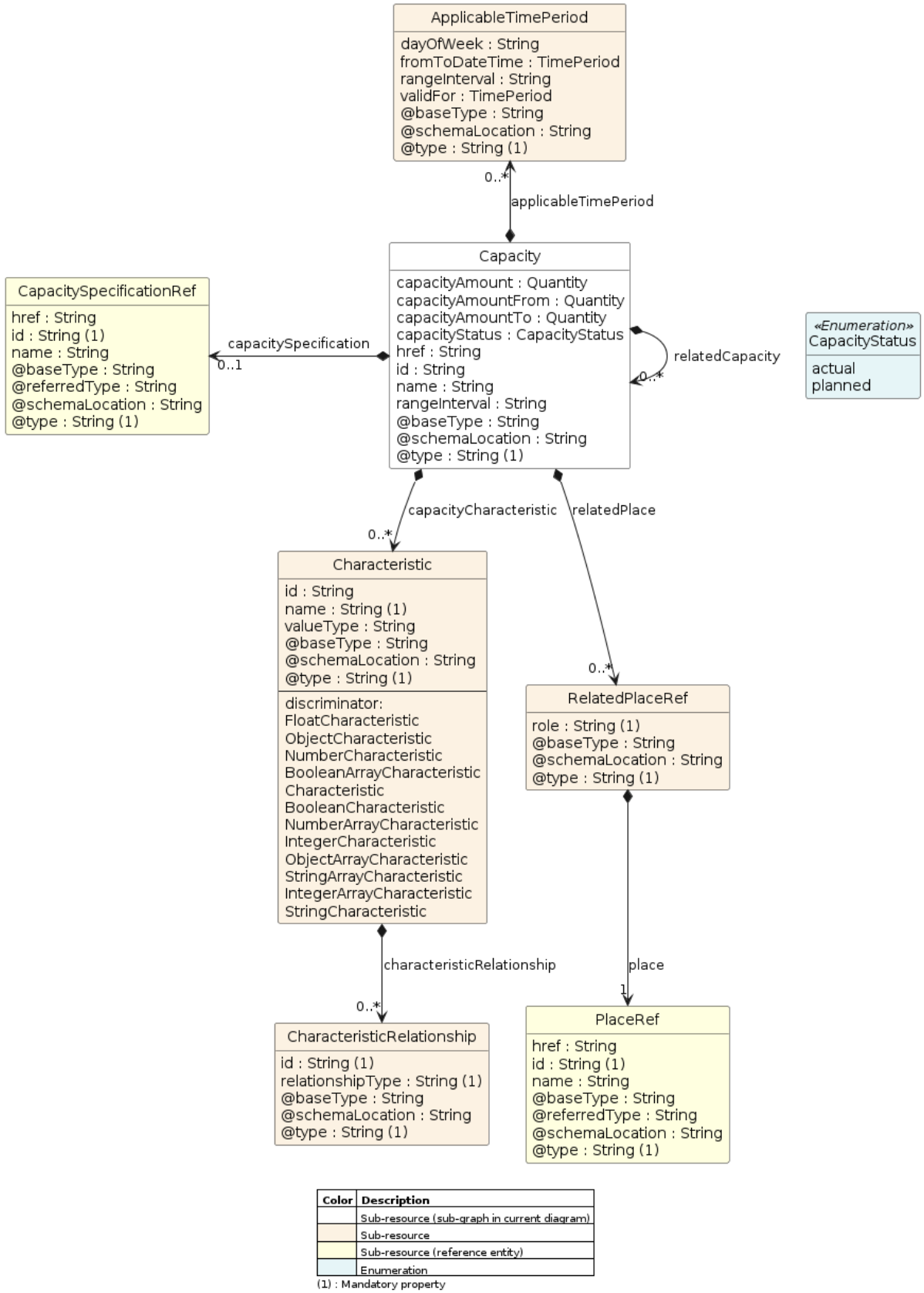


Figure 5 - Capacity

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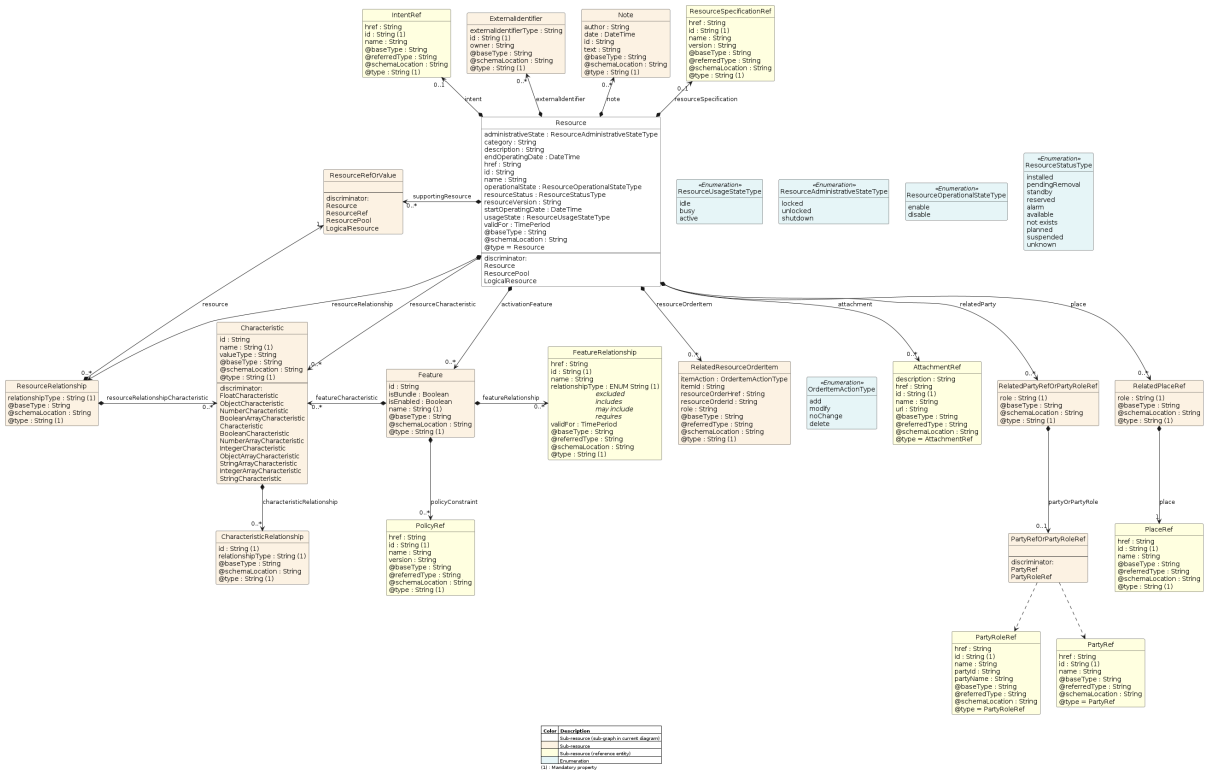


Figure 6 - Resource

Field descriptions

ResourcePool fields

activationFeature	A Feature. Configuration feature.
administrativeState	A ResourceAdministrativeStateType. ResourceAdministrativeStateType enumerations; values defined by ITU X.731: 'locked': The resource is administratively prohibited from performing services for its users; 'shutdown': Use of the resource is administratively permitted to existing instances of use only. While the system remains in the shutting down state the manager may at any time cause the managed object to revert to the unlocked state; 'unlocked': The resource is administratively permitted to perform services for its users. This is independent of its inherent operability. ENUMERATED with values: * locked * unlocked * shutdown
attachment	An AttachmentRef. Attachment reference. An attachment complements the description of an element (for instance a product) through video, pictures.
capacity	A Capacity. Specific ability of an entity measured in quantity and units of quantity over an extended period.
category	A String. Category of the concrete resource. e.g Gold, Silver for MSISDN concrete resource.

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description	A String. Free-text description of the resource.
endOperatingDate	A DateTime. A date time(DateTime). The date till the resource is operating.
externalIdentifier	An ExternalIdentifier. An identification of an entity that is owned by or originates in a software system different from the current system, for example a ProductOrder handed off from a commerce platform into an order handling system. The structure identifies the system itself, the nature of the entity within the system (e.g. class name) and the unique ID of the entity within the system. It is anticipated that multiple external IDs can be held for a single entity, e.g. if the entity passed through multiple systems on the way to the current system. In this case the consumer is expected to sequence the IDs in the array in reverse order of provenance, i.e. most recent system first in the list.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
intent	An IntentRef. Intent reference, for when Intent is used by other entities.
name	A String. The name of the resource.
note	A Note. Extra information about a given entity.
operationalState	A ResourceOperationalStateType. ResourceOperationalStateType enumerations; values defined by ITU X.731: 'disable': The resource is totally inoperable and unable to provide service to the user(s); 'enable': The resource is partially or fully operable and available for use. ENUMERATED with values: * enable * disable
place	A RelatedPlaceRef. Entity reference. The polymorphic attributes @type, @schemaLocation & @referredType are related to the RelatedPlace entity and not the RelatedPlaceRef class itself.
pooledResource	A ResourceRef. Resource reference, for when Resource is used by other entities.
pooledResourceSpecification	A ResourceSpecificationRef. Resources are physical or non-physical components (or some combination of these) within an enterprise's infrastructure or inventory. They are typically consumed or used by services (for example a physical port assigned to a service) or contribute to the realization of a Product (for example, a SIM card). They can be drawn from the Application, Computing and Network domains, and include, for example, Network Elements, software, IT systems, content and information, and technology components. A ResourceSpecification is an abstract base class for representing a generic means for implementing a particular type of Resource. In essence, a ResourceSpecification defines the common attributes and relationships of a set of related Resources, while Resource defines a specific instance that is based on a particular ResourceSpecification.

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relatedParty	A RelatedPartyRefOrPartyRoleRef. RelatedParty reference. A related party defines party or party role or its reference, linked to a specific entity.
resourceCharacteristic	A Characteristic. Describes a given characteristic of an object or entity through a name/value pair. This is an abstract base class, the actual value is in one of the strongly-typed subclasses : StringCharacteristic, ObjectCharacteristic, FloatCharacteristic, BooleanCharacteristic, NumberCharacteristic, IntegerCharacteristic, StringArrayCharacteristic, ObjectArrayCharacteristic, BooleanArrayCharacteristic, NumberArrayCharacteristic, IntegerArrayCharacteristic… Characteristic can be instanciated as * BooleanArrayCharacteristic * BooleanCharacteristic * FloatCharacteristic * IntegerArrayCharacteristic * IntegerCharacteristic * NumberArrayCharacteristic * NumberCharacteristic * ObjectArrayCharacteristic * ObjectCharacteristic * StringArrayCharacteristic * StringCharacteristic
resourceOrderItem	A RelatedResourceOrderItem. RelatedResourceOrderItem (a ResourceOrder item) .The resource order item which triggered resource creation/change/termination.
resourceRelationship	A ResourceRelationship. Linked resources to the one instantiate, such as [bundled] if the resource is a bundle and you want to describe the bundled resources inside this bundle; [reliesOn] if the resource needs another already owned resource to rely on (e.g. an option on an already owned mobile access resource) [targets] or [isTargeted] (depending on the way of expressing the link) for any other kind of links that may be useful.
resourceSpecification	A ResourceSpecificationRef. Resources are physical or non-physical components (or some combination of these) within an enterprise's infrastructure or inventory. They are typically consumed or used by services (for example a physical port assigned to a service) or contribute to the realization of a Product (for example, a SIM card). They can be drawn from the Application, Computing and Network domains, and include, for example, Network Elements, software, IT systems, content and information, and technology components. A ResourceSpecification is an abstract base class for representing a generic means for implementing a particular type of Resource. In essence, a ResourceSpecification defines the common attributes and relationships of a set of related Resources, while Resource defines a specific instance that is based on a particular ResourceSpecification.

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resourceStatus	A ResourceStatusType. ResourceStatusType enumerations. ENUMERATED with values: * alarm * available * installed * not exists * pendingRemoval * planned * reserved * standby * suspended * unknown
resourceVersion	A String. A field that identifies the specific version of an instance of a resource.
startOperatingDate	A DateTime. A date time(DateTime). The date from which the resource is operating.
supportingResource	A ResourceRefOrValue. The polymorphic attributes @type, @schemaLocation & @referredType are related to the Resource entity and not the ResourceRefOrValue class itself.
usageState	A ResourceUsageStateType. ResourceUsageStateType enumerations; values defined by ITU X.731: 'idle': The resource is not currently in use; 'active': The resource is in use, and has sufficient spare operating capacity to provide for additional users simultaneously; 'busy': The resource is in use, but it has no spare operating capacity to provide for additional users at this instant. ENUMERATED with values: * idle * active * busy
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
value	A String. The value of the logical resource. E.g '0746712345' for MSISDN's.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Capacity sub-resource fields

applicableTimePeriod	An ApplicableTimePeriod. The period of time for which Capacity or CapacityDemand applies.
capacityAmount	A Quantity. An amount in a given unit.
capacityAmountFrom	A Quantity. An amount in a given unit.
capacityAmountTo	A Quantity. An amount in a given unit.

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capacityCharacteristic	A Characteristic. Describes a given characteristic of an object or entity through a name/value pair. This is an abstract base class, the actual value is in one of the strongly-typed subclasses : StringCharacteristic, ObjectCharacteristic, FloatCharacteristic, BooleanCharacteristic, NumberCharacteristic, IntegerCharacteristic, StringArrayCharacteristic, ObjectArrayCharacteristic, BooleanArrayCharacteristic, NumberArrayCharacteristic, IntegerArrayCharacteristic… Characteristic can be instanciated as * BooleanArrayCharacteristic * BooleanCharacteristic * FloatCharacteristic * IntegerArrayCharacteristic * IntegerCharacteristic * NumberArrayCharacteristic * NumberCharacteristic * ObjectArrayCharacteristic * ObjectCharacteristic * StringArrayCharacteristic * StringCharacteristic
capacitySpecification	A CapacitySpecificationRef. CapacitySpecification reference, for when CapacitySpecification is used by other entities.
capacityStatus	A CapacityStatus. An indicator that specifies whether the capacity is planned or actual. ENUMERATED with values: * planned * actual
href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the Capacity; used to refer to the CapacitySpecification.
rangeInterval	A String. An indicator that specifies the inclusion or exclusion of the capacityAmount From and capacityAmountTo attributes. Possible values are "open", "closed", "closedBottom" and "closedTop".
relatedCapacity	A Capacity. Specific ability of an entity measured in quantity and units of quantity over an extended period.
relatedPlace	A RelatedPlaceRef. Entity reference. The polymorphic attributes @type, @schemaLocation & @referredType are related to the RelatedPlace entity and not the RelatedPlaceRef class itself.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ResourceRef sub-resource fields

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href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ResourceSpecificationRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
version	A String. Resource Specification version.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

CapacitySpecificationRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Characteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Another Characteristic that is related to the current Characteristic;.

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id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.
	Characteristic can be instanciated as * BooleanArrayCharacteristic * BooleanCharacteristic * FloatCharacteristic * IntegerArrayCharacteristic * IntegerCharacteristic * NumberArrayCharacteristic * NumberCharacteristic * ObjectArrayCharacteristic * ObjectCharacteristic * StringArrayCharacteristic * StringCharacteristic
value	This property is present in subclasses

FloatCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Another Characteristic that is related to the current Characteristic;.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	A Float. Value of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ObjectCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Another Characteristic that is related to the current Characteristic;.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	An object. Value of the characteristic.

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valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

NumberCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Another Characteristic that is related to the current Characteristic;.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	A Number. Value of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

BooleanArrayCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Another Characteristic that is related to the current Characteristic;.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	A Boolean. A characteristic which value is an array of Boolean(s).
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

BooleanCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Another Characteristic that is related to the current Characteristic;.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.

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value	A Boolean. Value of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

NumberArrayCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Another Characteristic that is related to the current Characteristic;.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	A Number. A characteristic which value is an array of Number(s).
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

IntegerCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Another Characteristic that is related to the current Characteristic;.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	An Integer. Value of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ObjectArrayCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Another Characteristic that is related to the current Characteristic;.
id	A String. Unique identifier of the characteristic.

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name	A String. Name of the characteristic.
value	An object. Collection of characteristic values.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

StringArrayCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Another Characteristic that is related to the current Characteristic;.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	A String. Collection of string characteristics.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

IntegerArrayCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Another Characteristic that is related to the current Characteristic;.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	An Integer. A characteristic which value is an array of Integer(s).
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

StringCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Another Characteristic that is related to the current Characteristic;.

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id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	A String. Value of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

RelatedPlaceRef sub-resource fields

place	A PlaceRef. Place reference.
role	A String.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ApplicableTimePeriod sub-resource fields

dayOfWeek	A String. A day or days representing when the schedule is applicable. For example 2, 3 represent Monday and Tuesday.
fromToDateTime	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
rangeInterval	A String. An indicator that specifies the inclusion or exclusion of the from and to DateTime attributes. Possible values are "open", "closed", "closedBottom" and "closedTop".
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

CharacteristicRelationship sub-resource fields

id	A String. Unique identifier of the characteristic.
relationshipType	A String. The type of relationship.
@baseType	A String. When sub-classing, this defines the super-class.

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@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

PlaceRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Resource sub-resource fields

activationFeature	A Feature. Configuration feature.
administrativeState	A ResourceAdministrativeStateType. ResourceAdministrativeStateType enumerations; values defined by ITU X.731: 'locked': The resource is administratively prohibited from performing services for its users; 'shutdown': Use of the resource is administratively permitted to existing instances of use only. While the system remains in the shutting down state the manager may at any time cause the managed object to revert to the unlocked state; 'unlocked': The resource is administratively permitted to perform services for its users. This is independent of its inherent operability. ENUMERATED with values: * locked * unlocked * shutdown
attachment	An AttachmentRef. Attachment reference. An attachment complements the description of an element (for instance a product) through video, pictures.
category	A String. Category of the concrete resource. e.g Gold, Silver for MSISDN concrete resource.
description	A String. Free-text description of the resource.
endOperatingDate	A DateTime. A date time(DateTime). The date till the resource is operating.

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externalIdentifier	An ExternalIdentifier. An identification of an entity that is owned by or originates in a software system different from the current system, for example a ProductOrder handed off from a commerce platform into an order handling system. The structure identifies the system itself, the nature of the entity within the system (e.g. class name) and the unique ID of the entity within the system. It is anticipated that multiple external IDs can be held for a single entity, e.g. if the entity passed through multiple systems on the way to the current system. In this case the consumer is expected to sequence the IDs in the array in reverse order of provenance, i.e. most recent system first in the list.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
intent	An IntentRef. Intent reference, for when Intent is used by other entities.
name	A String. The name of the resource.
note	A Note. Extra information about a given entity.
operationalState	A ResourceOperationalStateType. ResourceOperationalStateType enumerations; values defined by ITU X.731: 'disable': The resource is totally inoperable and unable to provide service to the user(s); 'enable': The resource is partially or fully operable and available for use. ENUMERATED with values: * enable * disable
place	A RelatedPlaceRef. Entity reference. The polymorphic attributes @type, @schemaLocation & @referredType are related to the RelatedPlace entity and not the RelatedPlaceRef class itself.
relatedParty	A RelatedPartyRefOrPartyRoleRef. RelatedParty reference. A related party defines party or party role or its reference, linked to a specific entity.

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resourceCharacteristic	A Characteristic. Describes a given characteristic of an object or entity through a name/value pair. This is an abstract base class, the actual value is in one of the strongly-typed subclasses : StringCharacteristic, ObjectCharacteristic, FloatCharacteristic, BooleanCharacteristic, NumberCharacteristic, IntegerCharacteristic, StringArrayCharacteristic, ObjectArrayCharacteristic, BooleanArrayCharacteristic, NumberArrayCharacteristic, IntegerArrayCharacteristic… Characteristic can be instanciated as * BooleanArrayCharacteristic * BooleanCharacteristic * FloatCharacteristic * IntegerArrayCharacteristic * IntegerCharacteristic * NumberArrayCharacteristic * NumberCharacteristic * ObjectArrayCharacteristic * ObjectCharacteristic * StringArrayCharacteristic * StringCharacteristic
resourceOrderItem	A RelatedResourceOrderItem. RelatedResourceOrderItem (a ResourceOrder item) .The resource order item which triggered resource creation/change/termination.
resourceRelationship	A ResourceRelationship. Linked resources to the one instantiate, such as [bundled] if the resource is a bundle and you want to describe the bundled resources inside this bundle; [reliesOn] if the resource needs another already owned resource to rely on (e.g. an option on an already owned mobile access resource) [targets] or [isTargeted] (depending on the way of expressing the link) for any other kind of links that may be useful.
resourceSpecification	A ResourceSpecificationRef. Resources are physical or non-physical components (or some combination of these) within an enterprise's infrastructure or inventory. They are typically consumed or used by services (for example a physical port assigned to a service) or contribute to the realization of a Product (for example, a SIM card). They can be drawn from the Application, Computing and Network domains, and include, for example, Network Elements, software, IT systems, content and information, and technology components. A ResourceSpecification is an abstract base class for representing a generic means for implementing a particular type of Resource. In essence, a ResourceSpecification defines the common attributes and relationships of a set of related Resources, while Resource defines a specific instance that is based on a particular ResourceSpecification.

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resourceStatus	A ResourceStatusType. ResourceStatusType enumerations. ENUMERATED with values: * alarm * available * installed * not exists * pendingRemoval * planned * reserved * standby * suspended * unknown
resourceVersion	A String. A field that identifies the specific version of an instance of a resource.
startOperatingDate	A DateTime. A date time(DateTime). The date from which the resource is operating.
supportingResource	A ResourceRefOrValue. The polymorphic attributes @type, @schemaLocation & @referredType are related to the Resource entity and not the ResourceRefOrValue class itself.
usageState	A ResourceUsageStateType. ResourceUsageStateType enumerations; values defined by ITU X.731: 'idle': The resource is not currently in use; 'active': The resource is in use, and has sufficient spare operating capacity to provide for additional users simultaneously; 'busy': The resource is in use, but it has no spare operating capacity to provide for additional users at this instant. ENUMERATED with values: * idle * active * busy
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.
	Resource can be instanciated as * LogicalResource
value	This property is present in subclasses

LogicalResource sub-resource fields

activationFeature	A Feature. Configuration feature.

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administrativeState	A ResourceAdministrativeStateType. ResourceAdministrativeStateType enumerations; values defined by ITU X.731: 'locked': The resource is administratively prohibited from performing services for its users; 'shutdown': Use of the resource is administratively permitted to existing instances of use only. While the system remains in the shutting down state the manager may at any time cause the managed object to revert to the unlocked state; 'unlocked': The resource is administratively permitted to perform services for its users. This is independent of its inherent operability. ENUMERATED with values: * locked * unlocked * shutdown
attachment	An AttachmentRef. Attachment reference. An attachment complements the description of an element (for instance a product) through video, pictures.
category	A String. Category of the concrete resource. e.g Gold, Silver for MSISDN concrete resource.
description	A String. Free-text description of the resource.
endOperatingDate	A DateTime. A date time(DateTime). The date till the resource is operating.
externalIdentifier	An ExternalIdentifier. An identification of an entity that is owned by or originates in a software system different from the current system, for example a ProductOrder handed off from a commerce platform into an order handling system. The structure identifies the system itself, the nature of the entity within the system (e.g. class name) and the unique ID of the entity within the system. It is anticipated that multiple external IDs can be held for a single entity, e.g. if the entity passed through multiple systems on the way to the current system. In this case the consumer is expected to sequence the IDs in the array in reverse order of provenance, i.e. most recent system first in the list.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
intent	An IntentRef. Intent reference, for when Intent is used by other entities.
name	A String. The name of the resource.
note	A Note. Extra information about a given entity.
operationalState	A ResourceOperationalStateType. ResourceOperationalStateType enumerations; values defined by ITU X.731: 'disable': The resource is totally inoperable and unable to provide service to the user(s); 'enable': The resource is partially or fully operable and available for use. ENUMERATED with values: * enable * disable

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place	A RelatedPlaceRef. Entity reference. The polymorphic attributes @type, @schemaLocation & @referredType are related to the RelatedPlace entity and not the RelatedPlaceRef class itself.
relatedParty	A RelatedPartyRefOrPartyRoleRef. RelatedParty reference. A related party defines party or party role or its reference, linked to a specific entity.
resourceCharacteristic	A Characteristic. Describes a given characteristic of an object or entity through a name/value pair. This is an abstract base class, the actual value is in one of the strongly-typed subclasses : StringCharacteristic, ObjectCharacteristic, FloatCharacteristic, BooleanCharacteristic, NumberCharacteristic, IntegerCharacteristic, StringArrayCharacteristic, ObjectArrayCharacteristic, BooleanArrayCharacteristic, NumberArrayCharacteristic, IntegerArrayCharacteristic… Characteristic can be instanciated as * BooleanArrayCharacteristic * BooleanCharacteristic * FloatCharacteristic * IntegerArrayCharacteristic * IntegerCharacteristic * NumberArrayCharacteristic * NumberCharacteristic * ObjectArrayCharacteristic * ObjectCharacteristic * StringArrayCharacteristic * StringCharacteristic
resourceOrderItem	A RelatedResourceOrderItem. RelatedResourceOrderItem (a ResourceOrder item) .The resource order item which triggered resource creation/change/termination.
resourceRelationship	A ResourceRelationship. Linked resources to the one instantiate, such as [bundled] if the resource is a bundle and you want to describe the bundled resources inside this bundle; [reliesOn] if the resource needs another already owned resource to rely on (e.g. an option on an already owned mobile access resource) [targets] or [isTargeted] (depending on the way of expressing the link) for any other kind of links that may be useful.
resourceSpecification	A ResourceSpecificationRef. Resources are physical or non-physical components (or some combination of these) within an enterprise's infrastructure or inventory. They are typically consumed or used by services (for example a physical port assigned to a service) or contribute to the realization of a Product (for example, a SIM card). They can be drawn from the Application, Computing and Network domains, and include, for example, Network Elements, software, IT systems, content and information, and technology components. A ResourceSpecification is an abstract base class for representing a generic means for implementing a particular type of Resource. In essence, a ResourceSpecification defines the common attributes and relationships of a set of related Resources, while Resource defines a specific instance that is based on a particular ResourceSpecification.

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resourceStatus	A ResourceStatusType. ResourceStatusType enumerations. ENUMERATED with values: * alarm * available * installed * not exists * pendingRemoval * planned * reserved * standby * suspended * unknown
resourceVersion	A String. A field that identifies the specific version of an instance of a resource.
startOperatingDate	A DateTime. A date time(DateTime). The date from which the resource is operating.
supportingResource	A ResourceRefOrValue. The polymorphic attributes @type, @schemaLocation & @referredType are related to the Resource entity and not the ResourceRefOrValue class itself.
usageState	A ResourceUsageStateType. ResourceUsageStateType enumerations; values defined by ITU X.731: 'idle': The resource is not currently in use; 'active': The resource is in use, and has sufficient spare operating capacity to provide for additional users simultaneously; 'busy': The resource is in use, but it has no spare operating capacity to provide for additional users at this instant. ENUMERATED with values: * idle * active * busy
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
value	A String. The value of the logical resource. E.g '0746712345' for MSISDN's.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.
	LogicalResource can be instanciated as * ResourcePool
capacity	This property is present in subclasses
pooledResource	This property is present in subclasses
pooledResourceSpecification	This property is present in subclasses

Json representation sample(s)

The following example demonstrates the details of the ResourcePool resource.

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```
{
    "@type": "ResourcePool",
    "id": "c3fef858-b864-4fac-971e-795b1195ded9",
    "href": "https://{server}:{port}/tmf-
api/resourcePoolManagement/v5/resourcePool/c3fef858-b864-4fac-971e-795b1195ded9",
    "name": "Phone Number block for partner MyPartner",
    "description": "This Pool collects the PhoneNumber resources assigned to the
partner MyPartner.",
    "pooledResourceSpecification":
    {
        "@type": "ResourceSpecificationRef",
        "@referredType": "ResourceSpecification",
        "id": "89afac38-32e2-40f8-8be2-7b86dd03c9c9",
        "href": "https://{server}:{port}/tmf-
api/resourceCatalogManagement/v5/resourceSpecification/89afac38-32e2-40f8-8be2-
7b86dd03c9c9",
        "name": "Phone Number"
    },
    "resourceCharacteristic":
    [
        {
            "@type": "StringCharacteristic",
            "@baseType": "Characteristic",
            "name": "cc",
            "valueType": "string",
            "value": "49"
        },
        {
            "@type": "StringCharacteristic",
            "@baseType": "Characteristic",
            "name": "ndc",
            "valueType": "string",
            "value": "228"
        }
    ],
    "capacity":
    [
        {
            "@type": "Capacity",
            "id": "cf8d1665-a3cc-470c-97eb-782f226075ce",
            "href": "https://{server}:{port}/tmf-
api/resourcePool/v5/capacity/cf8d1665-a3cc-470c-97eb-782f226075ce",
            "name": "Capacity of Phone Number pool for MyPartner",
            "capacitySpecification": {
                "id": "3d01344d-4409-4b15-8fcb-b8802314ff9c",
                "href": "https://{server}:{port}/tmf-
api/resourcePoolManagement/v5/capacitySpecification/3d01344d-4409-4b15-8fcb-
b8802314ff9c",
                "@type": "CapacitySpecificationRef",
                "@referredType": "CapacitySpecification"
            },
            "capacityStatus": "actual",
            "capacityAmount":
            {
                "amount": 1000,
                "units": "number"
            },
            "capacityCharacteristic":
            [
                {
                    "@type": "StringCharacteristic",
                    "@baseType": "Characteristic",
```


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```
                    "name": "lowerNumber",
                    "valueType": "string",
                    "value": "+49-228-181-000"
                },
                {
                    "@type": "StringCharacteristic",
                    "@baseType": "Characteristic",
                    "name": "upperNumber",
                    "valueType": "string",
                    "value": "+49-228-181-999"
                }
            ]
        }
    ],
    "resourceSpecification":
    {
        "id": "b0e0827c-5693-4e41-8445-22438abb33cb",
        "href": "https://{server}:{port}/tmf-
api/resourcePoolManagement/v5/resourcePoolSpecification/b0e0827c-5693-4e41-8445-
22438abb33cb",
        "@type": "ResourceSpecificationRef",
        "@referredType": "ResourcePoolSpecification"
    },
    "relatedParty":
    [
        {
            "@type": "RelatedPartyRefOrPartyRoleRef",
            "role": "Partner",
            "partyOrPartyRole":
            {
                "@type": "PartyRef",
                "@referredType": "Organization",
                "id": "47abd723-d3ca-49d8-ba53-76aa62e867e9",
                "href": "https://{server}:{port}/tmf-
api/partyManagement/v5/organization/47abd723-d3ca-49d8-ba53-76aa62e867e9",
                "name": "MyPartner"
            }
        }
    ]
}
```


AvailabilityCheck resource

A task resource that queries for a demanded capacity and returns the available capacities that can be used to fulfil the demand.

Resource model

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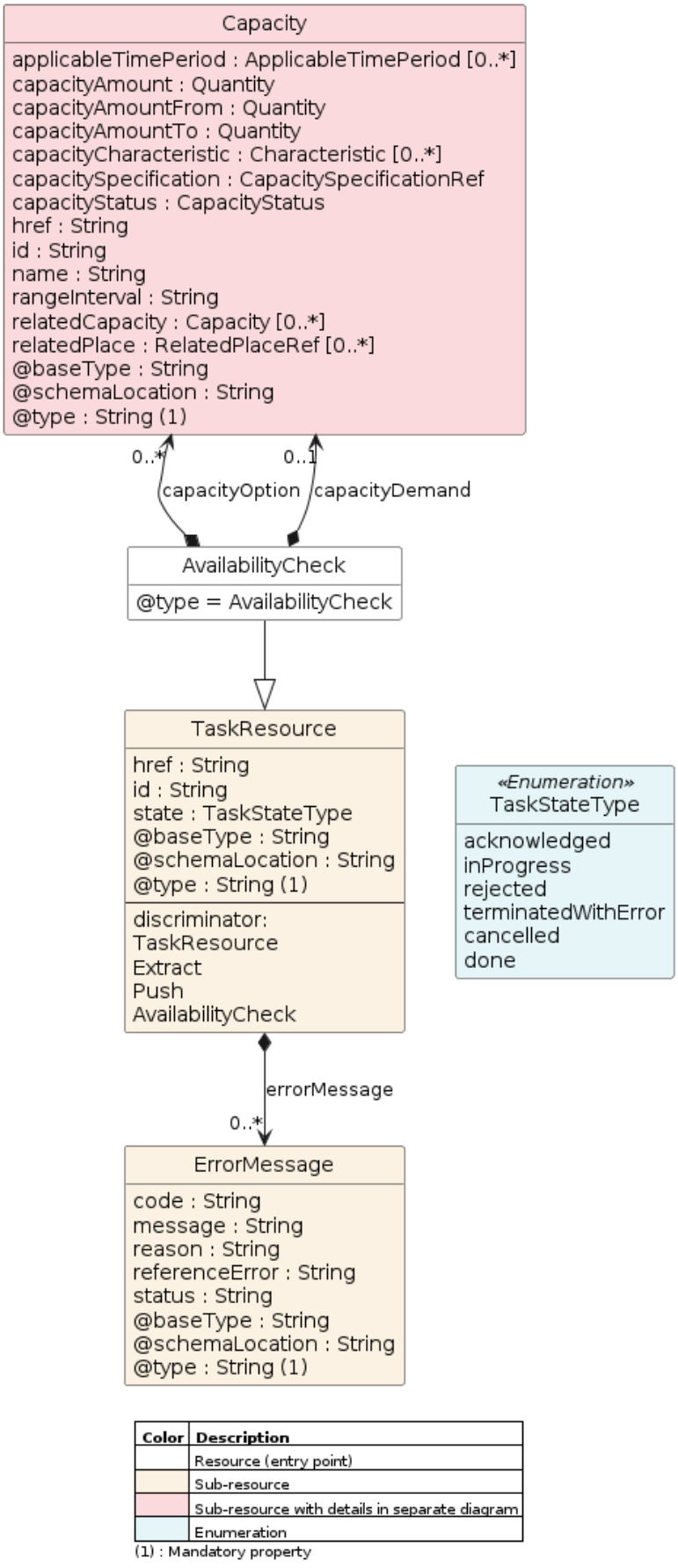


Figure 7 - AvailabilityCheck

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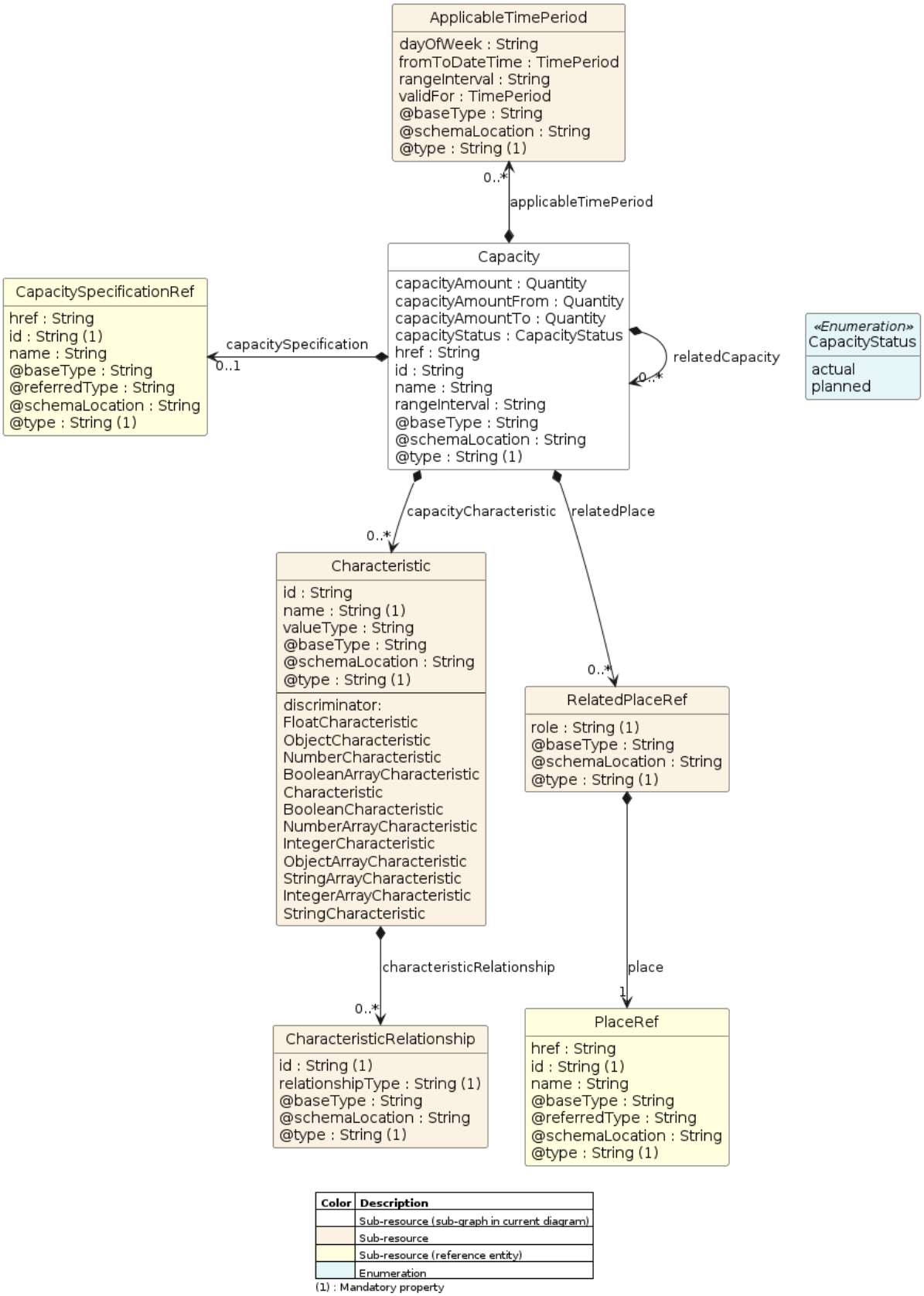


Figure 8 - Capacity

Field descriptions

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AvailabilityCheck fields

capacityDemand	A Capacity. Specific ability of an entity measured in quantity and units of quantity over an extended period.
capacityOption	A Capacity. Specific ability of an entity measured in quantity and units of quantity over an extended period.
errorMessage	An ErrorMessage. Represents an Error.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
state	A TaskStateType. Possible values for the state of a task. ENUMERATED with values: * acknowledged * rejected * inProgress * cancelled * done * terminatedWithError
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Capacity sub-resource fields

applicableTimePeriod	An ApplicableTimePeriod. The period of time for which Capacity or CapacityDemand applies.
capacityAmount	A Quantity. An amount in a given unit.
capacityAmountFrom	A Quantity. An amount in a given unit.
capacityAmountTo	A Quantity. An amount in a given unit.

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capacityCharacteristic	A Characteristic. Describes a given characteristic of an object or entity through a name/value pair. This is an abstract base class, the actual value is in one of the strongly-typed subclasses : StringCharacteristic, ObjectCharacteristic, FloatCharacteristic, BooleanCharacteristic, NumberCharacteristic, IntegerCharacteristic, StringArrayCharacteristic, ObjectArrayCharacteristic, BooleanArrayCharacteristic, NumberArrayCharacteristic, IntegerArrayCharacteristic… Characteristic can be instanciated as * BooleanArrayCharacteristic * BooleanCharacteristic * FloatCharacteristic * IntegerArrayCharacteristic * IntegerCharacteristic * NumberArrayCharacteristic * NumberCharacteristic * ObjectArrayCharacteristic * ObjectCharacteristic * StringArrayCharacteristic * StringCharacteristic
capacitySpecification	A CapacitySpecificationRef. CapacitySpecification reference, for when CapacitySpecification is used by other entities.
capacityStatus	A CapacityStatus. An indicator that specifies whether the capacity is planned or actual. ENUMERATED with values: * planned * actual
href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the Capacity; used to refer to the CapacitySpecification.
rangeInterval	A String. An indicator that specifies the inclusion or exclusion of the capacityAmount From and capacityAmountTo attributes. Possible values are "open", "closed", "closedBottom" and "closedTop".
relatedCapacity	A Capacity. Specific ability of an entity measured in quantity and units of quantity over an extended period.
relatedPlace	A RelatedPlaceRef. Entity reference. The polymorphic attributes @type, @schemaLocation & @referredType are related to the RelatedPlace entity and not the RelatedPlaceRef class itself.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

CapacitySpecificationRef sub-resource fields

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href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Characteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Another Characteristic that is related to the current Characteristic;.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.
	Characteristic can be instanciated as * BooleanArrayCharacteristic * BooleanCharacteristic * FloatCharacteristic * IntegerArrayCharacteristic * IntegerCharacteristic * NumberArrayCharacteristic * NumberCharacteristic * ObjectArrayCharacteristic * ObjectCharacteristic * StringArrayCharacteristic * StringCharacteristic
value	This property is present in subclasses

FloatCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Another Characteristic that is related to the current Characteristic;.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	A Float. Value of the characteristic.

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valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ObjectCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Another Characteristic that is related to the current Characteristic;.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	An object. Value of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

NumberCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Another Characteristic that is related to the current Characteristic;.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	A Number. Value of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

BooleanArrayCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Another Characteristic that is related to the current Characteristic;.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.

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value	A Boolean. A characteristic which value is an array of Boolean(s).
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

BooleanCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Another Characteristic that is related to the current Characteristic;.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	A Boolean. Value of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

NumberArrayCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Another Characteristic that is related to the current Characteristic;.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	A Number. A characteristic which value is an array of Number(s).
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

IntegerCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Another Characteristic that is related to the current Characteristic;.
id	A String. Unique identifier of the characteristic.

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name	A String. Name of the characteristic.
value	An Integer. Value of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ObjectArrayCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Another Characteristic that is related to the current Characteristic;.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	An object. Collection of characteristic values.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

StringArrayCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Another Characteristic that is related to the current Characteristic;.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	A String. Collection of string characteristics.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

IntegerArrayCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Another Characteristic that is related to the current Characteristic;.

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id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	An Integer. A characteristic which value is an array of Integer(s).
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

StringCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Another Characteristic that is related to the current Characteristic;.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	A String. Value of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

RelatedPlaceRef sub-resource fields

place	A PlaceRef. Place reference.
role	A String.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ApplicableTimePeriod sub-resource fields

dayOfWeek	A String. A day or days representing when the schedule is applicable. For example 2, 3 represent Monday and Tuesday.
fromToDateTime	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.

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rangeInterval	A String. An indicator that specifies the inclusion or exclusion of the from and to DateTime attributes. Possible values are "open", "closed", "closedBottom" and "closedTop".
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

CharacteristicRelationship sub-resource fields

id	A String. Unique identifier of the characteristic.
relationshipType	A String. The type of relationship.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

PlaceRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

TaskResource sub-resource fields

errorMessage	An ErrorMessage. Represents an Error.
href	A String. Hyperlink reference.
id	A String. Unique identifier.

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state	A TaskStateType. Possible values for the state of a task. ENUMERATED with values: * acknowledged * rejected * inProgress * cancelled * done * terminatedWithError
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.
	TaskResource can be instanciated as * AvailabilityCheck * Extract * Push
capacityDemand	This property is present in subclasses
capacityOption	This property is present in subclasses
extractedResource	This property is present in subclasses
pushedResource	This property is present in subclasses

Extract sub-resource fields

capacityDemand	A Capacity. Specific ability of an entity measured in quantity and units of quantity over an extended period.
errorMessage	An ErrorMessage. Represents an Error.
extractedResource	A ResourceRef. Resource reference, for when Resource is used by other entities.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
state	A TaskStateType. Possible values for the state of a task. ENUMERATED with values: * acknowledged * rejected * inProgress * cancelled * done * terminatedWithError
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

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Push sub-resource fields

errorMessage	An ErrorMessage. Represents an Error.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
pushedResource	A ResourceRef. Resource reference, for when Resource is used by other entities.
state	A TaskStateType. Possible values for the state of a task. ENUMERATED with values: * acknowledged * rejected * inProgress * cancelled * done * terminatedWithError
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Json representation sample(s)

The following example demonstrates the details of the AvailabilityCheck resource.

```
{
    "@type": "AvailabilityCheck",
    "id": "8cb645cb-279f-42f4-9d3b-72845bbfafbe",
    "href": "https://{server}:{port}/tmf-
api/resourcePoolManagement/v5/resourcePool/c3fef858-b864-4fac-971e-
795b1195ded9/availabilityCheck/8cb645cb-279f-42f4-9d3b-72845bbfafbe",
    "capacityDemand":
    {
        "@type": "Capacity",
        "capacityAmount":
        {
            "amount": 1,
            "units": "number"
        }
    },
    "state": "inProgress"
}
```

Push resource

A task resource that inserts a set of individual Resource instances into a Resource Pool and thus increases the capacity of the Pool.

Resource model

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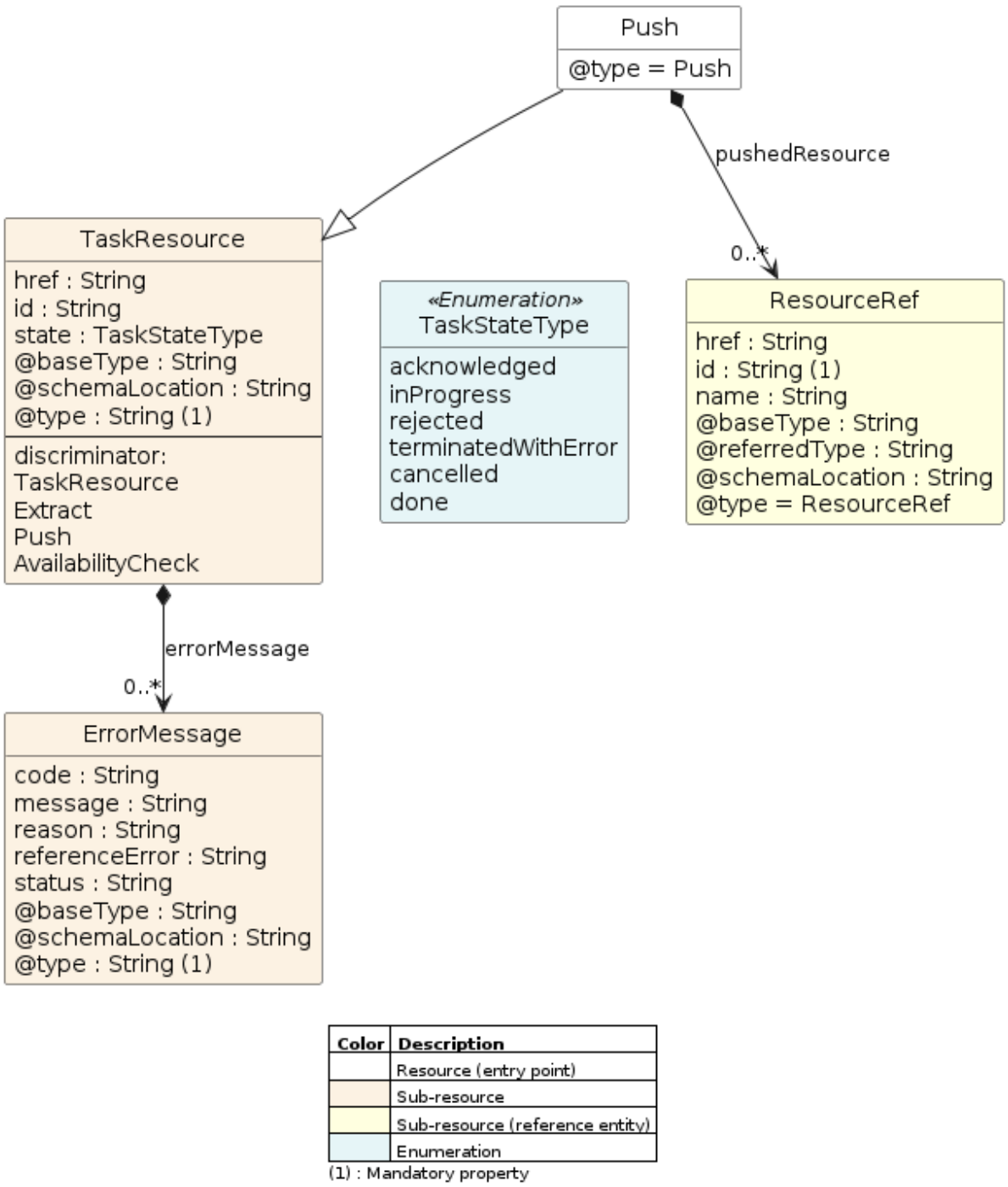


Figure 9 - Push

Field descriptions

Push fields

errorMessage	An ErrorMessage. Represents an Error.
href	A String. Hyperlink reference.
id	A String. Unique identifier.

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pushedResource	A ResourceRef. Resource reference, for when Resource is used by other entities.
state	A TaskStateType. Possible values for the state of a task. ENUMERATED with values: * acknowledged * rejected * inProgress * cancelled * done * terminatedWithError
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ResourceRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

TaskResource sub-resource fields

errorMessage	An ErrorMessage. Represents an Error.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
state	A TaskStateType. Possible values for the state of a task. ENUMERATED with values: * acknowledged * rejected * inProgress * cancelled * done * terminatedWithError
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.

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@type	A String. When sub-classing, this defines the sub-class Extensible name.
	TaskResource can be instanciated as * AvailabilityCheck * Extract * Push
capacityDemand	This property is present in subclasses
capacityOption	This property is present in subclasses
extractedResource	This property is present in subclasses
pushedResource	This property is present in subclasses

Extract sub-resource fields

capacityDemand	A Capacity. Specific ability of an entity measured in quantity and units of quantity over an extended period.
errorMessage	An ErrorMessage. Represents an Error.
extractedResource	A ResourceRef. Resource reference, for when Resource is used by other entities.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
state	A TaskStateType. Possible values for the state of a task. ENUMERATED with values: * acknowledged * rejected * inProgress * cancelled * done * terminatedWithError
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

AvailabilityCheck sub-resource fields

capacityDemand	A Capacity. Specific ability of an entity measured in quantity and units of quantity over an extended period.
capacityOption	A Capacity. Specific ability of an entity measured in quantity and units of quantity over an extended period.
errorMessage	An ErrorMessage. Represents an Error.
href	A String. Hyperlink reference.
id	A String. Unique identifier.

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state	A TaskStateType. Possible values for the state of a task. ENUMERATED with values: * acknowledged * rejected * inProgress * cancelled * done * terminatedWithError
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Json representation sample(s)

The following example demonstrates the details of the Push resource.

```
{
    "@type": "Push",
    "id": "ae63d76e-6d00-45aa-97a6-c8a2b54390e5",
    "href": "https://{server}:{port}/tmf-
api/resourcePoolManagement/v5/resourcePool/c3fef858-b864-4fac-971e-
795b1195ded9/push/ae63d76e-6d00-45aa-97a6-c8a2b54390e5",
    "pushedResource":
    [
        {
            "@type": "ResourceRef",
            "@referredType": "LogicalResource",
            "id": "4d179a07-1555-4e5f-a2de-0a51f5dd7e15",
            "href": "https://{server}:{port}/tmf-
api/resourceInventoryManagement/v5/resource/4d179a07-1555-4e5f-a2de-0a51f5dd7e15"
        },
        {
            "@type": "ResourceRef",
            "@referredType": "LogicalResource",
            "id": "15dfa8a7-f9db-43a8-8758-9d9df5a76264",
            "href": "https://{server}:{port}/tmf-
api/resourceInventoryManagement/v5/resource/15dfa8a7-f9db-43a8-8758-9d9df5a76264"
        },
        {
            "@type": "ResourceRef",
            "@referredType": "LogicalResource",
            "id": "195aa8ee-7560-4d97-9b9a-2add1a4e0372",
            "href": "https://{server}:{port}/tmf-
api/resourceInventoryManagement/v5/resource/195aa8ee-7560-4d97-9b9a-2add1a4e0372"
        }
    ],
    "state": "inProgress"
}
```

Extract resource

A task resource that extracts a demanded capacity of Resources from a Resource Pool and reduces the remaining capacity of the Pool.

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Resource model

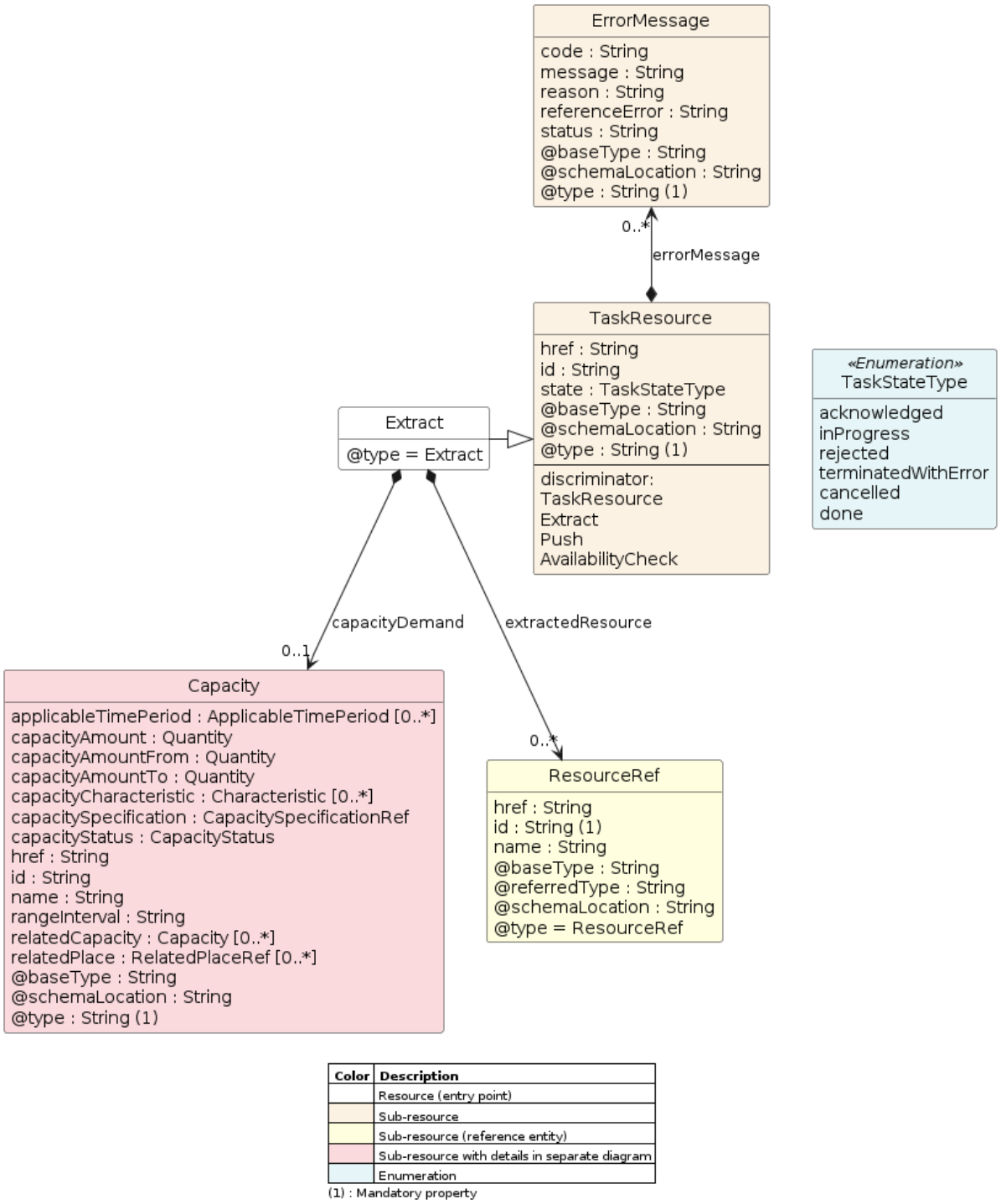


Figure 10 - Extract

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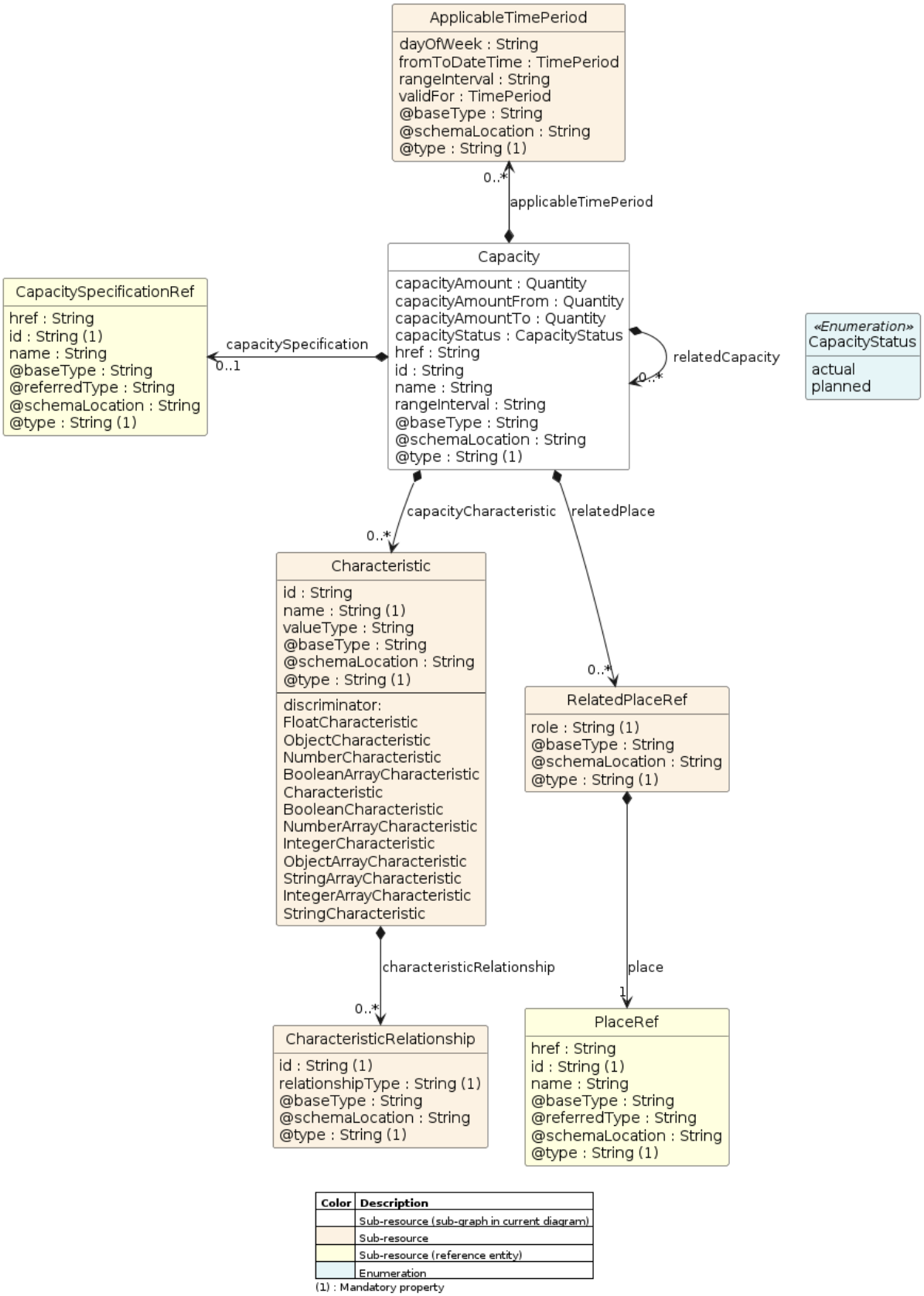


Figure 11 - Capacity

Field descriptions

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Extract fields

capacityDemand	A Capacity. Specific ability of an entity measured in quantity and units of quantity over an extended period.
errorMessage	An ErrorMessage. Represents an Error.
extractedResource	A ResourceRef. Resource reference, for when Resource is used by other entities.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
state	A TaskStateType. Possible values for the state of a task. ENUMERATED with values: * acknowledged * rejected * inProgress * cancelled * done * terminatedWithError
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Capacity sub-resource fields

applicableTimePeriod	An ApplicableTimePeriod. The period of time for which Capacity or CapacityDemand applies.
capacityAmount	A Quantity. An amount in a given unit.
capacityAmountFrom	A Quantity. An amount in a given unit.
capacityAmountTo	A Quantity. An amount in a given unit.

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capacityCharacteristic	A Characteristic. Describes a given characteristic of an object or entity through a name/value pair. This is an abstract base class, the actual value is in one of the strongly-typed subclasses : StringCharacteristic, ObjectCharacteristic, FloatCharacteristic, BooleanCharacteristic, NumberCharacteristic, IntegerCharacteristic, StringArrayCharacteristic, ObjectArrayCharacteristic, BooleanArrayCharacteristic, NumberArrayCharacteristic, IntegerArrayCharacteristic… Characteristic can be instanciated as * BooleanArrayCharacteristic * BooleanCharacteristic * FloatCharacteristic * IntegerArrayCharacteristic * IntegerCharacteristic * NumberArrayCharacteristic * NumberCharacteristic * ObjectArrayCharacteristic * ObjectCharacteristic * StringArrayCharacteristic * StringCharacteristic
capacitySpecification	A CapacitySpecificationRef. CapacitySpecification reference, for when CapacitySpecification is used by other entities.
capacityStatus	A CapacityStatus. An indicator that specifies whether the capacity is planned or actual. ENUMERATED with values: * planned * actual
href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the Capacity; used to refer to the CapacitySpecification.
rangeInterval	A String. An indicator that specifies the inclusion or exclusion of the capacityAmount From and capacityAmountTo attributes. Possible values are "open", "closed", "closedBottom" and "closedTop".
relatedCapacity	A Capacity. Specific ability of an entity measured in quantity and units of quantity over an extended period.
relatedPlace	A RelatedPlaceRef. Entity reference. The polymorphic attributes @type, @schemaLocation & @referredType are related to the RelatedPlace entity and not the RelatedPlaceRef class itself.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ResourceRef sub-resource fields

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href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

CapacitySpecificationRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Characteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Another Characteristic that is related to the current Characteristic;.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

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	Characteristic can be instanciated as * BooleanArrayCharacteristic * BooleanCharacteristic * FloatCharacteristic * IntegerArrayCharacteristic * IntegerCharacteristic * NumberArrayCharacteristic * NumberCharacteristic * ObjectArrayCharacteristic * ObjectCharacteristic * StringArrayCharacteristic * StringCharacteristic
value	This property is present in subclasses

FloatCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Another Characteristic that is related to the current Characteristic;.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	A Float. Value of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ObjectCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Another Characteristic that is related to the current Characteristic;.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	An object. Value of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

NumberCharacteristic sub-resource fields

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characteristicRelationship	A CharacteristicRelationship. Another Characteristic that is related to the current Characteristic;.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	A Number. Value of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

BooleanArrayCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Another Characteristic that is related to the current Characteristic;.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	A Boolean. A characteristic which value is an array of Boolean(s).
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

BooleanCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Another Characteristic that is related to the current Characteristic;.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	A Boolean. Value of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

NumberArrayCharacteristic sub-resource fields

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characteristicRelationship	A CharacteristicRelationship. Another Characteristic that is related to the current Characteristic;.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	A Number. A characteristic which value is an array of Number(s).
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

IntegerCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Another Characteristic that is related to the current Characteristic;.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	An Integer. Value of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ObjectArrayCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Another Characteristic that is related to the current Characteristic;.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	An object. Collection of characteristic values.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

StringArrayCharacteristic sub-resource fields

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characteristicRelationship	A CharacteristicRelationship. Another Characteristic that is related to the current Characteristic;.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	A String. Collection of string characteristics.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

IntegerArrayCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Another Characteristic that is related to the current Characteristic;.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	An Integer. A characteristic which value is an array of Integer(s).
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

StringCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Another Characteristic that is related to the current Characteristic;.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	A String. Value of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

RelatedPlaceRef sub-resource fields

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place	A PlaceRef. Place reference.
role	A String.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ApplicableTimePeriod sub-resource fields

dayOfWeek	A String. A day or days representing when the schedule is applicable. For example 2, 3 represent Monday and Tuesday.
fromToDateTime	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
rangeInterval	A String. An indicator that specifies the inclusion or exclusion of the from and to DateTime attributes. Possible values are "open", "closed", "closedBottom" and "closedTop".
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

CharacteristicRelationship sub-resource fields

id	A String. Unique identifier of the characteristic.
relationshipType	A String. The type of relationship.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

PlaceRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.

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@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

TaskResource sub-resource fields

errorMessage	An ErrorMessage. Represents an Error.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
state	A TaskStateType. Possible values for the state of a task. ENUMERATED with values: * acknowledged * rejected * inProgress * cancelled * done * terminatedWithError
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.
	TaskResource can be instanciated as * AvailabilityCheck * Extract * Push
capacityDemand	This property is present in subclasses
capacityOption	This property is present in subclasses
extractedResource	This property is present in subclasses
pushedResource	This property is present in subclasses

Push sub-resource fields

errorMessage	An ErrorMessage. Represents an Error.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
pushedResource	A ResourceRef. Resource reference, for when Resource is used by other entities.

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state	A TaskStateType. Possible values for the state of a task. ENUMERATED with values: * acknowledged * rejected * inProgress * cancelled * done * terminatedWithError
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

AvailabilityCheck sub-resource fields

capacityDemand	A Capacity. Specific ability of an entity measured in quantity and units of quantity over an extended period.
capacityOption	A Capacity. Specific ability of an entity measured in quantity and units of quantity over an extended period.
errorMessage	An ErrorMessage. Represents an Error.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
state	A TaskStateType. Possible values for the state of a task. ENUMERATED with values: * acknowledged * rejected * inProgress * cancelled * done * terminatedWithError
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Json representation sample(s)

The following example demonstrates the details of the Extract resource.

```
{
    "@type": "Extract",
    "id": "c6a1e831-ef5b-4d09-85a8-594fca73313b",
    "href": "https://{server}:{port}/tmf-
api/resourcePoolManagement/v5/resourcePool/c3fef858-b864-4fac-971e-
795b1195ded9/extract/c6a1e831-ef5b-4d09-85a8-594fca73313b",
    "capacityDemand":
```

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```
    {
        "@type": "Capacity",
        "capacityAmount":
        {
            "amount": 2,
            "units": "number"
        },
        "capacityStatus": "planned",
        "applicableTimePeriod":
        [
            {
                "@type": "ApplicableTimePeriod",
                "fromToDateTime":
                {
                    "startDateTime": "2024-05-13T00:00:00Z",
                    "endDateTime": "2024-09-20T00:00:00Z"
                }
            }
        ]
    },
    "state": "inProgress"
}
```


Notification Resource Models

20 notifications are defined for this API.

Notifications related to CapacitySpecification:

- Create Event
- Delete Event
- Attribute Value Change Event

Notifications related to ResourcePoolSpecification:

- Create Event
- Delete Event
- Attribute Value Change Event
- Status Change Event

Notifications related to ResourcePool:

- Create Event
- Delete Event
- Attribute Value Change Event
- Status Change Event

Notifications related to AvailabilityCheck:

- Create Event
- Delete Event

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- State Change Event

Notifications related to Push:

- Create Event
- Delete Event
- State Change Event

Notifications related to Extract:

- Create Event
- Delete Event
- State Change Event

The notification structure for all notifications in this API follow the pattern depicted by the figure below. A notification event resource (depicted by "SpecificEvent" placeholder) is a sub class of a generic Event structure containing at least an id of the event occurrence (eventId), an event timestamp (eventTime), and the name of the resource (eventType). This notification structure owns an event payload structure ("SpecificEventPayload" placeholder) linked to the resource concerned by the notification using the resource name as access field ("resourceName" placeholder).

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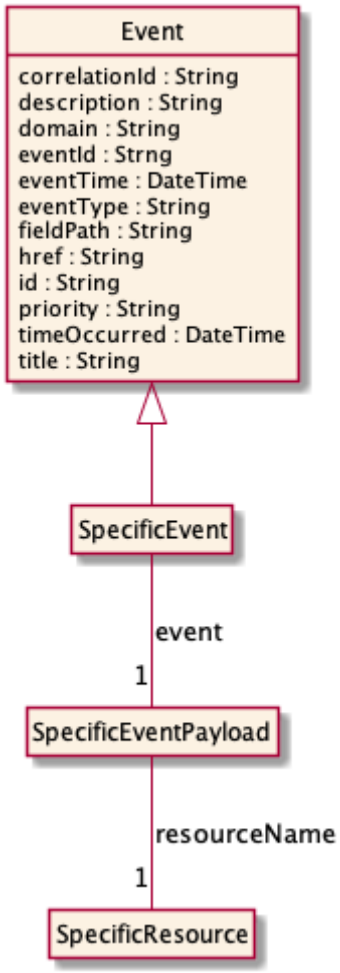


Figure 12 Notification Pattern

CapacitySpecification

Create Event

Message example for CapacitySpecificationCreateEvent event

```
Content-Type: application/json

{
    "@type": "CapacitySpecificationCreateEvent",
    "@baseType": "Event",
    "correlationId": "e4888f4d-71cd-43a4-a018-dd13f230935c",
    "description": "CapacitySpecificationCreateEvent illustration",
    "domain": "Commercial",
    "eventId": "3b63e628-0c66-4d57-abbf-2fb6aeb229fd",
    "eventTime": "2024-09-19T13:55:06.910Z",
    "eventType": "CapacitySpecificationCreateEvent",
    "priority": "3",
    "timeOccurred": "2024-09-19T13:55:03.868Z",
    "title": "CapacitySpecificationCreateEvent",
    "event": {
        "capacitySpecification": {
            "@type": "CapacitySpecification",
            "id": "3d01344d-4409-4b15-8fcb-b8802314ff9c",
```

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```
            "href": "https://{server}:{port}/tmf-
api/resourcePoolManagement/v5/capacitySpecification/3d01344d-4409-4b15-8fcb-
b8802314ff9c",
            "name": "Phone Number range",
            "description": "This CapacitySpecification defines the details of
capacity for Phone Number ranges.",
            "capacityCharacteristicSpecification":
            [
                {
                    "@type": "CharacteristicSpecification",
                    "name": "lowerNumber",
                    "description": "the lower boundary of a block of phone numbers;
phone numbers consist of a country code (cc), national destination code (ndc),
subscriber number (sn), and optionally a direct dial in (ddi), all sections
separated by '-'",
                    "valueType": "string",
                    "characteristicValueSpecification":
                    [
                        {
                            "@type": "CharacteristicValueSpecification",
                            "valueType": "string",
                            "regex": "^\\+(?<cc>[1-9]\\d{0,2})-(?<ndc>[1-9]\\d*)-
(?<sn>[1-9]\\d*)(?:-?(?<ddi>\\d+))?$"
                        }
                    ]
                },
                {
                    "@type": "CharacteristicSpecification",
                    "name": "upperNumber",
                    "description": "the upper boundary of a block of phone numbers;
phone numbers consist of a country code (cc), national destination code (ndc),
subscriber number (sn), and optionally a direct dial in (ddi), all sections
separated by '-'",
                    "valueType": "string",
                    "characteristicValueSpecification":
                    [
                        {
                            "@type": "CharacteristicValueSpecification",
                            "valueType": "string",
                            "regex": "^\\+(?<cc>[1-9]\\d{0,2})-(?<ndc>[1-9]\\d*)-
(?<sn>[1-9]\\d*)(?:-?(?<ddi>\\d+))?$"
                        }
                    ]
                }
            ]
        }
    },
    "reportingSystem": {
        "id": "109",
        "name": "APP-804",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "source": {
        "id": "8",
        "name": "APP-307",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    }
}
```


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Delete Event

Message example for CapacitySpecificationDeleteEvent event

```
Content-Type: application/json

{
    "@type": "CapacitySpecificationDeleteEvent",
    "@baseType": "Event",
    "correlationId": "5b2ca6b7-d69f-4ee7-b36e-5a4b442b14c2",
    "description": "CapacitySpecificationDeleteEvent illustration",
    "domain": "Commercial",
    "eventId": "4dfa961e-9dc4-4cea-8e5a-2d392f8076f1",
    "eventTime": "2024-09-19T13:55:06.910Z",
    "eventType": "CapacitySpecificationDeleteEvent",
    "priority": "3",
    "timeOccurred": "2024-09-19T13:55:03.868Z",
    "title": "CapacitySpecificationDeleteEvent",
    "event": {
        "capacitySpecification": {
            "@type": "CapacitySpecification",
            "id": "3d01344d-4409-4b15-8fcb-b8802314ff9c",
            "href": "https://{server}:{port}/tmf-
api/resourcePoolManagement/v5/capacitySpecification/3d01344d-4409-4b15-8fcb-
b8802314ff9c"
        }
    },
    "reportingSystem": {
        "id": "109",
        "name": "APP-804",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "source": {
        "id": "8",
        "name": "APP-307",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    }
}
```

Attribute Value Change Event

Message example for CapacitySpecificationAttributeValueChangeEvent event

```
Content-Type: application/json

{
    "@type": "CapacitySpecificationAttributeValueChangeEvent",
    "@baseType": "Event",
    "correlationId": "41a5ab1a-e6bd-4cfa-bd86-024dc2e13890",
    "description": "CapacitySpecificationAttributeValueChangeEvent illustration",
    "domain": "Commercial",
    "eventId": "5bb79084-116a-4db1-9cc3-4443f500a7e0",
    "eventTime": "2024-09-19T13:55:06.910Z",
    "eventType": "CapacitySpecificationAttributeValueChangeEvent",
    "priority": "3",
    "timeOccurred": "2024-09-19T13:55:03.868Z",
```

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```
    "title": "CapacitySpecificationAttributeValueChangeEvent",
    "event": {
        "capacitySpecification": {
            "@type": "CapacitySpecification",
            "id": "3d01344d-4409-4b15-8fcb-b8802314ff9c",
            "href": "https://{server}:{port}/tmf-
api/resourcePoolManagement/v5/capacitySpecification/3d01344d-4409-4b15-8fcb-
b8802314ff9c",
            "name": "Phone Number Range"
        }
    },
    "reportingSystem": {
        "id": "109",
        "name": "APP-804",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "source": {
        "id": "8",
        "name": "APP-307",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    }
}
```

ResourcePoolSpecification

Create Event

Message example for ResourcePoolSpecificationCreateEvent event

```
Content-Type: application/json

{
    "@type": "ResourcePoolSpecificationCreateEvent",
    "@baseType": "Event",
    "correlationId": "e4888f4d-71cd-43a4-a018-dd13f230935c",
    "description": "ResourcePoolSpecificationCreateEvent illustration",
    "domain": "Commercial",
    "eventId": "3b63e628-0c66-4d57-abbf-2fb6aeb229fd",
    "eventTime": "2024-09-19T13:55:06.910Z",
    "eventType": "ResourcePoolSpecificationCreateEvent",
    "priority": "3",
    "timeOccurred": "2024-09-19T13:55:03.868Z",
    "title": "ResourcePoolSpecificationCreateEvent",
    "event": {
        "resourcePoolSpecification": {
            "@type": "ResourcePoolSpecification",
            "id": "b0e0827c-5693-4e41-8445-22438abb33cb",
            "href": "https://{server}:{port}/tmf-
api/resourcePoolManagement/v5/resourcePoolSpecification/b0e0827c-5693-4e41-8445-
22438abb33cb",
            "name": "Phone Number pool",
            "description": "Specifies the details of ResourcePools for Phone Number
ranges",
            "capacitySpecification":
            [
                {
                    "@type": "CapacitySpecificationRef",
```

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```
                    "@referredType": "CapacitySpecification",
                    "id": "3d01344d-4409-4b15-8fcb-b8802314ff9c",
                    "href": "https://{server}:{port}/tmf-
api/resourcePoolManagement/v5/capacitySpecification/3d01344d-4409-4b15-8fcb-
b8802314ff9c",
                    "name": "Phone Number range"
                }
            ],
            "resourceSpecCharacteristic":
            [
                {
                    "@type": "CharacteristicSpecification",
                    "name": "cc",
                    "description": "Phone Number Country Code",
                    "valueType": "string",
                    "characteristicValueSpecification":
                    [
                        {
                            "@type": "CharacteristicValueSpecification",
                            "valueType": "string",
                            "regex": "^[1-9]\\d{0,2}$"
                        }
                    ]
                },
                {
                    "@type": "CharacteristicSpecification",
                    "name": "ndc",
                    "description": "Phone Number National Destination Code",
                    "valueType": "string",
                    "characteristicValueSpecification":
                    [
                        {
                            "@type": "CharacteristicValueSpecification",
                            "valueType": "string",
                            "regex": "^[1-9]\\d*?$"
                        }
                    ]
                }
            ]
        }
    },
    "reportingSystem": {
        "id": "109",
        "name": "APP-804",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "source": {
        "id": "8",
        "name": "APP-307",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    }
}
```


Delete Event

Message example for ResourcePoolSpecificationDeleteEvent event

```
Content-Type: application/json
```

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```
{
    "@type": "ResourcePoolSpecificationDeleteEvent",
    "@baseType": "Event",
    "correlationId": "5b2ca6b7-d69f-4ee7-b36e-5a4b442b14c2",
    "description": "ResourcePoolSpecificationDeleteEvent illustration",
    "domain": "Commercial",
    "eventId": "4dfa961e-9dc4-4cea-8e5a-2d392f8076f1",
    "eventTime": "2024-09-19T13:55:06.910Z",
    "eventType": "ResourcePoolSpecificationDeleteEvent",
    "priority": "3",
    "timeOccurred": "2024-09-19T13:55:03.868Z",
    "title": "ResourcePoolSpecificationDeleteEvent",
    "event": {
        "resourcePoolSpecification": {
            "@type": "ResourcePoolSpecification",
            "id": "b0e0827c-5693-4e41-8445-22438abb33cb",
            "href": "https://{server}:{port}/tmf-
api/resourcePoolManagement/v5/resourcePoolSpecification/b0e0827c-5693-4e41-8445-
22438abb33cb"
        }
    },
    "reportingSystem": {
        "id": "109",
        "name": "APP-804",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "source": {
        "id": "8",
        "name": "APP-307",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    }
}
```


Attribute Value Change Event

Message example for ResourcePoolSpecificationAttributeValueChangeEvent event


```
Content-Type: application/json

{
    "@type": "ResourcePoolSpecificationAttributeValueChangeEvent",
    "@baseType": "Event",
    "correlationId": "41a5ab1a-e6bd-4cfa-bd86-024dc2e13890",
    "description": "ResourcePoolSpecificationAttributeValueChangeEvent
illustration",
    "domain": "Commercial",
    "eventId": "5bb79084-116a-4db1-9cc3-4443f500a7e0",
    "eventTime": "2024-09-19T13:55:06.910Z",
    "eventType": "ResourcePoolSpecificationAttributeValueChangeEvent",
    "priority": "3",
    "timeOccurred": "2024-09-19T13:55:03.868Z",
    "title": "ResourcePoolSpecificationAttributeValueChangeEvent",
    "event": {
        "resourcePoolSpecification": {
            "@type": "ResourcePoolSpecification",
            "id": "b0e0827c-5693-4e41-8445-22438abb33cb",
            "href": "https://{server}:{port}/tmf-
```


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```
api/resourcePoolManagement/v5/resourcePoolSpecification/b0e0827c-5693-4e41-8445-
22438abb33cb",
            "validFor":
            {
                "startDateTime": "2024-05-13T00:00:00Z",
                "endDateTime": "2024-09-20T00:00:00Z"
            }
        }
    },
    "reportingSystem": {
        "id": "109",
        "name": "APP-804",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "source": {
        "id": "8",
        "name": "APP-307",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    }
}
```

Status Change Event

Message example for ResourcePoolSpecificationStatusChangeEvent event

```
Content-Type: application/json

{
    "@type": "ResourcePoolSpecificationStatusChangeEvent",
    "@baseType": "Event",
    "correlationId": "f162615b-69be-476f-bc99-440e9ff6b122",
    "description": "ResourcePoolSpecificationStatusChangeEvent illustration",
    "domain": "Commercial",
    "eventId": "dce63750-ee42-48c4-b220-85295f3830a6",
    "eventTime": "2024-09-19T13:55:06.910Z",
    "eventType": "ResourcePoolSpecificationStatusChangeEvent",
    "priority": "3",
    "timeOccurred": "2024-09-19T13:55:03.868Z",
    "title": "ResourcePoolSpecificationStatusChangeEvent",
    "event": {
        "resourcePoolSpecification": {
            "@type": "ResourcePoolSpecification",
            "id": "b0e0827c-5693-4e41-8445-22438abb33cb",
            "href": "https://{server}:{port}/tmf-
api/resourcePoolManagement/v5/resourcePoolSpecification/b0e0827c-5693-4e41-8445-
22438abb33cb",
            "lifecycleStatus": "active"
        }
    },
    "reportingSystem": {
        "id": "109",
        "name": "APP-804",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "source": {
        "id": "8",
        "name": "APP-307",
```

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```
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    }
}
```

ResourcePool

Create Event

Message example for ResourcePoolCreateEvent event

```
Content-Type: application/json

{
    "@type": "ResourcePoolCreateEvent",
    "@baseType": "Event",
    "correlationId": "e4888f4d-71cd-43a4-a018-dd13f230935c",
    "description": "ResourcePoolCreateEvent illustration",
    "domain": "Commercial",
    "eventId": "3b63e628-0c66-4d57-abbf-2fb6aeb229fd",
    "eventTime": "2024-09-19T13:55:06.910Z",
    "eventType": "ResourcePoolCreateEvent",
    "priority": "3",
    "timeOccurred": "2024-09-19T13:55:03.868Z",
    "title": "ResourcePoolCreateEvent",
    "event": {
        "resourcePool": {
            "@type": "ResourcePool",
            "id": "c3fef858-b864-4fac-971e-795b1195ded9",
            "href": "https://{server}:{port}/tmf-
api/resourcePoolManagement/v5/resourcePool/c3fef858-b864-4fac-971e-795b1195ded9",
            "name": "Phone Number block for partner MyPartner",
            "description": "This Pool collects the PhoneNumber resources assigned to
the partner MyPartner.",
            "pooledResourceSpecification":
            {
                "@type": "ResourceSpecificationRef",
                "@referredType": "ResourceSpecification",
                "id": "89afac38-32e2-40f8-8be2-7b86dd03c9c9",
                "href": "https://{server}:{port}/tmf-
api/resourceCatalogManagement/v5/resourceSpecification/89afac38-32e2-40f8-8be2-
7b86dd03c9c9",
                "name": "Phone Number"
            },
            "resourceCharacteristic":
            [
                {
                    "@type": "StringCharacteristic",
                    "@baseType": "Characteristic",
                    "name": "cc",
                    "valueType": "string",
                    "value": "49"
                },
                {
                    "@type": "StringCharacteristic",
                    "@baseType": "Characteristic",
                    "name": "ndc",
                    "valueType": "string",
                    "value": "228"
```

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```
                }
            ],
            "capacity":
            [
                {
                    "@type": "Capacity",
                    "id": "cf8d1665-a3cc-470c-97eb-782f226075ce",
                    "href": "https://{server}:{port}/tmf-
api/resourcePool/v5/capacity/cf8d1665-a3cc-470c-97eb-782f226075ce",
                    "name": "Capacity of Phone Number pool for MyPartner",
                    "capacitySpecification": {
                        "id": "3d01344d-4409-4b15-8fcb-b8802314ff9c",
                        "href": "https://{server}:{port}/tmf-
api/resourcePoolManagement/v5/capacitySpecification/3d01344d-4409-4b15-8fcb-
b8802314ff9c",
                        "@type": "CapacitySpecificationRef",
                        "@referredType": "CapacitySpecification"
                    },
                    "capacityStatus": "actual",
                    "capacityAmount":
                    {
                        "amount": 1000,
                        "units": "number"
                    },
                    "capacityCharacteristic":
                    [
                        {
                            "@type": "StringCharacteristic",
                            "@baseType": "Characteristic",
                            "name": "lowerNumber",
                            "valueType": "string",
                            "value": "+49-228-181-000"
                        },
                        {
                            "@type": "StringCharacteristic",
                            "@baseType": "Characteristic",
                            "name": "upperNumber",
                            "valueType": "string",
                            "value": "+49-228-181-999"
                        }
                    ]
                }
            ],
            "resourceSpecification":
            {
                "id": "b0e0827c-5693-4e41-8445-22438abb33cb",
                "href": "https://{server}:{port}/tmf-
api/resourcePoolManagement/v5/resourcePoolSpecification/b0e0827c-5693-4e41-8445-
22438abb33cb",
                "@type": "ResourceSpecificationRef",
                "@referredType": "ResourcePoolSpecification"
            },
            "relatedParty":
            [
                {
                    "@type": "RelatedPartyRefOrPartyRoleRef",
                    "role": "Partner",
                    "partyOrPartyRole":
                    {
                        "@type": "PartyRef",
                        "@referredType": "Organization",
                        "id": "47abd723-d3ca-49d8-ba53-76aa62e867e9",
                        "href": "https://{server}:{port}/tmf-
```


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```
api/partyManagement/v5/organization/47abd723-d3ca-49d8-ba53-76aa62e867e9",
                        "name": "MyPartner"
                    }
                }
            ]
        }
    },
    "reportingSystem": {
        "id": "109",
        "name": "APP-804",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "source": {
        "id": "8",
        "name": "APP-307",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    }
}
```

Delete Event

Message example for ResourcePoolDeleteEvent event

```
Content-Type: application/json

{
    "@type": "ResourcePoolDeleteEvent",
    "@baseType": "Event",
    "correlationId": "5b2ca6b7-d69f-4ee7-b36e-5a4b442b14c2",
    "description": "ResourcePoolDeleteEvent illustration",
    "domain": "Commercial",
    "eventId": "4dfa961e-9dc4-4cea-8e5a-2d392f8076f1",
    "eventTime": "2024-09-19T13:55:06.910Z",
    "eventType": "ResourcePoolDeleteEvent",
    "priority": "3",
    "timeOccurred": "2024-09-19T13:55:03.868Z",
    "title": "ResourcePoolDeleteEvent",
    "event": {
        "resourcePool": {
            "@type": "ResourcePool",
            "id": "c3fef858-b864-4fac-971e-795b1195ded9",
            "href": "https://{server}:{port}/tmf-
api/resourcePoolManagement/v5/resourcePool/c3fef858-b864-4fac-971e-795b1195ded9"
        }
    },
    "reportingSystem": {
        "id": "109",
        "name": "APP-804",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "source": {
        "id": "8",
        "name": "APP-307",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    }
```

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```
}
```

Attribute Value Change Event

Message example for ResourcePoolAttributeValueChangeEvent event

```
Content-Type: application/json

{
    "@type": "ResourcePoolAttributeValueChangeEvent",
    "@baseType": "Event",
    "correlationId": "41a5ab1a-e6bd-4cfa-bd86-024dc2e13890",
    "description": "ResourcePoolAttributeValueChangeEvent illustration",
    "domain": "Commercial",
    "eventId": "5bb79084-116a-4db1-9cc3-4443f500a7e0",
    "eventTime": "2024-09-19T13:55:06.910Z",
    "eventType": "ResourcePoolAttributeValueChangeEvent",
    "priority": "3",
    "timeOccurred": "2024-09-19T13:55:03.868Z",
    "title": "ResourcePoolAttributeValueChangeEvent",
    "event": {
        "resourcePool": {
            "@type": "ResourcePool",
            "id": "c3fef858-b864-4fac-971e-795b1195ded9",
            "href": "https://{server}:{port}/tmf-
api/resourcePoolManagement/v5/resourcePool/c3fef858-b864-4fac-971e-795b1195ded9",
            "name": "Phone Number block for partner 'MyPartner'",
            "description": "This Pool collects the PhoneNumber resources assigned to
the partner 'MyPartner'.",
            "validFor":
            {
                "startDateTime": "2024-05-13T00:00:00Z",
                "endDateTime": "2024-09-20T00:00:00Z"
            }
        }
    },
    "reportingSystem": {
        "id": "109",
        "name": "APP-804",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "source": {
        "id": "8",
        "name": "APP-307",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    }
}
```

Status Change Event

Message example for ResourcePoolStatusChangeEvent event

```
Content-Type: application/json

{
```

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```
    "@type": "ResourcePoolStatusChangeEvent",
    "@baseType": "Event",
    "correlationId": "f162615b-69be-476f-bc99-440e9ff6b122",
    "description": "ResourcePoolStatusChangeEvent illustration",
    "domain": "Commercial",
    "eventId": "dce63750-ee42-48c4-b220-85295f3830a6",
    "eventTime": "2024-09-19T13:55:06.910Z",
    "eventType": "ResourcePoolStatusChangeEvent",
    "priority": "3",
    "timeOccurred": "2024-09-19T13:55:03.868Z",
    "title": "ResourcePoolStatusChangeEvent",
    "event": {
        "resourcePool": {
            "@type": "ResourcePool",
            "id": "c3fef858-b864-4fac-971e-795b1195ded9",
            "href": "https://{server}:{port}/tmf-
api/resourcePoolManagement/v5/resourcePool/c3fef858-b864-4fac-971e-795b1195ded9",
            "resourceStatus": "planned"
        }
    },
    "reportingSystem": {
        "id": "109",
        "name": "APP-804",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "source": {
        "id": "8",
        "name": "APP-307",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    }
}
```


AvailabilityCheck

Create Event

Message example for AvailabilityCheckCreateEvent event


```
Content-Type: application/json

{
    "@type": "AvailabilityCheckCreateEvent",
    "@baseType": "Event",
    "correlationId": "e4888f4d-71cd-43a4-a018-dd13f230935c",
    "description": "AvailabilityCheckCreateEvent illustration",
    "domain": "Commercial",
    "eventId": "3b63e628-0c66-4d57-abbf-2fb6aeb229fd",
    "eventTime": "2024-09-19T13:55:06.910Z",
    "eventType": "AvailabilityCheckCreateEvent",
    "priority": "3",
    "timeOccurred": "2024-09-19T13:55:03.868Z",
    "title": "AvailabilityCheckCreateEvent",
    "event": {
        "availabilityCheck": {
            "@type": "AvailabilityCheck",
            "id": "8cb645cb-279f-42f4-9d3b-72845bbfafbe",
            "href": "https://{server}:{port}/tmf-
```


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```
api/resourcePoolManagement/v5/resourcePool/c3fef858-b864-4fac-971e-
795b1195ded9/availabilityCheck/8cb645cb-279f-42f4-9d3b-72845bbfafbe",
            "capacityDemand":
            {
                "@type": "Capacity",
                "capacityAmount":
                {
                    "amount": 1,
                    "units": "number"
                }
            }
        }
    },
    "reportingSystem": {
        "id": "109",
        "name": "APP-804",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "source": {
        "id": "8",
        "name": "APP-307",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    }
}
```

Delete Event

Message example for AvailabilityCheckDeleteEvent event

```
Content-Type: application/json

{
    "@type": "AvailabilityCheckDeleteEvent",
    "@baseType": "Event",
    "correlationId": "e4888f4d-71cd-43a4-a018-dd13f230935c",
    "description": "AvailabilityCheckDeleteEvent illustration",
    "domain": "Commercial",
    "eventId": "3b63e628-0c66-4d57-abbf-2fb6aeb229fd",
    "eventTime": "2024-09-19T13:55:06.910Z",
    "eventType": "AvailabilityCheckDeleteEvent",
    "priority": "3",
    "timeOccurred": "2024-09-19T13:55:03.868Z",
    "title": "AvailabilityCheckDeleteEvent",
    "event": {
        "availabilityCheck": {
            "@type": "AvailabilityCheck",
            "id": "8cb645cb-279f-42f4-9d3b-72845bbfafbe",
            "href": "https://{server}:{port}/tmf-
api/resourcePoolManagement/v5/resourcePool/c3fef858-b864-4fac-971e-
795b1195ded9/availabilityCheck/8cb645cb-279f-42f4-9d3b-72845bbfafbe"
        }
    },
    "reportingSystem": {
        "id": "109",
        "name": "APP-804",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
```

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```
    "source": {
        "id": "8",
        "name": "APP-307",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    }
}
```

State Change Event

Message example for AvailabilityCheckStateChangeEvent event

```
Content-Type: application/json

{
    "@type": "AvailabilityCheckStateChangeEvent",
    "@baseType": "Event",
    "correlationId": "e4888f4d-71cd-43a4-a018-dd13f230935c",
    "description": "AvailabilityCheckStateChangeEvent illustration",
    "domain": "Commercial",
    "eventId": "3b63e628-0c66-4d57-abbf-2fb6aeb229fd",
    "eventTime": "2024-09-19T13:55:06.910Z",
    "eventType": "AvailabilityCheckStateChangeEvent",
    "priority": "3",
    "timeOccurred": "2024-09-19T13:55:03.868Z",
    "title": "AvailabilityCheckStateChangeEvent",
    "event": {
        "availabilityCheck": {
            "@type": "AvailabilityCheck",
            "id": "8cb645cb-279f-42f4-9d3b-72845bbfafbe",
            "href": "https://{server}:{port}/tmf-
api/resourcePoolManagement/v5/resourcePool/c3fef858-b864-4fac-971e-
795b1195ded9/availabilityCheck/8cb645cb-279f-42f4-9d3b-72845bbfafbe",
            "state": "done"
        }
    },
    "reportingSystem": {
        "id": "109",
        "name": "APP-804",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "source": {
        "id": "8",
        "name": "APP-307",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    }
}
```

Push

Create Event

Message example for PushCreateEvent event

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```
Content-Type: application/json

{
    "@type": "PushCreateEvent",
    "@baseType": "Event",
    "correlationId": "e4888f4d-71cd-43a4-a018-dd13f230935c",
    "description": "PushCreateEvent illustration",
    "domain": "Commercial",
    "eventId": "3b63e628-0c66-4d57-abbf-2fb6aeb229fd",
    "eventTime": "2024-09-19T13:55:06.910Z",
    "eventType": "PushCreateEvent",
    "priority": "3",
    "timeOccurred": "2024-09-19T13:55:03.868Z",
    "title": "PushCreateEvent",
    "event": {
        "push": {
            "@type": "Push",
            "id": "ae63d76e-6d00-45aa-97a6-c8a2b54390e5",
            "href": "https://{server}:{port}/tmf-
api/resourcePoolManagement/v5/resourcePool/c3fef858-b864-4fac-971e-
795b1195ded9/push/ae63d76e-6d00-45aa-97a6-c8a2b54390e5",
            "pushedResource":
            [
                {
                    "@type": "ResourceRef",
                    "@referredType": "LogicalResource",
                    "id": "4d179a07-1555-4e5f-a2de-0a51f5dd7e15",
                    "href": "https://{server}:{port}/tmf-
api/resourceInventoryManagement/v5/resource/4d179a07-1555-4e5f-a2de-0a51f5dd7e15"
                },
                {
                    "@type": "ResourceRef",
                    "@referredType": "LogicalResource",
                    "id": "15dfa8a7-f9db-43a8-8758-9d9df5a76264",
                    "href": "https://{server}:{port}/tmf-
api/resourceInventoryManagement/v5/resource/15dfa8a7-f9db-43a8-8758-9d9df5a76264"
                },
                {
                    "@type": "ResourceRef",
                    "@referredType": "LogicalResource",
                    "id": "195aa8ee-7560-4d97-9b9a-2add1a4e0372",
                    "href": "https://{server}:{port}/tmf-
api/resourceInventoryManagement/v5/resource/195aa8ee-7560-4d97-9b9a-2add1a4e0372"
                }
            ]
        }
    },
    "reportingSystem": {
        "id": "109",
        "name": "APP-804",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "source": {
        "id": "8",
        "name": "APP-307",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    }
}
```

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Delete Event

Message example for PushDeleteEvent event

```
Content-Type: application/json

{
    "@type": "PushDeleteEvent",
    "@baseType": "Event",
    "correlationId": "e4888f4d-71cd-43a4-a018-dd13f230935c",
    "description": "PushDeleteEvent illustration",
    "domain": "Commercial",
    "eventId": "3b63e628-0c66-4d57-abbf-2fb6aeb229fd",
    "eventTime": "2024-09-19T13:55:06.910Z",
    "eventType": "PushDeleteEvent",
    "priority": "3",
    "timeOccurred": "2024-09-19T13:55:03.868Z",
    "title": "PushDeleteEvent",
    "event": {
        "push": {
            "@type": "Push",
            "id": "ae63d76e-6d00-45aa-97a6-c8a2b54390e5",
            "href": "https://{server}:{port}/tmf-
api/resourcePoolManagement/v5/resourcePool/c3fef858-b864-4fac-971e-
795b1195ded9/push/ae63d76e-6d00-45aa-97a6-c8a2b54390e5"
        }
    },
    "reportingSystem": {
        "id": "109",
        "name": "APP-804",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "source": {
        "id": "8",
        "name": "APP-307",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    }
}
```

State Change Event

Message example for PushStateChangeEvent event

```
Content-Type: application/json

{
    "@type": "PushStateChangeEvent",
    "@baseType": "Event",
    "correlationId": "e4888f4d-71cd-43a4-a018-dd13f230935c",
    "description": "PushStateChangeEvent illustration",
    "domain": "Commercial",
    "eventId": "3b63e628-0c66-4d57-abbf-2fb6aeb229fd",
    "eventTime": "2024-09-19T13:55:06.910Z",
    "eventType": "PushStateChangeEvent",
    "priority": "3",
    "timeOccurred": "2024-09-19T13:55:03.868Z",
```

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```
    "title": "PushStateChangeEvent",
    "event": {
        "push": {
            "@type": "Push",
            "id": "ae63d76e-6d00-45aa-97a6-c8a2b54390e5",
            "href": "https://{server}:{port}/tmf-
api/resourcePoolManagement/v5/resourcePool/c3fef858-b864-4fac-971e-
795b1195ded9/push/ae63d76e-6d00-45aa-97a6-c8a2b54390e5",
            "state": "done"
        }
    },
    "reportingSystem": {
        "id": "109",
        "name": "APP-804",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "source": {
        "id": "8",
        "name": "APP-307",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    }
}
```


Extract

Create Event

Message example for ExtractCreateEvent event


```
Content-Type: application/json

{
    "@type": "ExtractCreateEvent",
    "@baseType": "Event",
    "correlationId": "e4888f4d-71cd-43a4-a018-dd13f230935c",
    "description": "ExtractCreateEvent illustration",
    "domain": "Commercial",
    "eventId": "3b63e628-0c66-4d57-abbf-2fb6aeb229fd",
    "eventTime": "2024-09-19T13:55:06.910Z",
    "eventType": "ExtractCreateEvent",
    "priority": "3",
    "timeOccurred": "2024-09-19T13:55:03.868Z",
    "title": "ExtractCreateEvent",
    "event": {
        "extract": {
            "@type": "Extract",
            "id": "c6a1e831-ef5b-4d09-85a8-594fca73313b",
            "href": "https://{server}:{port}/tmf-
api/resourcePoolManagement/v5/resourcePool/c3fef858-b864-4fac-971e-
795b1195ded9/extract/c6a1e831-ef5b-4d09-85a8-594fca73313b",
            "capacityDemand":
            {
                "@type": "Capacity",
                "capacityAmount":
                {
                    "amount": 2,
                    "units": "number"
```


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```
                },
                "capacityStatus": "planned",
                "applicableTimePeriod":
                [
                    {
                        "@type": "ApplicableTimePeriod",
                        "fromToDateTime":
                        {
                            "startDateTime": "2024-05-13T00:00:00Z",
                            "endDateTime": "2024-09-20T00:00:00Z"
                        }
                    }
                ]
            }
        }
    },
    "reportingSystem": {
        "id": "109",
        "name": "APP-804",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "source": {
        "id": "8",
        "name": "APP-307",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    }
}
```


Delete Event

Message example for ExtractDeleteEvent event


```
Content-Type: application/json

{
    "@type": "ExtractDeleteEvent",
    "@baseType": "Event",
    "correlationId": "e4888f4d-71cd-43a4-a018-dd13f230935c",
    "description": "ExtractDeleteEvent illustration",
    "domain": "Commercial",
    "eventId": "3b63e628-0c66-4d57-abbf-2fb6aeb229fd",
    "eventTime": "2024-09-19T13:55:06.910Z",
    "eventType": "ExtractDeleteEvent",
    "priority": "3",
    "timeOccurred": "2024-09-19T13:55:03.868Z",
    "title": "ExtractDeleteEvent",
    "event": {
        "extract": {
            "@type": "Extract",
            "id": "c6a1e831-ef5b-4d09-85a8-594fca73313b",
            "href": "https://{server}:{port}/tmf-
api/resourcePoolManagement/v5/resourcePool/c3fef858-b864-4fac-971e-
795b1195ded9/extract/c6a1e831-ef5b-4d09-85a8-594fca73313b"
        }
    },
    "reportingSystem": {
        "id": "109",
        "name": "APP-804",
```


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```
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "source": {
        "id": "8",
        "name": "APP-307",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    }
}
```

State Change Event

Message example for ExtractStateChangeEvent event

```
Content-Type: application/json

{
    "@type": "ExtractStateChangeEvent",
    "@baseType": "Event",
    "correlationId": "e4888f4d-71cd-43a4-a018-dd13f230935c",
    "description": "ExtractStateChangeEvent illustration",
    "domain": "Commercial",
    "eventId": "3b63e628-0c66-4d57-abbf-2fb6aeb229fd",
    "eventTime": "2024-09-19T13:55:06.910Z",
    "eventType": "ExtractStateChangeEvent",
    "priority": "3",
    "timeOccurred": "2024-09-19T13:55:03.868Z",
    "title": "ExtractStateChangeEvent",
    "event": {
        "extract": {
            "@type": "Extract",
            "id": "c6a1e831-ef5b-4d09-85a8-594fca73313b",
            "href": "https://{server}:{port}/tmf-
api/resourcePoolManagement/v5/resourcePool/c3fef858-b864-4fac-971e-
795b1195ded9/extract/c6a1e831-ef5b-4d09-85a8-594fca73313b",
            "state": "done"
        }
    },
    "reportingSystem": {
        "id": "109",
        "name": "APP-804",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "source": {
        "id": "8",
        "name": "APP-307",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    }
}
```

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API OPERATIONS

Remember the following Uniform Contract:

Operation on Entities	Uniform API Operation	Description
Query Entities	GET Resource	GET must be used to retrieve a representation of a resource.
Create Entity	POST Resource	POST must be used to create a new resource
Partial Update of an Entity	PATCH Resource	PATCH must be used to partially update a resource
Remove an Entity	DELETE Resource	DELETE must be used to remove a resource
Execute an Action on an Entity	POST on TASK Resource	POST must be used to execute Task Resources
Other Request Methods	POST on TASK Resource	GET and POST must not be used to tunnel other request methods.

Filtering and attribute selection rules are described in the TMF REST Design Guidelines.

Notifications are also described in a subsequent section.

Operations on CapacitySpecification

Retrieves a CapacitySpecification by ID

`GET /capacitySpecification/{id}?fields=…`

Description

This operation retrieves a CapacitySpecification entity. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Here's an example of a request for retrieving a CapacitySpecification resource

Request

```
GET /capacitySpecification/3d01344d-4409-4b15-8fcb-b8802314ff9c
Content-Type: application/json
```

Response

```
200
```

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```
{
    "@type": "CapacitySpecification",
    "id": "3d01344d-4409-4b15-8fcb-b8802314ff9c",
    "href": "https://{server}:{port}/tmf-
api/resourcePoolManagement/v5/capacitySpecification/3d01344d-4409-4b15-8fcb-
b8802314ff9c",
    "name": "Phone Number range",
    "description": "This CapacitySpecification defines the details of capacity for
Phone Number ranges.",
    "capacityCharacteristicSpecification":
    [
        {
            "@type": "CharacteristicSpecification",
            "name": "lowerNumber",
            "description": "the lower boundary of a block of phone numbers; phone
numbers consist of a country code (cc), national destination code (ndc), subscriber
number (sn), and optionally a direct dial in (ddi), all sections separated by '-'",
            "valueType": "string",
            "characteristicValueSpecification":
            [
                {
                    "@type": "CharacteristicValueSpecification",
                    "valueType": "string",
                    "regex": "^\\+(?<cc>[1-9]\\d{0,2})-(?<ndc>[1-9]\\d*)-(?<sn>[1-
9]\\d*)(?:-?(?<ddi>\\d+))?$"
                }
            ]
        },
        {
            "@type": "CharacteristicSpecification",
            "name": "upperNumber",
            "description": "the upper boundary of a block of phone numbers; phone
numbers consist of a country code (cc), national destination code (ndc), subscriber
number (sn), and optionally a direct dial in (ddi), all sections separated by '-'",
            "valueType": "string",
            "characteristicValueSpecification":
            [
                {
                    "@type": "CharacteristicValueSpecification",
                    "valueType": "string",
                    "regex": "^\\+(?<cc>[1-9]\\d{0,2})-(?<ndc>[1-9]\\d*)-(?<sn>[1-
9]\\d*)(?:-?(?<ddi>\\d+))?$"
                }
            ]
        }
    ]
}
```

List or find CapacitySpecification objects

`GET /capacitySpecification?fields=…`

Description

This operation list CapacitySpecification entities. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

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Here's an example of a request for retrieving CapacitySpecification resources.

Request

```
GET /capacitySpecification?name=Phone Number range
Content-Type: application/json
```

Response

```
200

[
    {
        "@type": "CapacitySpecification",
        "id": "3d01344d-4409-4b15-8fcb-b8802314ff9c",
        "href": "https://{server}:{port}/tmf-
api/resourcePoolManagement/v5/capacitySpecification/3d01344d-4409-4b15-8fcb-
b8802314ff9c",
        "name": "Phone Number range",
        "description": "This CapacitySpecification defines the details of capacity
for Phone Number ranges.",
        "capacityCharacteristicSpecification":
        [
            {
                "@type": "CharacteristicSpecification",
                "name": "lowerNumber",
                "description": "the lower boundary of a block of phone numbers;
phone numbers consist of a country code (cc), national destination code (ndc),
subscriber number (sn), and optionally a direct dial in (ddi), all sections
separated by '-'",
                "valueType": "string",
                "characteristicValueSpecification":
                [
                    {
                        "@type": "CharacteristicValueSpecification",
                        "valueType": "string",
                        "regex": "^\\+(?<cc>[1-9]\\d{0,2})-(?<ndc>[1-9]\\d*)-
(?<sn>[1-9]\\d*)(?:-?(?<ddi>\\d+))?$"
                    }
                ]
            },
            {
                "@type": "CharacteristicSpecification",
                "name": "upperNumber",
                "description": "the upper boundary of a block of phone numbers;
phone numbers consist of a country code (cc), national destination code (ndc),
subscriber number (sn), and optionally a direct dial in (ddi), all sections
separated by '-'",
                "valueType": "string",
                "characteristicValueSpecification":
                [
                    {
                        "@type": "CharacteristicValueSpecification",
                        "valueType": "string",
                        "regex": "^\\+(?<cc>[1-9]\\d{0,2})-(?<ndc>[1-9]\\d*)-
(?<sn>[1-9]\\d*)(?:-?(?<ddi>\\d+))?$"
                    }
                ]
            }
```

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```
        ]
    },
    {
        "@type": "CapacitySpecification",
        "id": "3a2e4da3-ecf8-41d8-b379-e05998761f87",
        "href": "https://{server}:{port}/tmf-
api/resourcePoolManagement/v5/capacitySpecification/3a2e4da3-ecf8-41d8-b379-
e05998761f87",
        "name": "IPv4 Address range",
        "description": "This CapacitySpecification defines the details of capacity
for IP Address ranges.",
        "capacityCharacteristicSpecification":
        [
            {
                "@type": "CharacteristicSpecification",
                "name": "IPv4-PrefixSize",
                "description": "Number of bits to specify a range of IPv4 addresses,
e.g. the '8' in '192.0.0.0/8'",
                "valueType": "integer",
                "characteristicValueSpecification":
                [
                    {
                        "@type": "CharacteristicValueSpecification",
                        "valueType": "integer",
                        "valueFrom": 0,
                        "valueTo": 32
                    }
                ]
            },
            {
                "@type": "CharacteristicSpecification",
                "name": "IPv4-Address",
                "description": "a IPv4 address in decimal dot notation, e.g.
192.0.0.1",
                "valueType": "string",
                "characteristicValueSpecification":
                [
                    {
                        "@type": "CharacteristicValueSpecification",
                        "valueType": "string",
                        "regex": "^\\+(?<cc>[1-9]\\d{0,2})-(?<ndc>[1-9]\\d*)-
(?<sn>[1-9]\\d*)(?:-?(?<ddi>\\d+))?$"
                    }
                ]
            }
        ]
    }
]
```


Creates a CapacitySpecification

`POST /capacitySpecification?fields=…`

Description

This operation creates a CapacitySpecification entity.

Mandatory Attributes

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Mandatory Attributes	Rule
@type	

Usage samples

Here's an example of a request for creating a CapacitySpecification resources.

Request

```
POST /capacitySpecification
Content-Type: application/json

{
    "@type": "CapacitySpecification",
    "name": "Phone Number range",
    "description": "This CapacitySpecification defines the details of capacity for
Phone Number ranges.",
    "capacityCharacteristicSpecification":
    [
        {
            "@type": "CharacteristicSpecification",
            "name": "lowerNumber",
            "description": "the lower boundary of a block of phone numbers; phone
numbers consist of a country code (cc), national destination code (ndc), subscriber
number (sn), and optionally a direct dial in (ddi), all sections separated by '-'",
            "valueType": "string",
            "characteristicValueSpecification":
            [
                {
                    "@type": "CharacteristicValueSpecification",
                    "valueType": "string",
                    "regex": "^\\+(?<cc>[1-9]\\d{0,2})-(?<ndc>[1-9]\\d*)-(?<sn>[1-
9]\\d*)(?:-?(?<ddi>\\d+))?$"
                }
            ]
        },
        {
            "@type": "CharacteristicSpecification",
            "name": "upperNumber",
            "description": "the upper boundary of a block of phone numbers; phone
numbers consist of a country code (cc), national destination code (ndc), subscriber
number (sn), and optionally a direct dial in (ddi), all sections separated by '-'",
            "valueType": "string",
            "characteristicValueSpecification":
            [
                {
                    "@type": "CharacteristicValueSpecification",
                    "valueType": "string",
                    "regex": "^\\+(?<cc>[1-9]\\d{0,2})-(?<ndc>[1-9]\\d*)-(?<sn>[1-
9]\\d*)(?:-?(?<ddi>\\d+))?$"
                }
            ]
        }
    ]
}
```

Response

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```
201

{
    "@type": "CapacitySpecification",
    "id": "3d01344d-4409-4b15-8fcb-b8802314ff9c",
    "href": "https://{server}:{port}/tmf-
api/resourcePoolManagement/v5/capacitySpecification/3d01344d-4409-4b15-8fcb-
b8802314ff9c",
    "name": "Phone Number range",
    "description": "This CapacitySpecification defines the details of capacity for
Phone Number ranges.",
    "capacityCharacteristicSpecification":
    [
        {
            "@type": "CharacteristicSpecification",
            "name": "lowerNumber",
            "description": "the lower boundary of a block of phone numbers; phone
numbers consist of a country code (cc), national destination code (ndc), subscriber
number (sn), and optionally a direct dial in (ddi), all sections separated by '-'",
            "valueType": "string",
            "characteristicValueSpecification":
            [
                {
                    "@type": "CharacteristicValueSpecification",
                    "valueType": "string",
                    "regex": "^\\+(?<cc>[1-9]\\d{0,2})-(?<ndc>[1-9]\\d*)-(?<sn>[1-
9]\\d*)(?:-?(?<ddi>\\d+))?$"
                }
            ]
        },
        {
            "@type": "CharacteristicSpecification",
            "name": "upperNumber",
            "description": "the upper boundary of a block of phone numbers; phone
numbers consist of a country code (cc), national destination code (ndc), subscriber
number (sn), and optionally a direct dial in (ddi), all sections separated by '-'",
            "valueType": "string",
            "characteristicValueSpecification":
            [
                {
                    "@type": "CharacteristicValueSpecification",
                    "valueType": "string",
                    "regex": "^\\+(?<cc>[1-9]\\d{0,2})-(?<ndc>[1-9]\\d*)-(?<sn>[1-
9]\\d*)(?:-?(?<ddi>\\d+))?$"
                }
            ]
        }
    ]
}
```


Updates partially a CapacitySpecification

`PATCH /capacitySpecification/{id}?fields=…`

Description

This operation allows partial updates of a CapacitySpecification entity. Support of json/merge (https://tools.ietf.org/html/rfc7396) is mandatory, support of json/patch (http://tools.ietf.org/html/rfc5789) is optional. Note: If the update operation yields to the creation of sub-resources

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or relationships, the same rules concerning mandatory sub-resource attributes and default value settings in the POST operation applies to the PATCH operation. Hence these tables are not repeated here.

Patchable and Non Patchable Attributes

Non Patchable Attributes	Rule
href	
id	
@baseType	@baseType is immutable
@schemaLocation	@schemaLocation is immutable
@type	@type is immutable

Patchable Attributes	Rule
capacityCharacteristicSpecification	
description	
externalIdentifier	
name	
relatedCapacitySpecification	
validFor	

Usage samples

Here is an example of a request for patching a CapacitySpecification using APPLICATION-JSON syntax.

Request

```
PATCH /capacitySpecification/3d01344d-4409-4b15-8fcb-b8802314ff9c
Content-Type: application/json

{
    "@type": "CapacitySpecification",
    "name": "Phone Number Range"
}
```

Response

```
200

{
    "@type": "CapacitySpecification",
    "id": "3d01344d-4409-4b15-8fcb-b8802314ff9c",
    "href": "https://{server}:{port}/tmf-
api/resourcePoolManagement/v5/capacitySpecification/3d01344d-4409-4b15-8fcb-
b8802314ff9c",
```

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```
    "name": "Phone Number Range",
    "description": "This CapacitySpecification defines the details of capacity for
Phone Number ranges.",
    "capacityCharacteristicSpecification":
    [
        {
            "@type": "CharacteristicSpecification",
            "name": "lowerNumber",
            "description": "the lower boundary of a block of phone numbers; phone
numbers consist of a country code (cc), national destination code (ndc), subscriber
number (sn), and optionally a direct dial in (ddi), all sections separated by '-'",
            "valueType": "string",
            "characteristicValueSpecification":
            [
                {
                    "@type": "CharacteristicValueSpecification",
                    "valueType": "string",
                    "regex": "^\\+(?<cc>[1-9]\\d{0,2})-(?<ndc>[1-9]\\d*)-(?<sn>[1-
9]\\d*)(?:-?(?<ddi>\\d+))?$"
                }
            ]
        },
        {
            "@type": "CharacteristicSpecification",
            "name": "upperNumber",
            "description": "the upper boundary of a block of phone numbers; phone
numbers consist of a country code (cc), national destination code (ndc), subscriber
number (sn), and optionally a direct dial in (ddi), all sections separated by '-'",
            "valueType": "string",
            "characteristicValueSpecification":
            [
                {
                    "@type": "CharacteristicValueSpecification",
                    "valueType": "string",
                    "regex": "^\\+(?<cc>[1-9]\\d{0,2})-(?<ndc>[1-9]\\d*)-(?<sn>[1-
9]\\d*)(?:-?(?<ddi>\\d+))?$"
                }
            ]
        }
    ]
}
```

Here is an example of a request for patching a CapacitySpecification using MERGE syntax.

Request

```
PATCH /capacitySpecification/3d01344d-4409-4b15-8fcb-b8802314ff9c
Content-Type: application/merge-patch+json

{
    "@type": "CapacitySpecification",
    "name": "Phone Number Range"
}
```

Response

```
200
```

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```
{
    "@type": "CapacitySpecification",
    "id": "3d01344d-4409-4b15-8fcb-b8802314ff9c",
    "href": "https://{server}:{port}/tmf-
api/resourcePoolManagement/v5/capacitySpecification/3d01344d-4409-4b15-8fcb-
b8802314ff9c",
    "name": "Phone Number Range",
    "description": "This CapacitySpecification defines the details of capacity for
Phone Number ranges.",
    "capacityCharacteristicSpecification":
    [
        {
            "@type": "CharacteristicSpecification",
            "name": "lowerNumber",
            "description": "the lower boundary of a block of phone numbers; phone
numbers consist of a country code (cc), national destination code (ndc), subscriber
number (sn), and optionally a direct dial in (ddi), all sections separated by '-'",
            "valueType": "string",
            "characteristicValueSpecification":
            [
                {
                    "@type": "CharacteristicValueSpecification",
                    "valueType": "string",
                    "regex": "^\\+(?<cc>[1-9]\\d{0,2})-(?<ndc>[1-9]\\d*)-(?<sn>[1-
9]\\d*)(?:-?(?<ddi>\\d+))?$"
                }
            ]
        },
        {
            "@type": "CharacteristicSpecification",
            "name": "upperNumber",
            "description": "the upper boundary of a block of phone numbers; phone
numbers consist of a country code (cc), national destination code (ndc), subscriber
number (sn), and optionally a direct dial in (ddi), all sections separated by '-'",
            "valueType": "string",
            "characteristicValueSpecification":
            [
                {
                    "@type": "CharacteristicValueSpecification",
                    "valueType": "string",
                    "regex": "^\\+(?<cc>[1-9]\\d{0,2})-(?<ndc>[1-9]\\d*)-(?<sn>[1-
9]\\d*)(?:-?(?<ddi>\\d+))?$"
                }
            ]
        }
    ]
}
```

Here is an example of a request for patching a CapacitySpecification using JSON-PATCH syntax.

Request

```
PATCH /capacitySpecification/3d01344d-4409-4b15-8fcb-b8802314ff9c
Content-Type: application/json-patch+json

[
    {
        "op": "replace",
```

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```
        "path": "/name",
        "value": "Phone Number Range"
    },
    {
        "op": "add",
        "path": "/validFor",
        "value":
        {
            "startDateTime": "2024-05-13T00:00:00Z",
            "endDateTime": "2024-09-20T00:00:00Z"
        }
    }
]
```

Response

```
200

{
    "@type": "CapacitySpecification",
    "id": "3d01344d-4409-4b15-8fcb-b8802314ff9c",
    "href": "https://{server}:{port}/tmf-
api/resourcePoolManagement/v5/capacitySpecification/3d01344d-4409-4b15-8fcb-
b8802314ff9c",
    "name": "Phone Number Range",
    "description": "This CapacitySpecification defines the details of capacity for
Phone Number ranges.",
    "capacityCharacteristicSpecification":
    [
        {
            "@type": "CharacteristicSpecification",
            "name": "lowerNumber",
            "description": "the lower boundary of a block of phone numbers; phone
numbers consist of a country code (cc), national destination code (ndc), subscriber
number (sn), and optionally a direct dial in (ddi), all sections separated by '-'",
            "valueType": "string",
            "characteristicValueSpecification":
            [
                {
                    "@type": "CharacteristicValueSpecification",
                    "valueType": "string",
                    "regex": "^\\+(?<cc>[1-9]\\d{0,2})-(?<ndc>[1-9]\\d*)-(?<sn>[1-
9]\\d*)(?:-?(?<ddi>\\d+))?$"
                }
            ]
        },
        {
            "@type": "CharacteristicSpecification",
            "name": "upperNumber",
            "description": "the upper boundary of a block of phone numbers; phone
numbers consist of a country code (cc), national destination code (ndc), subscriber
number (sn), and optionally a direct dial in (ddi), all sections separated by '-'",
            "valueType": "string",
            "characteristicValueSpecification":
            [
                {
                    "@type": "CharacteristicValueSpecification",
                    "valueType": "string",
                    "regex": "^\\+(?<cc>[1-9]\\d{0,2})-(?<ndc>[1-9]\\d*)-(?<sn>[1-
9]\\d*)(?:-?(?<ddi>\\d+))?$"
```

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```
                }
            ]
        }
    ],
    "validFor":
    {
        "startDateTime": "2024-05-13T00:00:00Z",
        "endDateTime": "2024-09-20T00:00:00Z"
    }
}
```

Here is an example of a request for patching a CapacitySpecification using JSON-PATCH-QUERY syntax.

Request

```
PATCH /capacitySpecification/3d01344d-4409-4b15-8fcb-b8802314ff9c
Content-Type: application/json-patch-query+json


[
    {
        "op": "replace",
        "path": "$.capacityCharacteristicSpecification[0]/name",
        "value": "Lower Number"
    }
]
```

Response

```
200

{
    "@type": "CapacitySpecification",
    "id": "3d01344d-4409-4b15-8fcb-b8802314ff9c",
    "href": "https://{server}:{port}/tmf-
api/resourcePoolManagement/v5/capacitySpecification/3d01344d-4409-4b15-8fcb-
b8802314ff9c",
    "name": "Phone Number range",
    "description": "This CapacitySpecification defines the details of capacity for
Phone Number ranges.",
    "capacityCharacteristicSpecification":
    [
        {
            "@type": "CharacteristicSpecification",
            "name": "Lower Number",
            "description": "the lower boundary of a block of phone numbers; phone
numbers consist of a country code (cc), national destination code (ndc), subscriber
number (sn), and optionally a direct dial in (ddi), all sections separated by '-'",
            "valueType": "string",
            "characteristicValueSpecification":
            [
                {
                    "@type": "CharacteristicValueSpecification",
                    "valueType": "string",
                    "regex": "^\\+(?<cc>[1-9]\\d{0,2})-(?<ndc>[1-9]\\d*)-(?<sn>[1-
9]\\d*)(?:-?(?<ddi>\\d+))?$"
```

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```
                }
            ]
        },
        {
            "@type": "CharacteristicSpecification",
            "name": "upperNumber",
            "description": "the upper boundary of a block of phone numbers; phone
numbers consist of a country code (cc), national destination code (ndc), subscriber
number (sn), and optionally a direct dial in (ddi), all sections separated by '-'",
            "valueType": "string",
            "characteristicValueSpecification":
            [
                {
                    "@type": "CharacteristicValueSpecification",
                    "valueType": "string",
                    "regex": "^\\+(?<cc>[1-9]\\d{0,2})-(?<ndc>[1-9]\\d*)-(?<sn>[1-
9]\\d*)(?:-?(?<ddi>\\d+))?$"
                }
            ]
        }
    ]
}
```

Deletes a CapacitySpecification

`DELETE /capacitySpecification/{id}`

Description

This operation deletes a CapacitySpecification entity.

Operations on ResourcePoolSpecification

List or find ResourcePoolSpecification objects

`GET /resourcePoolSpecification?fields=…`

Description

This operation list ResourcePoolSpecification entities. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Here's an example of a request for retrieving ResourcePoolSpecification resources.

Request

```
GET /resourcePoolSpecification?name=Phone Number pool
Content-Type: application/json
```

Response

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```
200

[
    {
        "@type": "ResourcePoolSpecification",
        "id": "b0e0827c-5693-4e41-8445-22438abb33cb",
        "href": "https://{server}:{port}/tmf-
api/resourcePoolManagement/v5/resourcePoolSpecification/b0e0827c-5693-4e41-8445-
22438abb33cb",
        "name": "Phone Number pool",
        "description": "Specifies the details of ResourcePools for Phone Number
ranges",
        "capacitySpecification":
        [
            {
                "@type": "CapacitySpecificationRef",
                "@referredType": "CapacitySpecification",
                "id": "3d01344d-4409-4b15-8fcb-b8802314ff9c",
                "href": "https://{server}:{port}/tmf-
api/resourcePoolManagement/v5/capacitySpecification/3d01344d-4409-4b15-8fcb-
b8802314ff9c",
                "name": "Phone Number range"
            }
        ],
        "resourceSpecCharacteristic":
        [
            {
                "@type": "CharacteristicSpecification",
                "name": "cc",
                "description": "Phone Number Country Code",
                "valueType": "string",
                "characteristicValueSpecification":
                [
                    {
                        "@type": "CharacteristicValueSpecification",
                        "valueType": "string",
                        "regex": "^[1-9]\\d{0,2}$"
                    }
                ]
            },
            {
                "@type": "CharacteristicSpecification",
                "name": "ndc",
                "description": "Phone Number National Destination Code",
                "valueType": "string",
                "characteristicValueSpecification":
                [
                    {
                        "@type": "CharacteristicValueSpecification",
                        "valueType": "string",
                        "regex": "^[1-9]\\d*?$"
                    }
                ]
            }
        ]
    }
]
```

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Retrieves a ResourcePoolSpecification by ID

`GET /resourcePoolSpecification/{id}?fields=…`

Description

This operation retrieves a ResourcePoolSpecification entity. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Here's an example of a request for retrieving a ResourcePoolSpecification resource

Request

```
GET /resourcePoolSpecification/b0e0827c-5693-4e41-8445-22438abb33cb
Content-Type: application/json
```

Response

```
200

{
    "@type": "ResourcePoolSpecification",
    "id": "b0e0827c-5693-4e41-8445-22438abb33cb",
    "href": "https://{server}:{port}/tmf-
api/resourcePoolManagement/v5/resourcePoolSpecification/b0e0827c-5693-4e41-8445-
22438abb33cb",
    "name": "Phone Number pool",
    "description": "Specifies the details of ResourcePools for Phone Number ranges",
    "capacitySpecification":
    [
        {
            "@type": "CapacitySpecificationRef",
            "@referredType": "CapacitySpecification",
            "id": "3d01344d-4409-4b15-8fcb-b8802314ff9c",
            "href": "https://{server}:{port}/tmf-
api/resourcePoolManagement/v5/capacitySpecification/3d01344d-4409-4b15-8fcb-
b8802314ff9c",
            "name": "Phone Number range"
        }
    ],
    "resourceSpecCharacteristic":
    [
        {
            "@type": "CharacteristicSpecification",
            "name": "cc",
            "description": "Phone Number Country Code",
            "valueType": "string",
            "characteristicValueSpecification":
            [
                {
                    "@type": "CharacteristicValueSpecification",
                    "valueType": "string",
                    "regex": "^[1-9]\\d{0,2}$"
                }
```

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```
            ]
        },
        {
            "@type": "CharacteristicSpecification",
            "name": "ndc",
            "description": "Phone Number National Destination Code",
            "valueType": "string",
            "characteristicValueSpecification":
            [
                {
                    "@type": "CharacteristicValueSpecification",
                    "valueType": "string",
                    "regex": "^[1-9]\\d*?$"
                }
            ]
        }
    ]
}
```

Creates a ResourcePoolSpecification

POST /resourcePoolSpecification?fields=…

Description

This operation creates a ResourcePoolSpecification entity.

Mandatory Attributes

Mandatory Attributes	Rule
@type	

Usage samples

Here's an example of a request for creating a ResourcePoolSpecification resources.

Request

```
POST /resourcePoolSpecification
Content-Type: application/json

{
    "@type": "ResourcePoolSpecification",
    "name": "Phone Number pool",
    "description": "Specifies the details of ResourcePools for Phone Number ranges",
    "capacitySpecification":
    [
        {
            "@type": "CapacitySpecificationRef",
            "@referredType": "CapacitySpecification",
            "id": "3d01344d-4409-4b15-8fcb-b8802314ff9c",
            "href": "https://{server}:{port}/tmf-
api/resourcePoolManagement/v5/capacitySpecification/3d01344d-4409-4b15-8fcb-
b8802314ff9c",
            "name": "Phone Number range"
        }
```

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```
    ],
    "resourceSpecCharacteristic":
    [
        {
            "@type": "CharacteristicSpecification",
            "name": "cc",
            "description": "Phone Number Country Code",
            "valueType": "string",
            "characteristicValueSpecification":
            [
                {
                    "@type": "CharacteristicValueSpecification",
                    "valueType": "string",
                    "regex": "^[1-9]\\d{0,2}$"
                }
            ]
        },
        {
            "@type": "CharacteristicSpecification",
            "name": "ndc",
            "description": "Phone Number National Destination Code",
            "valueType": "string",
            "characteristicValueSpecification":
            [
                {
                    "@type": "CharacteristicValueSpecification",
                    "valueType": "string",
                    "regex": "^[1-9]\\d*?$"
                }
            ]
        }
    ]
}
```

Response

```
201

{
    "@type": "ResourcePoolSpecification",
    "id": "b0e0827c-5693-4e41-8445-22438abb33cb",
    "href": "https://{server}:{port}/tmf-
api/resourcePoolManagement/v5/resourcePoolSpecification/b0e0827c-5693-4e41-8445-
22438abb33cb",
    "name": "Phone Number pool",
    "description": "Specifies the details of ResourcePools for Phone Number ranges",
    "capacitySpecification":
    [
        {
            "@type": "CapacitySpecificationRef",
            "@referredType": "CapacitySpecification",
            "id": "3d01344d-4409-4b15-8fcb-b8802314ff9c",
            "href": "https://{server}:{port}/tmf-
api/resourcePoolManagement/v5/capacitySpecification/3d01344d-4409-4b15-8fcb-
b8802314ff9c",
            "name": "Phone Number range"
        }
    ],
    "resourceSpecCharacteristic":
    [
```

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```
        {
            "@type": "CharacteristicSpecification",
            "name": "cc",
            "description": "Phone Number Country Code",
            "valueType": "string",
            "characteristicValueSpecification":
            [
                {
                    "@type": "CharacteristicValueSpecification",
                    "valueType": "string",
                    "regex": "^[1-9]\\d{0,2}$"
                }
            ]
        },
        {
            "@type": "CharacteristicSpecification",
            "name": "ndc",
            "description": "Phone Number National Destination Code",
            "valueType": "string",
            "characteristicValueSpecification":
            [
                {
                    "@type": "CharacteristicValueSpecification",
                    "valueType": "string",
                    "regex": "^[1-9]\\d*?$"
                }
            ]
        }
    ]
}
```

Updates partially a ResourcePoolSpecification

`PATCH /resourcePoolSpecification/{id}?fields=…`

Description

This operation allows partial updates of a ResourcePoolSpecification entity. Support of json/merge (https://tools.ietf.org/html/rfc7396) is mandatory, support of json/patch (http://tools.ietf.org/html/rfc5789) is optional. Note: If the update operation yields to the creation of sub-resources or relationships, the same rules concerning mandatory sub-resource attributes and default value settings in the POST operation applies to the PATCH operation. Hence these tables are not repeated here.

Patchable and Non Patchable Attributes

Non Patchable Attributes	Rule
href	
id	
@baseType	@baseType is immutable
@schemaLocation	@schemaLocation is immutable
@type	@type is immutable

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Patchable Attributes	Rule
attachment	
capacitySpecification	
category	
description	
externalIdentifier	
featureSpecification	
intentSpecification	
isBundle	
lastUpdate	
lifecycleStatus	
name	
relatedParty	
resourceSpecCharacteristic	
resourceSpecRelationship	
targetResourceSchema	
validFor	
version	

Usage samples

Here is an example of a request for patching a ResourcePoolSpecification using APPLICATION-JSON syntax.

Request

```
PATCH /resourcePoolSpecification/b0e0827c-5693-4e41-8445-22438abb33cb
Content-Type: application/json

{
    "@type": "ResourcePoolSpecification",
    "validFor":
    {
        "startDateTime": "2024-05-13T00:00:00Z",
        "endDateTime": "2024-09-20T00:00:00Z"
    }
}
```

Response

```
200

{
    "@type": "ResourcePoolSpecification",
```

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```
    "id": "b0e0827c-5693-4e41-8445-22438abb33cb",
    "href": "https://{server}:{port}/tmf-
api/resourcePoolManagement/v5/resourcePoolSpecification/b0e0827c-5693-4e41-8445-
22438abb33cb",
    "name": "Phone Number pool",
    "description": "Specifies the details of ResourcePools for Phone Number ranges",
    "capacitySpecification":
    [
        {
            "@type": "CapacitySpecificationRef",
            "@referredType": "CapacitySpecification",
            "id": "3d01344d-4409-4b15-8fcb-b8802314ff9c",
            "href": "https://{server}:{port}/tmf-
api/resourcePoolManagement/v5/capacitySpecification/3d01344d-4409-4b15-8fcb-
b8802314ff9c",
            "name": "Phone Number range"
        }
    ],
    "resourceSpecCharacteristic":
    [
        {
            "@type": "CharacteristicSpecification",
            "name": "cc",
            "description": "Phone Number Country Code",
            "valueType": "string",
            "characteristicValueSpecification":
            [
                {
                    "@type": "CharacteristicValueSpecification",
                    "valueType": "string",
                    "regex": "^[1-9]\\d{0,2}$"
                }
            ]
        },
        {
            "@type": "CharacteristicSpecification",
            "name": "ndc",
            "description": "Phone Number National Destination Code",
            "valueType": "string",
            "characteristicValueSpecification":
            [
                {
                    "@type": "CharacteristicValueSpecification",
                    "valueType": "string",
                    "regex": "^[1-9]\\d*?$"
                }
            ]
        }
    ],
    "validFor":
    {
        "startDateTime": "2024-05-13T00:00:00Z",
        "endDateTime": "2024-09-20T00:00:00Z"
    }
}
```


Here is an example of a request for patching a ResourcePoolSpecification using MERGE syntax.

Request

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```
PATCH /resourcePoolSpecification/b0e0827c-5693-4e41-8445-22438abb33cb
Content-Type: application/merge-patch+json

{
    "@type": "ResourcePoolSpecification",
    "validFor":
    {
        "startDateTime": "2024-05-13T00:00:00Z",
        "endDateTime": "2024-09-20T00:00:00Z"
    }
}
```

Response

```
200

{
    "@type": "ResourcePoolSpecification",
    "id": "b0e0827c-5693-4e41-8445-22438abb33cb",
    "href": "https://{server}:{port}/tmf-
api/resourcePoolManagement/v5/resourcePoolSpecification/b0e0827c-5693-4e41-8445-
22438abb33cb",
    "name": "Phone Number pool",
    "description": "Specifies the details of ResourcePools for Phone Number ranges",
    "capacitySpecification":
    [
        {
            "@type": "CapacitySpecificationRef",
            "@referredType": "CapacitySpecification",
            "id": "3d01344d-4409-4b15-8fcb-b8802314ff9c",
            "href": "https://{server}:{port}/tmf-
api/resourcePoolManagement/v5/capacitySpecification/3d01344d-4409-4b15-8fcb-
b8802314ff9c",
            "name": "Phone Number range"
        }
    ],
    "resourceSpecCharacteristic":
    [
        {
            "@type": "CharacteristicSpecification",
            "name": "cc",
            "description": "Phone Number Country Code",
            "valueType": "string",
            "characteristicValueSpecification":
            [
                {
                    "@type": "CharacteristicValueSpecification",
                    "valueType": "string",
                    "regex": "^[1-9]\\d{0,2}$"
                }
            ]
        },
        {
            "@type": "CharacteristicSpecification",
            "name": "ndc",
            "description": "Phone Number National Destination Code",
            "valueType": "string",
            "characteristicValueSpecification":
            [
```

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```
                {
                    "@type": "CharacteristicValueSpecification",
                    "valueType": "string",
                    "regex": "^[1-9]\\d*?$"
                }
            ]
        }
    ],
    "validFor":
    {
        "startDateTime": "2024-05-13T00:00:00Z",
        "endDateTime": "2024-09-20T00:00:00Z"
    }
}
```

Here is an example of a request for patching a ResourcePoolSpecification using JSON-PATCH syntax.

Request

```
PATCH /resourcePoolSpecification/b0e0827c-5693-4e41-8445-22438abb33cb
Content-Type: application/json-patch+json

[
    {
        "op": "replace",
        "path": "/name",
        "value": "Phone Number pool (single subsubriber)"
    },
    {
        "op": "add",
        "path": "/resourceSpecCharacteristic",
        "value":
        {
            "@type": "CharacteristicSpecification",
            "name": "sn",
            "description": "Phone Number Subscriber Number",
            "valueType": "string",
            "characteristicValueSpecification":
            [
                {
                    "@type": "CharacteristicValueSpecification",
                    "valueType": "string",
                    "regex": "^[1-9]\\d*$"
                }
            ]
        }
    }
]
```

Response

```
200

{
    "@type": "ResourcePoolSpecification",
    "id": "b0e0827c-5693-4e41-8445-22438abb33cb",
```

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```
    "href": "https://{server}:{port}/tmf-
api/resourcePoolManagement/v5/resourcePoolSpecification/b0e0827c-5693-4e41-8445-
22438abb33cb",
    "name": "Phone Number pool (single subsubriber)",
    "description": "Specifies the details of ResourcePools for Phone Number ranges",
    "capacitySpecification":
    [
        {
            "@type": "CapacitySpecificationRef",
            "@referredType": "CapacitySpecification",
            "id": "3d01344d-4409-4b15-8fcb-b8802314ff9c",
            "href": "https://{server}:{port}/tmf-
api/resourcePoolManagement/v5/capacitySpecification/3d01344d-4409-4b15-8fcb-
b8802314ff9c",
            "name": "Phone Number range"
        }
    ],
    "resourceSpecCharacteristic":
    [
        {
            "@type": "CharacteristicSpecification",
            "name": "cc",
            "description": "Phone Number Country Code",
            "valueType": "string",
            "characteristicValueSpecification":
            [
                {
                    "@type": "CharacteristicValueSpecification",
                    "valueType": "string",
                    "regex": "^[1-9]\\d{0,2}$"
                }
            ]
        },
        {
            "@type": "CharacteristicSpecification",
            "name": "ndc",
            "description": "Phone Number National Destination Code",
            "valueType": "string",
            "characteristicValueSpecification":
            [
                {
                    "@type": "CharacteristicValueSpecification",
                    "valueType": "string",
                    "regex": "^[1-9]\\d*?$"
                }
            ]
        },
        {
            "@type": "CharacteristicSpecification",
            "name": "sn",
            "description": "Phone Number Subscriber Number",
            "valueType": "string",
            "characteristicValueSpecification":
            [
                {
                    "@type": "CharacteristicValueSpecification",
                    "valueType": "string",
                    "regex": "^[1-9]\\d*$"
                }
            ]
        }
    ]
```


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```
}
```

Here is an example of a request for patching a ResourcePoolSpecification using JSON-PATCH-QUERY syntax.

Request

```
PATCH /resourcePoolSpecification/b0e0827c-5693-4e41-8445-22438abb33cb
Content-Type: application/json-patch-query+json

[
    {
        "op": "replace",
        "path": "$.resourceSpecCharacteristic[?(@.name == 'cc')].description",
        "value": "Phone Number Country Code (CC)"
    },
    {
        "op": "replace",
        "path": "$.resourceSpecCharacteristic[?(@.name == 'ndc')].description",
        "value": "Phone Number National Destination Code (NDC)"
    }
]
```

Response

```
200

{
    "@type": "ResourcePoolSpecification",
    "id": "b0e0827c-5693-4e41-8445-22438abb33cb",
    "href": "https://{server}:{port}/tmf-
api/resourcePoolManagement/v5/resourcePoolSpecification/b0e0827c-5693-4e41-8445-
22438abb33cb",
    "name": "Phone Number pool",
    "description": "Specifies the details of ResourcePools for Phone Number ranges",
    "capacitySpecification":
    [
        {
            "@type": "CapacitySpecificationRef",
            "@referredType": "CapacitySpecification",
            "id": "3d01344d-4409-4b15-8fcb-b8802314ff9c",
            "href": "https://{server}:{port}/tmf-
api/resourcePoolManagement/v5/capacitySpecification/3d01344d-4409-4b15-8fcb-
b8802314ff9c",
            "name": "Phone Number range"
        }
    ],
    "resourceSpecCharacteristic":
    [
        {
            "@type": "CharacteristicSpecification",
            "name": "cc",
            "description": "Phone Number Country Code (CC)",
            "valueType": "string",
            "characteristicValueSpecification":
            [
                {
```

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```
                    "@type": "CharacteristicValueSpecification",
                    "valueType": "string",
                    "regex": "^[1-9]\\d{0,2}$"
                }
            ]
        },
        {
            "@type": "CharacteristicSpecification",
            "name": "ndc",
            "description": "Phone Number National Destination Code (NDC)",
            "valueType": "string",
            "characteristicValueSpecification":
            [
                {
                    "@type": "CharacteristicValueSpecification",
                    "valueType": "string",
                    "regex": "^[1-9]\\d*?$"
                }
            ]
        }
    ]
}
```

Deletes a ResourcePoolSpecification

`DELETE /resourcePoolSpecification/{id}`

Description

This operation deletes a ResourcePoolSpecification entity.

Operations on ResourcePool

List or find ResourcePool objects

`GET /resourcePool?fields=…`

Description

This operation list ResourcePool entities. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Here's an example of a request for retrieving ResourcePool resources.

Request

```
GET /resourcePool?name=Phone Number block for partner MyPartner
Content-Type: application/json
```

Response

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```
200

[
    {
        "@type": "ResourcePool",
        "id": "c3fef858-b864-4fac-971e-795b1195ded9",
        "href": "https://{server}:{port}/tmf-
api/resourcePoolManagement/v5/resourcePool/c3fef858-b864-4fac-971e-795b1195ded9",
        "name": "Phone Number block for partner MyPartner",
        "description": "This Pool collects the PhoneNumber resources assigned to the
partner MyPartner.",
        "pooledResourceSpecification":
        {
            "@type": "ResourceSpecificationRef",
            "@referredType": "ResourceSpecification",
            "id": "89afac38-32e2-40f8-8be2-7b86dd03c9c9",
            "href": "https://{server}:{port}/tmf-
api/resourceCatalogManagement/v5/resourceSpecification/89afac38-32e2-40f8-8be2-
7b86dd03c9c9",
            "name": "Phone Number"
        },
        "resourceCharacteristic":
        [
            {
                "@type": "StringCharacteristic",
                "@baseType": "Characteristic",
                "name": "cc",
                "valueType": "string",
                "value": "49"
            },
            {
                "@type": "StringCharacteristic",
                "@baseType": "Characteristic",
                "name": "ndc",
                "valueType": "string",
                "value": "228"
            }
        ],
        "capacity":
        [
            {
                "@type": "Capacity",
                "id": "cf8d1665-a3cc-470c-97eb-782f226075ce",
                "href": "https://{server}:{port}/tmf-
api/resourcePool/v5/capacity/cf8d1665-a3cc-470c-97eb-782f226075ce",
                "name": "Capacity of Phone Number pool for MyPartner",
                "capacitySpecification": {
                    "id": "3d01344d-4409-4b15-8fcb-b8802314ff9c",
                    "href": "https://{server}:{port}/tmf-
api/resourcePoolManagement/v5/capacitySpecification/3d01344d-4409-4b15-8fcb-
b8802314ff9c",
                    "@type": "CapacitySpecificationRef",
                    "@referredType": "CapacitySpecification"
                },
                "capacityStatus": "actual",
                "capacityAmount":
                {
                    "amount": 1000,
                    "units": "number"
                },
                "capacityCharacteristic":
                [
```


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```
                    {
                        "@type": "StringCharacteristic",
                        "@baseType": "Characteristic",
                        "name": "lowerNumber",
                        "valueType": "string",
                        "value": "+49-228-181-000"
                    },
                    {
                        "@type": "StringCharacteristic",
                        "@baseType": "Characteristic",
                        "name": "upperNumber",
                        "valueType": "string",
                        "value": "+49-228-181-999"
                    }
                ]
            }
        ],
        "resourceSpecification":
        {
            "id": "b0e0827c-5693-4e41-8445-22438abb33cb",
            "href": "https://{server}:{port}/tmf-
api/resourcePoolManagement/v5/resourcePoolSpecification/b0e0827c-5693-4e41-8445-
22438abb33cb",
            "@type": "ResourceSpecificationRef",
            "@referredType": "ResourcePoolSpecification"
        },
        "relatedParty":
        [
            {
                "@type": "RelatedPartyRefOrPartyRoleRef",
                "role": "Partner",
                "partyOrPartyRole":
                {
                    "@type": "PartyRef",
                    "@referredType": "Organization",
                    "id": "47abd723-d3ca-49d8-ba53-76aa62e867e9",
                    "href": "https://{server}:{port}/tmf-
api/partyManagement/v5/organization/47abd723-d3ca-49d8-ba53-76aa62e867e9",
                    "name": "MyPartner"
                }
            }
        ]
    }
]
```


Retrieves a ResourcePool by ID

`GET /resourcePool/{id}?fields=…`

Description

This operation retrieves a ResourcePool entity. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Here's an example of a request for retrieving a ResourcePool resource

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Request

```
GET /resourcePool/c3fef858-b864-4fac-971e-795b1195ded9
Content-Type: application/json
```

Response

```
200

{
    "@type": "ResourcePool",
    "id": "c3fef858-b864-4fac-971e-795b1195ded9",
    "href": "https://{server}:{port}/tmf-
api/resourcePoolManagement/v5/resourcePool/c3fef858-b864-4fac-971e-795b1195ded9",
    "name": "Phone Number block for partner MyPartner",
    "description": "This Pool collects the PhoneNumber resources assigned to the
partner MyPartner.",
    "pooledResourceSpecification":
    {
        "@type": "ResourceSpecificationRef",
        "@referredType": "ResourceSpecification",
        "id": "89afac38-32e2-40f8-8be2-7b86dd03c9c9",
        "href": "https://{server}:{port}/tmf-
api/resourceCatalogManagement/v5/resourceSpecification/89afac38-32e2-40f8-8be2-
7b86dd03c9c9",
        "name": "Phone Number"
    },
    "resourceCharacteristic":
    [
        {
            "@type": "StringCharacteristic",
            "@baseType": "Characteristic",
            "name": "cc",
            "valueType": "string",
            "value": "49"
        },
        {
            "@type": "StringCharacteristic",
            "@baseType": "Characteristic",
            "name": "ndc",
            "valueType": "string",
            "value": "228"
        }
    ],
    "capacity":
    [
        {
            "@type": "Capacity",
            "id": "cf8d1665-a3cc-470c-97eb-782f226075ce",
            "href": "https://{server}:{port}/tmf-
api/resourcePool/v5/capacity/cf8d1665-a3cc-470c-97eb-782f226075ce",
            "name": "Capacity of Phone Number pool for MyPartner",
            "capacitySpecification": {
                "id": "3d01344d-4409-4b15-8fcb-b8802314ff9c",
                "href": "https://{server}:{port}/tmf-
api/resourcePoolManagement/v5/capacitySpecification/3d01344d-4409-4b15-8fcb-
b8802314ff9c",
                "@type": "CapacitySpecificationRef",
                "@referredType": "CapacitySpecification"
```

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```
            },
            "capacityStatus": "actual",
            "capacityAmount":
            {
                "amount": 1000,
                "units": "number"
            },
            "capacityCharacteristic":
            [
                {
                    "@type": "StringCharacteristic",
                    "@baseType": "Characteristic",
                    "name": "lowerNumber",
                    "valueType": "string",
                    "value": "+49-228-181-000"
                },
                {
                    "@type": "StringCharacteristic",
                    "@baseType": "Characteristic",
                    "name": "upperNumber",
                    "valueType": "string",
                    "value": "+49-228-181-999"
                }
            ]
        }
    ],
    "resourceSpecification":
    {
        "id": "b0e0827c-5693-4e41-8445-22438abb33cb",
        "href": "https://{server}:{port}/tmf-
api/resourcePoolManagement/v5/resourcePoolSpecification/b0e0827c-5693-4e41-8445-
22438abb33cb",
        "@type": "ResourceSpecificationRef",
        "@referredType": "ResourcePoolSpecification"
    },
    "relatedParty":
    [
        {
            "@type": "RelatedPartyRefOrPartyRoleRef",
            "role": "Partner",
            "partyOrPartyRole":
            {
                "@type": "PartyRef",
                "@referredType": "Organization",
                "id": "47abd723-d3ca-49d8-ba53-76aa62e867e9",
                "href": "https://{server}:{port}/tmf-
api/partyManagement/v5/organization/47abd723-d3ca-49d8-ba53-76aa62e867e9",
                "name": "MyPartner"
            }
        }
    ]
}
```


Creates a ResourcePool

`POST /resourcePool?fields=…`

Description

This operation creates a ResourcePool entity.

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Mandatory Attributes

Mandatory Attributes	Rule
@type	

Usage samples

Here's an example of a request for creating a ResourcePool resources.

Request

```
POST /resourcePool
Content-Type: application/json

{
    "@type": "ResourcePool",
    "name": "Phone Number block for partner MyPartner",
    "description": "This Pool collects the PhoneNumber resources assigned to the
partner MyPartner.",
    "pooledResourceSpecification":
    {
        "@type": "ResourceSpecificationRef",
        "@referredType": "ResourceSpecification",
        "id": "89afac38-32e2-40f8-8be2-7b86dd03c9c9",
        "href": "https://{server}:{port}/tmf-
api/resourceCatalogManagement/v5/resourceSpecification/89afac38-32e2-40f8-8be2-
7b86dd03c9c9",
        "name": "Phone Number"
    },
    "resourceCharacteristic":
    [
        {
            "@type": "StringCharacteristic",
            "@baseType": "Characteristic",
            "name": "cc",
            "valueType": "string",
            "value": "49"
        },
        {
            "@type": "StringCharacteristic",
            "@baseType": "Characteristic",
            "name": "ndc",
            "valueType": "string",
            "value": "228"
        }
    ],
    "capacity":
    [
        {
            "@type": "Capacity",
            "name": "Capacity of Phone Number pool for MyPartner",
            "capacitySpecification": {
                "id": "3d01344d-4409-4b15-8fcb-b8802314ff9c",
                "href": "https://{server}:{port}/tmf-
api/resourcePoolManagement/v5/capacitySpecification/3d01344d-4409-4b15-8fcb-
b8802314ff9c",
                "@type": "CapacitySpecificationRef",
                "@referredType": "CapacitySpecification"
            },
```

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```
            "capacityStatus": "actual",
            "capacityAmount":
            {
                "amount": 1000,
                "units": "number"
            },
            "capacityCharacteristic":
            [
                {
                    "@type": "StringCharacteristic",
                    "@baseType": "Characteristic",
                    "id": "bde67cfd-8d4a-4a22-a948-7565978aa0ed",
                    "name": "lowerNumber",
                    "valueType": "string",
                    "value": "+49-228-181-000"
                },
                {
                    "@type": "StringCharacteristic",
                    "@baseType": "Characteristic",
                    "id": "2612c070-d3a9-4cd8-9765-55309d5e1d16",
                    "name": "upperNumber",
                    "valueType": "string",
                    "value": "+49-228-181-999"
                }
            ]
        }
    ],
    "resourceSpecification":
    {
        "id": "b0e0827c-5693-4e41-8445-22438abb33cb",
        "href": "https://{server}:{port}/tmf-
api/resourcePoolManagement/v5/resourcePoolSpecification/b0e0827c-5693-4e41-8445-
22438abb33cb",
        "@type": "ResourceSpecificationRef",
        "@referredType": "ResourcePoolSpecification"
    },
    "relatedParty":
    [
        {
            "@type": "RelatedPartyRefOrPartyRoleRef",
            "role": "Partner",
            "partyOrPartyRole":
            {
                "@type": "PartyRef",
                "@referredType": "Organization",
                "id": "47abd723-d3ca-49d8-ba53-76aa62e867e9",
                "href": "https://{server}:{port}/tmf-
api/partyManagement/v5/organization/47abd723-d3ca-49d8-ba53-76aa62e867e9",
                "name": "MyPartner"
            }
        }
    ]
}
```


Response


```
201

{
    "@type": "ResourcePool",
```


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```
    "id": "c3fef858-b864-4fac-971e-795b1195ded9",
    "href": "https://{server}:{port}/tmf-
api/resourcePoolManagement/v5/resourcePool/c3fef858-b864-4fac-971e-795b1195ded9",
    "name": "Phone Number block for partner MyPartner",
    "description": "This Pool collects the PhoneNumber resources assigned to the
partner MyPartner.",
    "pooledResourceSpecification":
    {
        "@type": "ResourceSpecificationRef",
        "@referredType": "ResourceSpecification",
        "id": "89afac38-32e2-40f8-8be2-7b86dd03c9c9",
        "href": "https://{server}:{port}/tmf-
api/resourceCatalogManagement/v5/resourceSpecification/89afac38-32e2-40f8-8be2-
7b86dd03c9c9",
        "name": "Phone Number"
    },
    "resourceCharacteristic":
    [
        {
            "@type": "StringCharacteristic",
            "@baseType": "Characteristic",
            "name": "cc",
            "valueType": "string",
            "value": "49"
        },
        {
            "@type": "StringCharacteristic",
            "@baseType": "Characteristic",
            "name": "ndc",
            "valueType": "string",
            "value": "228"
        }
    ],
    "capacity":
    [
        {
            "@type": "Capacity",
            "id": "cf8d1665-a3cc-470c-97eb-782f226075ce",
            "href": "https://{server}:{port}/tmf-
api/resourcePool/v5/capacity/cf8d1665-a3cc-470c-97eb-782f226075ce",
            "name": "Capacity of Phone Number pool for MyPartner",
            "capacitySpecification": {
                "id": "3d01344d-4409-4b15-8fcb-b8802314ff9c",
                "href": "https://{server}:{port}/tmf-
api/resourcePoolManagement/v5/capacitySpecification/3d01344d-4409-4b15-8fcb-
b8802314ff9c",
                "@type": "CapacitySpecificationRef",
                "@referredType": "CapacitySpecification"
            },
            "capacityStatus": "actual",
            "capacityAmount":
            {
                "amount": 1000,
                "units": "number"
            },
            "capacityCharacteristic":
            [
                {
                    "@type": "StringCharacteristic",
                    "@baseType": "Characteristic",
                    "name": "lowerNumber",
                    "valueType": "string",
                    "value": "+49-228-181-000"
```


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```
                },
                {
                    "@type": "StringCharacteristic",
                    "@baseType": "Characteristic",
                    "name": "upperNumber",
                    "valueType": "string",
                    "value": "+49-228-181-999"
                }
            ]
        }
    ],
    "resourceSpecification":
    {
        "id": "b0e0827c-5693-4e41-8445-22438abb33cb",
        "href": "https://{server}:{port}/tmf-
api/resourcePoolManagement/v5/resourcePoolSpecification/b0e0827c-5693-4e41-8445-
22438abb33cb",
        "@type": "ResourceSpecificationRef",
        "@referredType": "ResourcePoolSpecification"
    },
    "relatedParty":
    [
        {
            "@type": "RelatedPartyRefOrPartyRoleRef",
            "role": "Partner",
            "partyOrPartyRole":
            {
                "@type": "PartyRef",
                "@referredType": "Organization",
                "id": "47abd723-d3ca-49d8-ba53-76aa62e867e9",
                "href": "https://{server}:{port}/tmf-
api/partyManagement/v5/organization/47abd723-d3ca-49d8-ba53-76aa62e867e9",
                "name": "MyPartner"
            }
        }
    ]
}
```


Updates partially a ResourcePool

`PATCH /resourcePool/{id}?fields=…`

Description

This operation allows partial updates of a ResourcePool entity. Support of json/merge (https://tools.ietf.org/html/rfc7396) is mandatory, support of json/patch (http://tools.ietf.org/html/rfc5789) is optional. Note: If the update operation yields to the creation of sub-resources or relationships, the same rules concerning mandatory sub-resource attributes and default value settings in the POST operation applies to the PATCH operation. Hence these tables are not repeated here.

Patchable and Non Patchable Attributes

Non Patchable Attributes	Rule
href	
id	

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Non Patchable Attributes	Rule
@baseType	@baseType is immutable
@schemaLocation	@schemaLocation is immutable
@type	@type is immutable

Patchable Attributes	Rule
activationFeature	
administrativeState	
attachment	
capacity	
category	
description	
endOperatingDate	
externalIdentifier	
intent	
name	
note	
operationalState	
place	
pooledResource	
pooledResourceSpecification	
relatedParty	
resourceCharacteristic	
resourceOrderItem	
resourceRelationship	
resourceSpecification	
resourceStatus	
resourceVersion	
startOperatingDate	
supportingResource	
usageState	
validFor	
value	

Usage samples

Here is an example of a request for patching a ResourcePool using APPLICATION-JSON

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syntax.

Request

```
PATCH /resourcePool/c3fef858-b864-4fac-971e-795b1195ded9
Content-Type: application/json

{
    "@type": "ResourcePool",
    "name": "Phone Number block for partner 'MyPartner'",
    "description": "This Pool collects the PhoneNumber resources assigned to the
partner 'MyPartner'.",
    "validFor":
    {
        "startDateTime": "2024-05-13T00:00:00Z",
        "endDateTime": "2024-09-20T00:00:00Z"
    }
}
```

Response

```
200

{
    "@type": "ResourcePool",
    "id": "c3fef858-b864-4fac-971e-795b1195ded9",
    "href": "https://{server}:{port}/tmf-
api/resourcePoolManagement/v5/resourcePool/c3fef858-b864-4fac-971e-795b1195ded9",
    "name": "Phone Number block for partner 'MyPartner'",
    "description": "This Pool collects the PhoneNumber resources assigned to the
partner 'MyPartner'.",
    "pooledResourceSpecification":
    {
        "@type": "ResourceSpecificationRef",
        "@referredType": "ResourceSpecification",
        "id": "89afac38-32e2-40f8-8be2-7b86dd03c9c9",
        "href": "https://{server}:{port}/tmf-
api/resourceCatalogManagement/v5/resourceSpecification/89afac38-32e2-40f8-8be2-
7b86dd03c9c9",
        "name": "Phone Number"
    },
    "resourceCharacteristic":
    [
        {
            "@type": "StringCharacteristic",
            "@baseType": "Characteristic",
            "name": "cc",
            "valueType": "string",
            "value": "49"
        },
        {
            "@type": "StringCharacteristic",
            "@baseType": "Characteristic",
            "name": "ndc",
            "valueType": "string",
            "value": "228"
        }
    ],
    "capacity":
```

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```
    [
        {
            "@type": "Capacity",
            "id": "cf8d1665-a3cc-470c-97eb-782f226075ce",
            "href": "https://{server}:{port}/tmf-
api/resourcePool/v5/capacity/cf8d1665-a3cc-470c-97eb-782f226075ce",
            "name": "Capacity of Phone Number pool for MyPartner",
            "capacitySpecification": {
                "id": "3d01344d-4409-4b15-8fcb-b8802314ff9c",
                "href": "https://{server}:{port}/tmf-
api/resourcePoolManagement/v5/capacitySpecification/3d01344d-4409-4b15-8fcb-
b8802314ff9c",
                "@type": "CapacitySpecificationRef",
                "@referredType": "CapacitySpecification"
            },
            "capacityStatus": "actual",
            "capacityAmount":
            {
                "amount": 1000,
                "units": "number"
            },
            "capacityCharacteristic":
            [
                {
                    "@type": "StringCharacteristic",
                    "@baseType": "Characteristic",
                    "name": "lowerNumber",
                    "valueType": "string",
                    "value": "+49-228-181-000"
                },
                {
                    "@type": "StringCharacteristic",
                    "@baseType": "Characteristic",
                    "name": "upperNumber",
                    "valueType": "string",
                    "value": "+49-228-181-999"
                }
            ]
        }
    ],
    "resourceSpecification":
    {
        "id": "b0e0827c-5693-4e41-8445-22438abb33cb",
        "href": "https://{server}:{port}/tmf-
api/resourcePoolManagement/v5/resourcePoolSpecification/b0e0827c-5693-4e41-8445-
22438abb33cb",
        "@type": "ResourceSpecificationRef",
        "@referredType": "ResourcePoolSpecification"
    },
    "relatedParty":
    [
        {
            "@type": "RelatedPartyRefOrPartyRoleRef",
            "role": "Partner",
            "partyOrPartyRole":
            {
                "@type": "PartyRef",
                "@referredType": "Organization",
                "id": "47abd723-d3ca-49d8-ba53-76aa62e867e9",
                "href": "https://{server}:{port}/tmf-
api/partyManagement/v5/organization/47abd723-d3ca-49d8-ba53-76aa62e867e9",
                "name": "MyPartner"
            }
```


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```
        }
    ],
    "validFor":
    {
        "startDateTime": "2024-05-13T00:00:00Z",
        "endDateTime": "2024-09-20T00:00:00Z"
    }
}
```

Here is an example of a request for patching a ResourcePool using MERGE syntax.

Request

```
PATCH /resourcePool/c3fef858-b864-4fac-971e-795b1195ded9
Content-Type: application/merge-patch+json

{
    "@type": "ResourcePool",
    "name": "Phone Number block for partner 'MyPartner'",
    "description": "This Pool collects the PhoneNumber resources assigned to the
partner 'MyPartner'.",
    "validFor":
    {
        "startDateTime": "2024-05-13T00:00:00Z",
        "endDateTime": "2024-09-20T00:00:00Z"
    }
}
```

Response

```
200

{
    "@type": "ResourcePool",
    "id": "c3fef858-b864-4fac-971e-795b1195ded9",
    "href": "https://{server}:{port}/tmf-
api/resourcePoolManagement/v5/resourcePool/c3fef858-b864-4fac-971e-795b1195ded9",
    "name": "Phone Number block for partner 'MyPartner'",
    "description": "This Pool collects the PhoneNumber resources assigned to the
partner 'MyPartner'.",
    "pooledResourceSpecification":
    {
        "@type": "ResourceSpecificationRef",
        "@referredType": "ResourceSpecification",
        "id": "89afac38-32e2-40f8-8be2-7b86dd03c9c9",
        "href": "https://{server}:{port}/tmf-
api/resourceCatalogManagement/v5/resourceSpecification/89afac38-32e2-40f8-8be2-
7b86dd03c9c9",
        "name": "Phone Number"
    },
    "resourceCharacteristic":
    [
        {
            "@type": "StringCharacteristic",
            "@baseType": "Characteristic",
            "name": "cc",
            "valueType": "string",
```

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```
            "value": "49"
        },
        {
            "@type": "StringCharacteristic",
            "@baseType": "Characteristic",
            "name": "ndc",
            "valueType": "string",
            "value": "228"
        }
    ],
    "capacity":
    [
        {
            "@type": "Capacity",
            "id": "cf8d1665-a3cc-470c-97eb-782f226075ce",
            "href": "https://{server}:{port}/tmf-
api/resourcePool/v5/capacity/cf8d1665-a3cc-470c-97eb-782f226075ce",
            "name": "Capacity of Phone Number pool for MyPartner",
            "capacitySpecification": {
                "id": "3d01344d-4409-4b15-8fcb-b8802314ff9c",
                "href": "https://{server}:{port}/tmf-
api/resourcePoolManagement/v5/capacitySpecification/3d01344d-4409-4b15-8fcb-
b8802314ff9c",
                "@type": "CapacitySpecificationRef",
                "@referredType": "CapacitySpecification"
            },
            "capacityStatus": "actual",
            "capacityAmount":
            {
                "amount": 1000,
                "units": "number"
            },
            "capacityCharacteristic":
            [
                {
                    "@type": "StringCharacteristic",
                    "@baseType": "Characteristic",
                    "name": "lowerNumber",
                    "valueType": "string",
                    "value": "+49-228-181-000"
                },
                {
                    "@type": "StringCharacteristic",
                    "@baseType": "Characteristic",
                    "name": "upperNumber",
                    "valueType": "string",
                    "value": "+49-228-181-999"
                }
            ]
        }
    ],
    "resourceSpecification":
    {
        "id": "b0e0827c-5693-4e41-8445-22438abb33cb",
        "href": "https://{server}:{port}/tmf-
api/resourcePoolManagement/v5/resourcePoolSpecification/b0e0827c-5693-4e41-8445-
22438abb33cb",
        "@type": "ResourceSpecificationRef",
        "@referredType": "ResourcePoolSpecification"
    },
    "relatedParty":
    [
        {
```


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```
            "@type": "RelatedPartyRefOrPartyRoleRef",
            "role": "Partner",
            "partyOrPartyRole":
            {
                "@type": "PartyRef",
                "@referredType": "Organization",
                "id": "47abd723-d3ca-49d8-ba53-76aa62e867e9",
                "href": "https://{server}:{port}/tmf-
api/partyManagement/v5/organization/47abd723-d3ca-49d8-ba53-76aa62e867e9",
                "name": "MyPartner"
            }
        }
    ],
    "validFor":
    {
        "startDateTime": "2024-05-13T00:00:00Z",
        "endDateTime": "2024-09-20T00:00:00Z"
    }
}
```

Here is an example of a request for patching a ResourcePool using JSON-PATCH syntax.

Request

```
PATCH /resourcePool/c3fef858-b864-4fac-971e-795b1195ded9
Content-Type: application/json-patch+json

[
    {
        "op": "add",
        "path": "/resourceCharacteristic",
        "value":
        {
            "@type": "StringCharacteristic",
            "@baseType": "Characteristic",
            "name": "sn",
            "valueType": "string",
            "value": "181"
        }
    }
]
```

Response

```
200

{
    "@type": "ResourcePool",
    "id": "c3fef858-b864-4fac-971e-795b1195ded9",
    "href": "https://{server}:{port}/tmf-
api/resourcePoolManagement/v5/resourcePool/c3fef858-b864-4fac-971e-795b1195ded9",
    "name": "Phone Number block for partner MyPartner",
    "description": "This Pool collects the PhoneNumber resources assigned to the
partner MyPartner.",
    "pooledResourceSpecification":
    {
        "@type": "ResourceSpecificationRef",
```

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```
        "@referredType": "ResourceSpecification",
        "id": "89afac38-32e2-40f8-8be2-7b86dd03c9c9",
        "href": "https://{server}:{port}/tmf-
api/resourceCatalogManagement/v5/resourceSpecification/89afac38-32e2-40f8-8be2-
7b86dd03c9c9",
        "name": "Phone Number"
    },
    "resourceCharacteristic":
    [
        {
            "@type": "StringCharacteristic",
            "@baseType": "Characteristic",
            "name": "cc",
            "valueType": "string",
            "value": "49"
        },
        {
            "@type": "StringCharacteristic",
            "@baseType": "Characteristic",
            "name": "ndc",
            "valueType": "string",
            "value": "228"
        },
        {
            "@type": "StringCharacteristic",
            "@baseType": "Characteristic",
            "name": "sn",
            "valueType": "string",
            "value": "181"
        }
    ],
    "capacity":
    [
        {
            "@type": "Capacity",
            "id": "cf8d1665-a3cc-470c-97eb-782f226075ce",
            "href": "https://{server}:{port}/tmf-
api/resourcePool/v5/capacity/cf8d1665-a3cc-470c-97eb-782f226075ce",
            "name": "Capacity of Phone Number pool for MyPartner",
            "capacitySpecification": {
                "id": "3d01344d-4409-4b15-8fcb-b8802314ff9c",
                "href": "https://{server}:{port}/tmf-
api/resourcePoolManagement/v5/capacitySpecification/3d01344d-4409-4b15-8fcb-
b8802314ff9c",
                "@type": "CapacitySpecificationRef",
                "@referredType": "CapacitySpecification"
            },
            "capacityStatus": "actual",
            "capacityAmount":
            {
                "amount": 1000,
                "units": "number"
            },
            "capacityCharacteristic":
            [
                {
                    "@type": "StringCharacteristic",
                    "@baseType": "Characteristic",
                    "name": "lowerNumber",
                    "valueType": "string",
                    "value": "+49-228-181-000"
                },
                {
```


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```
                    "@type": "StringCharacteristic",
                    "@baseType": "Characteristic",
                    "name": "upperNumber",
                    "valueType": "string",
                    "value": "+49-228-181-999"
                }
            ]
        }
    ],
    "resourceSpecification":
    {
        "id": "b0e0827c-5693-4e41-8445-22438abb33cb",
        "href": "https://{server}:{port}/tmf-
api/resourcePoolManagement/v5/resourcePoolSpecification/b0e0827c-5693-4e41-8445-
22438abb33cb",
        "@type": "ResourceSpecificationRef",
        "@referredType": "ResourcePoolSpecification"
    },
    "relatedParty":
    [
        {
            "@type": "RelatedPartyRefOrPartyRoleRef",
            "role": "Partner",
            "partyOrPartyRole":
            {
                "@type": "PartyRef",
                "@referredType": "Organization",
                "id": "47abd723-d3ca-49d8-ba53-76aa62e867e9",
                "href": "https://{server}:{port}/tmf-
api/partyManagement/v5/organization/47abd723-d3ca-49d8-ba53-76aa62e867e9",
                "name": "MyPartner"
            }
        }
    ]
}
```

Here is an example of a request for patching a ResourcePool using JSON-PATCH-QUERY syntax.

Request

```
PATCH /resourcePool/c3fef858-b864-4fac-971e-795b1195ded9
Content-Type: application/json-patch-query+json


[
    {
        "op": "replace",
        "path": "/name",
        "value": "Phone Number block for partner 'MyPartner'"
    },
    {
        "op": "replace",
        "path": "/description",
        "value": "This Pool collects the PhoneNumber resources assigned to the
partner 'MyPartner'."
    },
    {
        "op": "add",
        "path": "/validFor",
```

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```
        "value":
        {
            "startDateTime": "2024-05-13T00:00:00Z",
            "endDateTime": "2024-09-20T00:00:00Z"
        }
    },
    {
        "op": "replace",
        "path": "$.relatedParty[?(@.partyOrPartyRole.name == 'MyPartner')].role",
        "value": "Business Customer"
    }
]
```

Response

```
200

{
    "@type": "ResourcePool",
    "id": "c3fef858-b864-4fac-971e-795b1195ded9",
    "href": "https://{server}:{port}/tmf-
api/resourcePoolManagement/v5/resourcePool/c3fef858-b864-4fac-971e-795b1195ded9",
    "name": "Phone Number block for partner 'MyPartner'",
    "description": "This Pool collects the PhoneNumber resources assigned to the
partner 'MyPartner'.",
    "pooledResourceSpecification":
    {
        "@type": "ResourceSpecificationRef",
        "@referredType": "ResourceSpecification",
        "id": "89afac38-32e2-40f8-8be2-7b86dd03c9c9",
        "href": "https://{server}:{port}/tmf-
api/resourceCatalogManagement/v5/resourceSpecification/89afac38-32e2-40f8-8be2-
7b86dd03c9c9",
        "name": "Phone Number"
    },
    "resourceCharacteristic":
    [
        {
            "@type": "StringCharacteristic",
            "@baseType": "Characteristic",
            "name": "cc",
            "valueType": "string",
            "value": "49"
        },
        {
            "@type": "StringCharacteristic",
            "@baseType": "Characteristic",
            "name": "ndc",
            "valueType": "string",
            "value": "228"
        },
        {
            "@type": "StringCharacteristic",
            "@baseType": "Characteristic",
            "name": "sn",
            "valueType": "string",
            "value": "181"
        }
    ],
    "capacity":
```

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```
    [
        {
            "@type": "Capacity",
            "id": "cf8d1665-a3cc-470c-97eb-782f226075ce",
            "href": "https://{server}:{port}/tmf-
api/resourcePool/v5/capacity/cf8d1665-a3cc-470c-97eb-782f226075ce",
            "name": "Capacity of Phone Number pool for MyPartner",
            "capacitySpecification": {
                "id": "3d01344d-4409-4b15-8fcb-b8802314ff9c",
                "href": "https://{server}:{port}/tmf-
api/resourcePoolManagement/v5/capacitySpecification/3d01344d-4409-4b15-8fcb-
b8802314ff9c",
                "@type": "CapacitySpecificationRef",
                "@referredType": "CapacitySpecification"
            },
            "capacityStatus": "actual",
            "capacityAmount":
            {
                "amount": 1000,
                "units": "number"
            },
            "capacityCharacteristic":
            [
                {
                    "@type": "StringCharacteristic",
                    "@baseType": "Characteristic",
                    "name": "lowerNumber",
                    "valueType": "string",
                    "value": "+49-228-181-000"
                },
                {
                    "@type": "StringCharacteristic",
                    "@baseType": "Characteristic",
                    "name": "upperNumber",
                    "valueType": "string",
                    "value": "+49-228-181-999"
                }
            ]
        }
    ],
    "resourceSpecification":
    {
        "id": "b0e0827c-5693-4e41-8445-22438abb33cb",
        "href": "https://{server}:{port}/tmf-
api/resourcePoolManagement/v5/resourcePoolSpecification/b0e0827c-5693-4e41-8445-
22438abb33cb",
        "@type": "ResourceSpecificationRef",
        "@referredType": "ResourcePoolSpecification"
    },
    "relatedParty":
    [
        {
            "@type": "RelatedPartyRefOrPartyRoleRef",
            "role": "Business Customer",
            "partyOrPartyRole":
            {
                "@type": "PartyRef",
                "@referredType": "Organization",
                "id": "47abd723-d3ca-49d8-ba53-76aa62e867e9",
                "href": "https://{server}:{port}/tmf-
api/partyManagement/v5/organization/47abd723-d3ca-49d8-ba53-76aa62e867e9",
                "name": "MyPartner"
            }
```


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```
        }
    ],
    "validFor":
    {
        "startDateTime": "2024-05-13T00:00:00Z",
        "endDateTime": "2024-09-20T00:00:00Z"
    }
}
```

Deletes a ResourcePool

`DELETE /resourcePool/{id}`

Description

This operation deletes a ResourcePool entity.

Operations on AvailabilityCheck

List or find AvailabilityCheck objects

`GET /resourcePool/{resourcePoolId}/availabilityCheck?fields=…`

Description

This operation list AvailabilityCheck entities. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Here's an example of a request for retrieving AvailabilityCheck resources.

Request

```
GET /resourcePool/c3fef858-b864-4fac-971e-795b1195ded9/availabilityCheck
Content-Type: application/json
```

Response

```
200

[
    {
        "@type": "AvailabilityCheck",
        "id": "8cb645cb-279f-42f4-9d3b-72845bbfafbe",
        "href": "https://{server}:{port}/tmf-
api/resourcePoolManagement/v5/resourcePool/c3fef858-b864-4fac-971e-
795b1195ded9/availabilityCheck/8cb645cb-279f-42f4-9d3b-72845bbfafbe",
        "capacityDemand":
        {
            "@type": "Capacity",
            "capacityAmount":
```

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```
            {
                "amount": 1,
                "units": "number"
            }
        },
        "state": "inProgress"
    }
]
```

Retrieves a AvailabilityCheck by ID

`GET /resourcePool/{resourcePoolId}/availabilityCheck/{id}?fields=…`

Description

This operation retrieves an AvailabilityCheck entity. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Here's an example of a request for retrieving a AvailabilityCheck resource

Request

```
GET /resourcePool/c3fef858-b864-4fac-971e-795b1195ded9/availabilityCheck/8cb645cb-
279f-42f4-9d3b-72845bbfafbe
Content-Type: application/json
```

Response

```
200

{
    "@type": "AvailabilityCheck",
    "id": "8cb645cb-279f-42f4-9d3b-72845bbfafbe",
    "href": "https://{server}:{port}/tmf-
api/resourcePoolManagement/v5/resourcePool/c3fef858-b864-4fac-971e-
795b1195ded9/availabilityCheck/8cb645cb-279f-42f4-9d3b-72845bbfafbe",
    "capacityDemand":
    {
        "@type": "Capacity",
        "capacityAmount":
        {
            "amount": 1,
            "units": "number"
        }
    },
    "state": "inProgress"
}
```

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Creates a AvailabilityCheck

POST /resourcePool/{resourcePoolId}/availabilityCheck?fields=…

Description

This operation creates an AvailabilityCheck entity.

Mandatory Attributes

Mandatory Attributes	Rule
@type	

Usage samples

Here's an example of a request for creating a AvailabilityCheck resources.

Request

```
POST /resourcePool/c3fef858-b864-4fac-971e-795b1195ded9/availabilityCheck
Content-Type: application/json

{
    "@type": "AvailabilityCheck",
    "capacityDemand":
    {
        "@type": "Capacity",
        "capacityAmount":
        {
            "amount": 1,
            "units": "number"
        }
    }
}
```

Response

```
201

{
    "@type": "AvailabilityCheck",
    "id": "8cb645cb-279f-42f4-9d3b-72845bbfafbe",
    "href": "https://{server}:{port}/tmf-
api/resourcePoolManagement/v5/resourcePool/c3fef858-b864-4fac-971e-
795b1195ded9/availabilityCheck/8cb645cb-279f-42f4-9d3b-72845bbfafbe",
    "capacityDemand":
    {
        "@type": "Capacity",
        "capacityAmount":
        {
            "amount": 10,
            "units": "number"
        }
    },
    "capacityOption":
```

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```
    [
        {
            "@type": "Capacity",
            "capacityAmount":
            {
                "amount": 6,
                "units": "number"
            }
        },
        {
            "@type": "Capacity",
            "capacityAmount":
            {
                "amount": 4,
                "units": "number"
            }
        }
    ],
    "state": "inProgress"
}
```

Operations on Push

Retrieves a Push by ID

`GET /resourcePool/{resourcePoolId}/push/{id}?fields=…`

Description

This operation retrieves a Push entity. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Here's an example of a request for retrieving a Push resource

Request

```
GET /resourcePool/c3fef858-b864-4fac-971e-795b1195ded9/push/ae63d76e-6d00-45aa-97a6-
c8a2b54390e5
Content-Type: application/json
```

Response

```
200

{
    "@type": "Push",
    "id": "ae63d76e-6d00-45aa-97a6-c8a2b54390e5",
    "href": "https://{server}:{port}/tmf-
api/resourcePoolManagement/v5/resourcePool/c3fef858-b864-4fac-971e-
795b1195ded9/push/ae63d76e-6d00-45aa-97a6-c8a2b54390e5",
    "pushedResource":
    [
```

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```
        {
            "@type": "ResourceRef",
            "@referredType": "LogicalResource",
            "id": "4d179a07-1555-4e5f-a2de-0a51f5dd7e15",
            "href": "https://{server}:{port}/tmf-
api/resourceInventoryManagement/v5/resource/4d179a07-1555-4e5f-a2de-0a51f5dd7e15"
        },
        {
            "@type": "ResourceRef",
            "@referredType": "LogicalResource",
            "id": "15dfa8a7-f9db-43a8-8758-9d9df5a76264",
            "href": "https://{server}:{port}/tmf-
api/resourceInventoryManagement/v5/resource/15dfa8a7-f9db-43a8-8758-9d9df5a76264"
        },
        {
            "@type": "ResourceRef",
            "@referredType": "LogicalResource",
            "id": "195aa8ee-7560-4d97-9b9a-2add1a4e0372",
            "href": "https://{server}:{port}/tmf-
api/resourceInventoryManagement/v5/resource/195aa8ee-7560-4d97-9b9a-2add1a4e0372"
        }
    ],
    "state": "inProgress"
}
```

List or find Push objects

`GET /resourcePool/{resourcePoolId}/push?fields=…`

Description

This operation list Push entities. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Here's an example of a request for retrieving Push resources.

Request

```
GET /resourcePool/c3fef858-b864-4fac-971e-795b1195ded9/push
Content-Type: application/json
```

Response

```
200

[
    {
        "@type": "Push",
        "id": "ae63d76e-6d00-45aa-97a6-c8a2b54390e5",
        "href": "https://{server}:{port}/tmf-
api/resourcePoolManagement/v5/resourcePool/c3fef858-b864-4fac-971e-
795b1195ded9/push/ae63d76e-6d00-45aa-97a6-c8a2b54390e5",
```

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```
        "pushedResource":
        [
            {
                "@type": "ResourceRef",
                "@referredType": "LogicalResource",
                "id": "4d179a07-1555-4e5f-a2de-0a51f5dd7e15",
                "href": "https://{server}:{port}/tmf-
api/resourceInventoryManagement/v5/resource/4d179a07-1555-4e5f-a2de-0a51f5dd7e15"
            },
            {
                "@type": "ResourceRef",
                "@referredType": "LogicalResource",
                "id": "15dfa8a7-f9db-43a8-8758-9d9df5a76264",
                "href": "https://{server}:{port}/tmf-
api/resourceInventoryManagement/v5/resource/15dfa8a7-f9db-43a8-8758-9d9df5a76264"
            },
            {
                "@type": "ResourceRef",
                "@referredType": "LogicalResource",
                "id": "195aa8ee-7560-4d97-9b9a-2add1a4e0372",
                "href": "https://{server}:{port}/tmf-
api/resourceInventoryManagement/v5/resource/195aa8ee-7560-4d97-9b9a-2add1a4e0372"
            }
        ],
        "state": "inProgress"
    }
]
```

Creates a Push

POST /resourcePool/{resourcePoolId}/push?fields=…

Description

This operation creates a Push entity.

Mandatory Attributes

Mandatory Attributes	Rule
@type	

Usage samples

Here's an example of a request for creating a Push resources.

Request

```
POST /resourcePool/c3fef858-b864-4fac-971e-795b1195ded9/push
Content-Type: application/json

{
    "@type": "Push",
    "pushedResource":
    [
        {
            "@type": "ResourceRef",
```

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```
            "@referredType": "LogicalResource",
            "id": "4d179a07-1555-4e5f-a2de-0a51f5dd7e15",
            "href": "https://{server}:{port}/tmf-
api/resourceInventoryManagement/v5/resource/4d179a07-1555-4e5f-a2de-0a51f5dd7e15"
        },
        {
            "@type": "ResourceRef",
            "@referredType": "LogicalResource",
            "id": "15dfa8a7-f9db-43a8-8758-9d9df5a76264",
            "href": "https://{server}:{port}/tmf-
api/resourceInventoryManagement/v5/resource/15dfa8a7-f9db-43a8-8758-9d9df5a76264"
        },
        {
            "@type": "ResourceRef",
            "@referredType": "LogicalResource",
            "id": "195aa8ee-7560-4d97-9b9a-2add1a4e0372",
            "href": "https://{server}:{port}/tmf-
api/resourceInventoryManagement/v5/resource/195aa8ee-7560-4d97-9b9a-2add1a4e0372"
        }
    ]
}
```

Response

```
201

{
    "@type": "Push",
    "id": "ae63d76e-6d00-45aa-97a6-c8a2b54390e5",
    "href": "https://{server}:{port}/tmf-
api/resourcePoolManagement/v5/resourcePool/c3fef858-b864-4fac-971e-
795b1195ded9/push/ae63d76e-6d00-45aa-97a6-c8a2b54390e5",
    "pushedResource":
    [
        {
            "@type": "ResourceRef",
            "@referredType": "LogicalResource",
            "id": "4d179a07-1555-4e5f-a2de-0a51f5dd7e15",
            "href": "https://{server}:{port}/tmf-
api/resourceInventoryManagement/v5/resource/4d179a07-1555-4e5f-a2de-0a51f5dd7e15"
        },
        {
            "@type": "ResourceRef",
            "@referredType": "LogicalResource",
            "id": "15dfa8a7-f9db-43a8-8758-9d9df5a76264",
            "href": "https://{server}:{port}/tmf-
api/resourceInventoryManagement/v5/resource/15dfa8a7-f9db-43a8-8758-9d9df5a76264"
        },
        {
            "@type": "ResourceRef",
            "@referredType": "LogicalResource",
            "id": "195aa8ee-7560-4d97-9b9a-2add1a4e0372",
            "href": "https://{server}:{port}/tmf-
api/resourceInventoryManagement/v5/resource/195aa8ee-7560-4d97-9b9a-2add1a4e0372"
        }
    ],
    "state": "inProgress"
}
```

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Operations on Extract

Retrieves a Extract by ID

`GET /resourcePool/{resourcePoolId}/extract/{id}?fields=…`

Description

This operation retrieves an Extract entity. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Here's an example of a request for retrieving a Extract resource

Request

```
GET /resourcePool/c3fef858-b864-4fac-971e-795b1195ded9/extract/c6a1e831-ef5b-4d09-
85a8-594fca73313b
Content-Type: application/json
```

Response

```
200

{
    "@type": "Extract",
    "id": "c6a1e831-ef5b-4d09-85a8-594fca73313b",
    "href": "https://{server}:{port}/tmf-
api/resourcePoolManagement/v5/resourcePool/c3fef858-b864-4fac-971e-
795b1195ded9/extract/c6a1e831-ef5b-4d09-85a8-594fca73313b",
    "capacityDemand":
    {
        "@type": "Capacity",
        "capacityAmount":
        {
            "amount": 2,
            "units": "number"
        },
        "capacityStatus": "planned",
        "applicableTimePeriod":
        [
            {
                "@type": "ApplicableTimePeriod",
                "fromToDateTime":
                {
                    "startDateTime": "2024-05-13T00:00:00Z",
                    "endDateTime": "2024-09-20T00:00:00Z"
                }
            }
        ]
    },
    "state": "inProgress"
}
```

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List or find Extract objects

GET /resourcePool/{resourcePoolId}/extract?fields=…

Description

This operation list Extract entities. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Here's an example of a request for retrieving Extract resources.

Request

```
GET /resourcePool/c3fef858-b864-4fac-971e-795b1195ded9/extract
Content-Type: application/json
```

Response

```
200

[
    {
        "@type": "Extract",
        "id": "c6a1e831-ef5b-4d09-85a8-594fca73313b",
        "href": "https://{server}:{port}/tmf-
api/resourcePoolManagement/v5/resourcePool/c3fef858-b864-4fac-971e-
795b1195ded9/extract/c6a1e831-ef5b-4d09-85a8-594fca73313b",
        "capacityDemand":
        {
            "@type": "Capacity",
            "capacityAmount":
            {
                "amount": 2,
                "units": "number"
            },
            "capacityStatus": "planned",
            "applicableTimePeriod":
            [
                {
                    "@type": "ApplicableTimePeriod",
                    "fromToDateTime":
                    {
                        "startDateTime": "2024-05-13T00:00:00Z",
                        "endDateTime": "2024-09-20T00:00:00Z"
                    }
                }
            ]
        },
        "state": "inProgress"
    }
]
```

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Creates a Extract

POST /resourcePool/{resourcePoolId}/extract?fields=…

Description

This operation creates an Extract entity.

Mandatory Attributes

Mandatory Attributes	Rule
@type	

Usage samples

Here's an example of a request for creating a Extract resources.

Request

```
POST /resourcePool/c3fef858-b864-4fac-971e-795b1195ded9/extract
Content-Type: application/json

{
    "@type": "Extract",
    "capacityDemand":
    {
        "@type": "Capacity",
        "capacityAmount":
        {
            "amount": 2,
            "units": "number"
        },
        "capacityStatus": "planned",
        "applicableTimePeriod":
        [
            {
                "@type": "ApplicableTimePeriod",
                "fromToDateTime":
                {
                    "startDateTime": "2024-05-13T00:00:00Z",
                    "endDateTime": "2024-09-20T00:00:00Z"
                }
            }
        ]
    }
}
```

Response

```
201

{
    "@type": "Extract",
    "id": "c6a1e831-ef5b-4d09-85a8-594fca73313b",
    "href": "https://{server}:{port}/tmf-
```

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```
api/resourcePoolManagement/v5/resourcePool/c3fef858-b864-4fac-971e-
795b1195ded9/extract/c6a1e831-ef5b-4d09-85a8-594fca73313b",
    "capacityDemand":
    {
        "@type": "Capacity",
        "capacityAmount":
        {
            "amount": 2,
            "units": "number"
        },
        "capacityStatus": "planned",
        "applicableTimePeriod":
        [
            {
                "@type": "ApplicableTimePeriod",
                "fromToDateTime":
                {
                    "startDateTime": "2024-05-13T00:00:00Z",
                    "endDateTime": "2024-09-20T00:00:00Z"
                }
            }
        ]
    },
    "extractedResource":
    [
        {
            "@type": "ResourceRef",
            "@referredType": "LogicalResource",
            "id": "4d179a07-1555-4e5f-a2de-0a51f5dd7e15",
            "href": "https://{server}:{port}/tmf-
api/resourceInventoryManagement/v5/resource/4d179a07-1555-4e5f-a2de-0a51f5dd7e15"
        },
        {
            "@type": "ResourceRef",
            "@referredType": "LogicalResource",
            "id": "15dfa8a7-f9db-43a8-8758-9d9df5a76264",
            "href": "https://{server}:{port}/tmf-
api/resourceInventoryManagement/v5/resource/15dfa8a7-f9db-43a8-8758-9d9df5a76264"
        }
    ],
    "state": "inProgress"
}
```


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API NOTIFICATIONS

For every single of operation on the entities use the following templates and provide sample REST notification POST calls.

It is assumed that the Pub/Sub uses the Register and UnRegister mechanisms described in the REST Guidelines reproduced below.

Register listener

POST /hub

Description

Sets the communication endpoint address the service instance must use to deliver information about its health state, execution state, failures and metrics. Subsequent POST calls will be rejected by the service if it does not support multiple listeners. In this case DELETE /api/hub/{id} must be called before an endpoint can be created again.

Behavior

Returns HTTP/1.1 status code 204 if the request was successful.

Returns HTTP/1.1 status code 409 if request is not successful.

Usage Samples

Here's an example of a request for registering a listener.

Request

```
POST /api/hub

Accept: application/json

{
    "callback": "http://in.listener.com"
}
```

Response

```
201

Content-Type: application/json

Location: /api/hub/42

{
    "id":"42",
    "callback": "http://in.listener.com",
    "query":""
}
```

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Unregister listener

DELETE /hub/{id}

Description

Clears the communication endpoint address that was set by creating the Hub..

Behavior

Returns HTTP/1.1 status code 204 if the request was successful.

Returns HTTP/1.1 status code 404 if the resource is not found.

Usage Samples

Here's an example of a request for un-registering a listener.

Request

```
DELETE /api/hub/42

Accept: application/json
```

Response

```
204
```

Publish Event to listener

POST /client/listener

Description

Clears the communication endpoint address that was set by creating the Hub.

Provides to a registered listener the description of the event that was raised. The /client/listener url is the callback url passed when registering the listener.

Behavior

Returns HTTP/1.1 status code 201 if the service is able to set the configuration.

Usage Samples

Here's an example of a notification received by the listener. In this example "EVENT TYPE" should be replaced by one of the notification types supported by this API (see Notification resources Models section) and EVENT BODY refers to the data structure of the given notification type.

Request

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```
POST /client/listener

Accept: application/json

{
    "event": {
        EVENT BODY
    },
    "eventType": "EVENT_TYPE"
}
```

Response

```
201
```

For detailed examples on the general TM Forum notification mechanism, see the TMF REST Design Guidelines.

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Acknowledgements

Release History

Release Number	Date	Release led by:	Description
Release 18.0.0	June 2018	sousuke izumida izumida.sousuke@nttcom.co.jp	Initial Release
Version 5.0.0	September 2023	Lutz Bettge Deutsche Telekom Lutz.Bettge@telekom.de	Upgrade to V5 design patterns and OAS3

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